

Every smooth complex Fano threefold is Oka

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October 6, 2026

Abstract

We establish convex approximation in arbitrary source dimension for smooth complex Fano threefolds. A single finite spectral construction treats the toric and rank-two matrix presentations of fourteen families, including sextic double solids, Gushel–Mukai threefolds, and the genus-eight models. It produces dominating holomorphic deformations without a global holomorphic selection of very free curves. A common analytic transfer theorem converts these scalar deformations into convex approximation. The classified models are organized by common mechanisms of holomorphic flexibility: the spectral transfer developed here, uniform rationality, the algebraic ellipticity of cubic hypersurfaces, and the passage from rationality to algebraic ellipticity.

Contents

1	Introduction	2
1.1	From the Oka–Grauert principle to Oka manifolds	3
1.2	Fano geometry and the approximation problem	3
1.3	The geometric and analytic ideas	4
1.4	Organization and reading guide	5
2	Oka-theoretic foundations	5
2.1	Stein spaces, approximation, and linear extension	5
2.2	Oka properties and convex approximation	6
2.3	Sprays on a target and sprays along a map	7
2.4	Sections, gluing, and localization	7
2.5	Finite entire scalar families	8
3	Fano threefolds and the guide to the geometric cases	8
3.1	Classification background and conventions	9
3.2	Weighted models, covers, and their blowups	9
3.3	Quadric, Grassmannian, and conic models	10

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2020 Mathematics Subject Classification. Primary 32Q56; Secondary 14J45, 14J30, 32E30.

3.4	The families supplied by external theorems	10
4	Finite entire directions imply convex approximation	11
4.1	Hypothesis and conclusion	11
4.2	Relative charts on a whole old compact	11
4.3	Scalar propagation with explicit choice order	12
4.4	One invocation of property P and the exact intermediate cores	16
4.5	Direct convex-cut passage to CAP	17
5	Spectral transfer for toric and matrix presentations	18
5.1	The two quotient presentations	18
5.2	Rooted coefficient maps and finite selectors	20
5.3	The spectral coefficient algebra	24
5.4	Moment weights and quotient gauges	30
5.5	The sufficient full residual system	34
5.6	The projective value ideal and actual source infinity	37
5.7	Completion	42
6	Fano models in smooth toric ambient varieties	42
6.1	Ordinary complete intersections and Verra threefolds	42
6.2	Double covers in smooth toric projective bundles	43
6.3	The double cover of the point blowup	45
6.4	The double flag as a regular codimension-two intersection	46
6.5	Weighted models and smooth toric resolutions	47
6.6	The elliptic-centre graph models	49
7	Balanced curves on the matrix models	50
7.1	Balanced curves through a prescribed point	50
7.2	The genus-eight tautological bundle	52
7.3	Gushel–Mukai threefolds	53
8	Completion of the classification argument	55
8.1	The classified families	55
8.2	Uniformly rational families	55
8.3	Cubic and rational families	57
8.4	The remaining families and the main theorem	58

1 Introduction

A folklore conjecture in Oka theory asserts that every Fano manifold is Oka. Forstnerič [33, Section 1] records this question. It asks whether the positivity of the anticanonical bundle entails holomorphic flexibility for maps from Stein spaces, with no restriction on the dimension of the source. We establish the three-dimensional case.

Theorem 1.1. *Every smooth complex Fano threefold is an Oka manifold.*

Here the Oka property includes homotopy, approximation, and interpolation for holomorphic maps from Stein spaces. An initial continuous map supplies the necessary topological compatibility; holomorphicity can then be achieved while preserving prescribed holomorphic data. Section 2 recalls the precise formulations used in the proof. Theorem 1.1 concerns every smooth member of the classified families, not only a general member. It makes no assertion about singular Fano varieties or Fano manifolds of higher dimension.

The proof separates a common analytic construction from the geometric verification of its hypotheses. A single finite spectral method treats fourteen family labels in seventeen toric or rank-two matrix presentations. The other families are supplied by the external results specified in Section 8. The classification identifies the actual models and completes the list; the analytic passage from rooted curves to convex approximation is proved once.

1.1 From the Oka–Grauert principle to Oka manifolds

The classical starting point is Oka’s work on the second Cousin problem [58]: on domains of holomorphy, a topological obstruction governs a problem of holomorphic trivialization. Grauert [36, 37] extended this principle to holomorphic bundles with complex Lie groups as structure groups over Stein spaces. These results explain why continuous bundle data can control a holomorphic classification. They do not, by themselves, provide approximation of maps into an arbitrary complex target.

Gromov [39] brought geometric deformation methods into this picture through dominating sprays and elliptic submersions. A spray supplies holomorphic directions in which a map or section can move; domination ensures that these directions span the tangent directions required for approximation. Forstnerič and Prezelj [35] developed the spray-gluing method, and Forstnerič [27] established the Oka principle for subelliptic submersions, allowing finitely many sprays to supply domination jointly.

Forstnerič’s convex approximation theorem [28] makes the modern theory particularly useful for geometric applications: approximation on compact convex subsets of Euclidean spaces, in every dimension, characterizes the Oka property. His subsequent work [29, 30] establishes parametric and section versions of the principle. Kusakabe [50] gives an elliptic characterization relative to holomorphic maps and proves Zariski localization of the Oka property. These results turn explicit deformations on a target into global approximation theorems. For a broader account, including the historical development and its applications, we refer to Forstnerič [31, 32, 33]. The foundations needed here are recalled in Section 2; their general proofs remain in the cited works.

1.2 Fano geometry and the approximation problem

The classification of smooth Fano threefolds, developed by Iskovskikh [44, 45] and Mori and Mukai [56], reduces the theorem to 105 families. Mukai’s models [57] and their subsequent developments provide geometric realizations suited to the constructions below. Section 3 recalls this classification history and locates the verification for every model treated by the common construction. The final classification step is in Section 8; it does not use deformation invariance of the Oka property.

The rational-connectedness theorems of Campana [9] and Kollár, Miyaoka, and Mori [48] make Fano manifolds natural candidates for holomorphic flexibility. In 2023, Alarcón and Forstnerič [1] introduced Oka–1 manifolds, restricting approximation and interpolation to maps from open Riemann surfaces, and conjectured that every rationally connected smooth projective manifold is Oka–1. Forstnerič and Lárusson [34] subsequently introduced the algebraic Oka–1 property. Du, Guo, and Xie [25] prove analytic and algebraic Oka–1 approximation results

for smooth projective morphisms with rationally connected fibres, including the conjectured conclusion when the base is a point.

This one-dimensional viewpoint also informs the analytic construction of rational curves on Fano manifolds by Du, Guo, and Xie [26]: successive deformation and enlargement of holomorphic disks suggest how flexibility can produce rational curves. Rational curves through points nevertheless do not automatically supply holomorphic deformations of a preassigned map from an arbitrary Stein source. The problem addressed here retains that unrestricted source dimension.

1.3 The geometric and analytic ideas

The central difficulty is to deform a preassigned holomorphic map over a neighborhood of its entire compact source domain. We call these source variables old, to distinguish them from the fresh scalar deformation parameters introduced below. Through each value of the old map, very free curves furnish local deformation directions, but these curves need not admit a single global holomorphic selection. We instead retain finite curve data in a spectral algebra and solve the defining equations before specialization, keeping the nonreduced factors. The resulting scalar families have the exact old map as their common core.

The sextic double solid motivates the toric part of this construction. Theorem 5.1 applies to a smooth rationally connected projective threefold meeting the dense torus of a smooth projective toric n -fold and cut out scheme-theoretically by $n - 3$ Cox-homogeneous equations. Ordinary complete intersections, double covers, and weighted models enter through their actual projective-bundle and graph presentations. In the weighted cases, the Cartier and smooth-locus checks precede the ambient resolution, so that the target itself is unchanged. The construction therefore includes ramification and exceptional loci, rather than replacing the given threefold by a birational model.

The analytic output is weaker than a jointly entire dominating spray but stronger than a collection of local deformation germs. Each family is holomorphic on one fixed old neighborhood and entire in a fresh scalar; the scalar derivatives jointly span the tangent directions. For a target of dimension d , Theorem 4.1 proves that the corresponding property $\mathcal{P}_{k+d}$ of Definition 2.1 implies CAP_k . Its proof follows Kusakabe’s convex-cut method [50], with scalar propagation and Forstnerič’s Cartan-pair splitting theorem [28]. This bridge retains arbitrary source dimension without requiring a global holomorphic choice of one rational curve.

A rooted rank-two balancing argument brings the Gushel–Mukai and genus-eight presentations into the same spectral construction. Under the hypotheses of Theorem 7.1, suitable free teeth attached to a very free curve reduce the splitting gap while preserving its prescribed root and very freeness. The equal splitting obtained in Section 7 gives an actual tautological matrix lift with equal column degrees. The ordered matrix gauge, and the determinant-valued covering coordinate in the special Gushel–Mukai case, are then incorporated into the common coefficient calculation. The special case includes the ramification locus.

The shared calculation retains all normal equations and controls the actual projective value ideals, including at source infinity. These are identities in the full coefficient rings, not merely statements about reduced points. The dimension-three incidence argument leaves at most two inactive characteristic values near a chosen anchor. A pointed section of the corresponding punctured rational line avoids the remaining raw sites on the entire old domain while preserving the required derivative.

The model verifications and this common deformation construction are distinct from the external results on which they rely: classification and very-free-curve theorems, restriction and connected-evaluation results for balancing, and Oka-theoretic splitting and section principles.

The final classification step also uses uniform-rationality, cubic, and rationality results. The dependence on Banecki's preprint for seven families is stated explicitly in Section 8.

Projection methods provide a complementary geometric perspective. The proper-integration formulation of Xie and Zhao [66, Lemma 3.1] underlies constructions by vertical flows and exceptional rescaling. The normal-ratio comparison theorem of Wang, Xie, and Zhao [64, Theorem 2.1], which concerns simple normal crossing divisors, provides a related method. These constructions form part of the background to our approach; the proof below uses the common spectral criterion for the actual toric and matrix presentations.

Related work supplies both results and methods, with different roles. Guo and Xie's quartic hypersurface theorem [41] gives an independent result in intrinsic dimensions at least three; its threefold case also follows from the complete-intersection construction here. Xie's cubic-complement construction [65] provides a precedent for projection sprays and exceptional rescaling. Xie and Zhao [66] study Oka K3 surfaces and prove density of Oka periods in local Kuranishi spaces; their genus-one motions are also relevant to the constructions considered here. K3 surfaces are not Fano: this relationship concerns deformation methods, not the classification step proving Theorem 1.1.

1.4 Organization and reading guide

Sections 2 and 3 recall the Oka-theoretic foundations and the classification conventions. The latter section also locates the argument for each geometric model. Readers familiar with either subject may use these sections to enter the other.

Section 4 proves the common scalar-family approximation theorem. Section 5 constructs the required families for toric and matrix presentations. Sections 6 and 7 verify the geometric hypotheses, the latter through rooted balancing. Section 8 combines these constructions with the external results and completes the proof of Theorem 1.1 for every classified smooth member.

2 Oka-theoretic foundations

The geometric constructions in this paper move a prescribed holomorphic map in enough tangent directions. Oka theory converts those motions into approximation on an entire Stein source. This section recalls the standard results that make the conversion possible and distinguishes the three kinds of deformation used later: sprays on the target, sprays along a map on a whole Stein source, and scalar families over a neighborhood of an old compact. The recalled general theorems are external results; the finite scalar approximation theorem is proved in Section 4.

2.1 Stein spaces, approximation, and linear extension

A Stein manifold is a complex manifold on which global holomorphic functions separate points, furnish local coordinates, and have compact holomorphic hulls of compact sets. For a compact subset K of a complex space S , its holomorphic hull is

$$\widehat{K}_{\mathcal{O}(S)} = \{x \in S : |h(x)| \leq \sup_K |h| \text{ for every } h \in \mathcal{O}(S)\}.$$

The set K is $\mathcal{O}(S)$ -convex when this hull equals K . Euclidean spaces, domains of holomorphy, and open Riemann surfaces are basic Stein examples. In particular, the convex neighborhoods of compact convex sets used below are Stein.

The Oka–Weil theorem permits uniform approximation of holomorphic functions near an $\mathcal{O}(S)$ -convex compact by global holomorphic functions on S . Cartan’s Theorems A and B provide the corresponding sheaf-theoretic tools. For a coherent analytic sheaf $\mathcal{F}$ on a Stein space, its stalks are generated by germs of global sections, and

$$H^q(S, \mathcal{F}) = 0, \quad q \geq 1.$$

We refer to Cartan [10, Section 4] for these statements, to [11] for their proofs, and to Forstnerič [32] for a modern treatment of approximation and extension on Stein spaces.

Here is the extension mechanism used for parameter sections and jets. For a holomorphic vector bundle $E \rightarrow S$, a closed complex subspace $A \subset S$ with ideal sheaf $\mathcal{I}_A$, and an integer $r \geq 0$, the exact sequence

$$0 \longrightarrow \mathcal{I}_A^{r+1} \mathcal{O}(E) \longrightarrow \mathcal{O}(E) \longrightarrow \mathcal{O}(E)/\mathcal{I}_A^{r+1} \mathcal{O}(E) \longrightarrow 0$$

has vanishing first cohomology for its left-hand sheaf. Consequently, every global section of the quotient lifts to a global holomorphic section of E . Such a quotient section is a compatible bundle jet along the specified subspace. This is a linear extension statement; it does not assert extension of arbitrary maps into a nonlinear target. Likewise, Theorem A and compactness give finitely many sections spanning a vector bundle near a specified compact. The whole-Stein parameter constructions later require the global generation and jet-extension statements used in their actual proofs, rather than a substitution of compact-local generation for them.

2.2 Oka properties and convex approximation

Let S be a reduced Stein space, $K \subset S$ an $\mathcal{O}(S)$ -convex compact, and $A \subset S$ a closed complex subvariety. The basic Oka property with approximation and interpolation for a complex manifold Y says that a continuous map $f : S \rightarrow Y$, holomorphic near K and on A , admits a homotopy to a holomorphic map on S . The homotopy is fixed on A , remains holomorphic on a common neighborhood of K , and stays uniformly as close to f on K as prescribed. When f is holomorphic near A , interpolation can preserve any prescribed finite jet order there.

The parametric property concerns a continuous family of such maps, indexed by a compact parameter set $P \subset \mathbb{R}^m$. If the maps are already holomorphic on all of S for parameters in a compact subset $P_0 \subset P$, the homotopy can also be fixed on P_0 . The neighborhood of K and the approximation bound work uniformly in the parameters. Forstnerič [29, Theorem 1.1 and Definition 1.2] establishes these formulations for Oka fibres; his stratified section theorem [30] supplies the section formulations used below. The initial continuous map or section is part of the hypothesis: the Oka principle does not remove topological incompatibility.

For a compact convex set $Q \subset \mathbb{C}^k$, the notation $\mathcal{O}(Q, Y)$ means restrictions to Q of maps holomorphic on neighborhoods of Q , with the uniform topology. A manifold Y has CAP_k when every map in $\mathcal{O}(Q, Y)$ can be approximated uniformly on Q by holomorphic maps $\mathbb{C}^k \rightarrow Y$. The conjunction of CAP_k for all $k \geq 1$ is the convex approximation property, CAP. Forstnerič’s theorem [28, Theorem 0.1], together with the parametric formulation above, identifies CAP with the Oka property. Thus approximation on a restricted class of source compacts recovers a principle for general Stein spaces.

All source dimensions matter in this characterization. The existence of entire curves through points is not CAP, nor is a statement confined to one-dimensional sources. Oka–1 manifolds satisfy the analogous approximation and finite-jet interpolation principle for maps from open Riemann surfaces, with interpolation at a closed discrete set; see Alarcón and Forstnerič [1, Definition 1.1]. The rationally connected fibre results of Du, Guo, and Xie [25] belong to this one-dimensional theory. They motivate the geometric approximation problem without replacing the arbitrary-dimensional constructions required here.

2.3 Sprays on a target and sprays along a map

For a holomorphic vector bundle $p : E \rightarrow Y$, a holomorphic spray is a map $s : E \rightarrow Y$ on the entire total space satisfying $s(0_y) = y$. Its anchor at y is

$$a_{s,y} : E_y \longrightarrow T_y Y, \quad a_{s,y}(v) = \left. \frac{d}{dt} \right|_{t=0} s(tv).$$

The spray dominates at y if this linear map is surjective. A manifold admitting one spray dominating everywhere is elliptic. It is subelliptic if a finite family of whole-bundle sprays has anchor images jointly spanning $T_y Y$ at every point. Gromov's elliptic principle [39] and Forstnerič's subelliptic principle [27, Definition 2 and Theorem 1.1] make these manifolds Oka. The whole-bundle domination hypothesis is stronger than the scalar-family property used below.

The simplest geometric example is a complex homogeneous manifold $Y = G/H$. If $\mathfrak{g}$ is the Lie algebra of G , the map

$$Y \times \mathfrak{g} \longrightarrow Y, \quad (y, v) \longmapsto \exp(v) \cdot y$$

is an entire spray. The orbit map has surjective differential because the action is transitive, so its anchor dominates. In less symmetric examples, a single global action can be replaced by finitely many line-bundle sprays.

A spray along a specified map $f : S \rightarrow Y$ is a holomorphic map

$$F : S \times \mathbb{C}^m \longrightarrow Y, \quad F(s, 0) = f(s).$$

Domination means surjectivity of $\partial_t F(s, 0)$ at each s . The condition that such a dominating spray exists for every map from every Stein manifold is usually denoted Ell_1 . Kusakabe [50, Definition 1.2 and Theorem 1.3] characterizes Oka manifolds by relative ellipticity. His finite-family criterion [50, Corollary 4.1] allows finitely many sprays along the same map to dominate at any specified source point; the finite family may depend on that point. A spray on $S \times \mathbb{B}_\epsilon^m$ alone does not satisfy this condition: the fresh parameters must range over their whole Euclidean space. The scalar families used in this paper satisfy the compact-local condition stated below; Theorem 4.1 supplies their separate passage to CAP.

These principles have uses beyond the classification problem. Guo and Xie study universal holomorphic maps with growth control [40] and frequent translation universality for maps into connected Oka targets [42]. The latter assumes that the target is already Oka. Du, Guo, Wang, and Xie [24] construct Oka complements for specified periodic sets, using transport and Fatou–Bieberbach constructions. Du, Guo, Kusakabe, and Xie [23] investigate Oka geometry in the proposed torus-fibred complex six-sphere setting (see also the independent proof of the Oka property by Chen [15]). Du, Guo, Kusakabe, and Xie also treat deformations, automorphisms, and biholomorphism classes of these complex threefolds. These works concern different targets and approximation questions; none is used as a substitute for the geometric verification of the Fano cases below.

2.4 Sections, gluing, and localization

An auxiliary holomorphic bundle with Oka fibres admits an Oka principle for its sections over a Stein base [29, Theorem 1.1]. Once a continuous section with the required compatibility is available, it can be deformed to a holomorphic section with approximation and interpolation. This is why rational lines with at most two points removed can serve as auxiliary targets in the spectral construction. Their fibres are $\mathbb{P}^1$, $\mathbb{C}$, or $\mathbb{C}^*$, all Oka independently of the

threefold under consideration. The number of removed points can vary. Subsection 5.6.1 establishes the finite-map stratification and continuous section before applying the stratified section theorem [30, Theorem 1.4(C) and Corollary 1.6]. The interpolation conditions are checked there as well. A family of nonempty fibres by itself is not a section.

For a holomorphic submersion $Z \rightarrow S$, a vertical spray takes its values in the same fibre as its starting point. Domination for the section principle is domination of the vertical tangent bundle. A spray into Z that passes through critical points of a projection does not automatically give a dominating spray on its regular submersion locus. This distinction is relevant whenever a geometric projection has singular fibres.

Gluing is performed on Cartan pairs of compacts with suitable Stein neighborhoods and separated differences. In the convex-cut argument, both members and their intersection are convex, and the separation comes from a strict affine gap. Two relative sprays close on the overlap are compared in a parameter chart. Forstnerič's nonlinear splitting lemma [28, Lemma 2.1] splits a transition map close to the identity into changes of parameters on the two members. Section 4 supplies the domains, smallness conditions, and estimates for the specific splitting used here.

Localization in the target is a different issue. Kusakabe's theorem [50, Theorem 1.4] says that a complex manifold covered by Zariski open Oka subsets is Oka. Ordinary coordinate neighborhoods do not provide such a criterion. The toric applications below retain the actual global equations and zero schemes. They do not infer the Oka property from local charts or from a deformation of the target alone.

2.5 Finite entire scalar families

The approximation argument needs entire scalar deformations of a specified old map on one neighborhood of its whole compact domain. The precise condition is the following.

Definition 2.1. For $N \geq 1$, a complex manifold Y satisfies $\mathcal{P}_N$ if the following holds. Given a compact convex set $Q \subset \mathbb{C}^N$, a neighborhood $W \supset Q$, and a holomorphic map $f : W \rightarrow Y$, there are an open set U with $Q \subset U \subset W$ and finitely many holomorphic maps

$$H_j : U \times \mathbb{C} \longrightarrow Y$$

such that $H_j(s, 0) = f(s)$ on U and

$$\text{span}_j \partial_t H_j(s, 0) = T_{f(s)} Y, \quad s \in Q.$$

The same neighborhood U is required for every value of the fresh scalar t .

The old variables range on U , while the fresh scalar ranges on all of $\mathbb{C}$. Both the exact core identity and a single old neighborhood are essential. For a target of dimension d , Theorem 4.1 proves that $\mathcal{P}_{k+d}$ implies CAP_k . Hence $\mathcal{P}_N$ for every N suffices for the full Oka property. This is not a claim of joint entire dependence on the old variables; it is the precise bridge needed by the spectral construction.

3 Fano threefolds and the guide to the geometric cases

The classification has two roles in this paper. It supplies a finite list of smooth targets, and it realizes those targets by equations, covers, homogeneous models, and blowups on which holomorphic motions can be constructed. The list alone does not prove an Oka property. This section recalls the classification conventions and locates the arguments that use each model. The verification that every family has been covered is deferred to Section 8.

3.1 Classification background and conventions

A smooth complex Fano threefold is a smooth projective threefold X with ample anticanonical bundle $-K_X$. Its Picard rank is $\rho(X) = \text{rank Pic}(X)$, and its index is the largest positive integer r for which $-K_X = rH$ with $H \in \text{Pic}(X)$. In Picard rank one, the numerical data distinguish familiar projective and weighted models. Index four gives $\mathbb{P}^3$, index three gives the smooth quadric, and the five index-two families have $H^3 = 1, \dots, 5$. For index one, the anticanonical degree is $(-K_X)^3 = 2g - 2$, with

$$g = 2, 3, 4, 5, 6, 7, 8, 9, 10, 12.$$

A numerical label does not require that the anticanonical morphism be an embedding: double-cover models occur among these families.

The subject begins with Gino Fano’s study of three-dimensional projective varieties; Iskovskikh’s account [44] places that earlier work in the modern classification. Iskovskikh [44, 45] developed the classification of the Picard-rank-one threefolds and their geometric descriptions. Mori and Mukai [56] classified the higher-Picard-rank cases, using contractions and explicit constructions. The erratum to their classification is included in the cited reference. These are complementary parts of the classification, rather than interchangeable sources for every model.

Gushel [43] gave the genus-six construction; Mukai [57] subsequently provided homogeneous realizations for the higher-genus index-one varieties. Debarre and Kuznetsov [20] develop the Gushel–Mukai and Lagrangian descriptions used in Section 7. The models of De Biase, Fatighenti, and Tanturri [22] give further useful realizations by homogeneous vector bundles. More recently, Bayer, Kuznetsov, and Macrì [6, 7] establish the Mukai-bundle and homogeneous-model results in the all-member forms needed here. These geometric descriptions are credited separately from the analytic deformations constructed in the present paper.

We use the labels and model descriptions in the account by Araujo and collaborators [2, Table 6.1]. A label $\rho.j$ denotes the j th family of Picard rank ρ in that account. The family counts in ranks one through ten are

$$17, 36, 31, 13, 3, 1, 1, 1, 1, 1,$$

whose sum is 105. Three families require two distinct presentations: 1.2 has quartic and double-quadric presentations, 1.5 has ordinary and special Gushel–Mukai presentations, and 2.6 has Verra and double-flag presentations. Thus the proofs address 108 presentations of 105 families.

Every family statement below concerns all its smooth complex members. General projection centres may be selected on a fixed smooth member; this does not restrict the member itself to be general. In blowup descriptions, the centre is the actual smooth centre satisfying the classification’s position conditions, not an arbitrary curve with the same degree and genus. Smoothness of the threefold also does not exclude singular auxiliary quadrics or degenerate fibres of a projection.

3.2 Weighted models, covers, and their blowups

Family 1.1 consists of sextic double solids. Its Oka property is proved in Corollary 6.6. The weighted double-cover equation has rational curves available, but these must deform a specified map holomorphically over its whole old compact. The finite spectral construction replaces a global choice of curves, retains the nilpotent coefficient algebra, and yields entire scalar families with the prescribed core.

The elliptic-centre blowup in family 2.1 is treated in Corollary 6.14. Its base is the index-two, degree-one threefold, not a sextic double solid. The graph model couples the weighted sextic

equation to an incidence equation for the blowup. These are the two global equations to which the common spectral theorem applies. Theorem 6.13 treats the degree-one base itself, family 1.11. Its half-anticanonical base point lies in the smooth weighted locus because the nonzero weight-two and weight-three coordinates have coprime weights. The resolution in Subsection 6.5 therefore preserves this point and every other point of the threefold.

Quartic double solids, family 1.12, are covered by Corollary 6.7. The toric projective-bundle construction also treats family 2.2, with branch degrees $(2, 4)$ on $\mathbb{P}^1 \times \mathbb{P}^2$, and family 3.1, with branch degrees $(2, 2, 2)$ on $(\mathbb{P}^1)^3$; see Corollaries 6.8 and 6.9. The hypersurface equation excludes the fibre infinity divisor scheme-theoretically. Its local double-cover equation gives the required smoothness, including along the branch divisor.

Theorem 6.15 treats family 2.3, the blowup of a quartic double solid along the smooth preimage of a line. Its actual graph has two global equations, whose common zero scheme is unchanged by the specified smooth toric resolution. Thus the common theorem covers the exceptional divisor as well. Theorem 6.10 treats family 2.8, the double cover of $\mathrm{Bl}_p \mathbb{P}^3$ branched in a smooth anticanonical surface. The exceptional quadric can be a cone and its branch conic can be reducible. The smoothness required by the toric model concerns the total branch surface, so both exceptional possibilities remain included.

3.3 Quadric, Grassmannian, and conic models

For family 1.2, Theorems 6.1 and 6.12 cover the quartic and double-quadric presentations. In the latter, the branch divisor has class $4H_Q$ on the smooth quadric threefold, hence projective degree eight. Theorem 6.2 treats family 1.3, the quadric–cubic intersections in $\mathbb{P}^5$, without assuming the containing quadric is smooth. Corollary 6.3 applies the same toric theorem to family 1.4, the intersections of three quadrics in $\mathbb{P}^6$.

Family 1.5 has ordinary and special Gushel–Mukai presentations, and family 1.7 is the genus-eight Grassmannian section. Section 7 supplies equal-split very free curves through every point of these models. Their actual matrix equations then satisfy the common transfer criterion.

Family 2.6 has two presentations in Section 6. Theorem 6.4 applies the toric theorem to every smooth Verra hypersurface. Theorem 6.11 treats the double flag: the incidence equation and the covering equation form a regular sequence in a smooth toric projective bundle. Lifting the branch section and excluding the infinity divisor identify the exact double-cover scheme; no deformation invariance of the Oka property is used.

3.4 The families supplied by external theorems

The fourteen family labels just discussed are the cases treated by the geometric constructions in this paper. Twelve admit the displayed toric complete-intersection presentations; the remaining Gushel–Mukai and genus-eight cases use the matrix presentation of the same transfer theorem. The remaining 91 are treated in Section 8: 81 by uniform rationality and algebraic ellipticity, three by the cubic theorem and its blowup applications, and seven by rationality together with an algebraic ellipticity theorem. This is a partition by the proofs used, not a claim that the geometric properties are disjoint.

The uniform-rationality arguments rely on the smooth-centre blowup and ellipticity results of Arzhantsev, Kaliman, and Zaidenberg [3], with homogeneous examples and the affine-space coverage results of [4]. The algebraic-to-holomorphic implication is the Oka principle developed by Gromov and Forstnerič [39, 27]. The cubic cases use Kaliman and Zaidenberg [46] and Kusakabe’s algebraic Ell_1 blowup theorem [49].

For the seven remaining rational families, the rationality statements come from Kuznetsov and Prokhorov [51] and Prokhorov [59]. The implication from smooth rational projective varieties to algebraic ellipticity is the theorem in Banecki's preprint [5]. Its preprint status is retained in the bibliography and at the application. Rationality alone is not being equated with Oka flexibility, nor is arbitrary birational invariance asserted. Section 8 checks the required hypotheses on every classified model and completes the count.

4 Finite entire directions imply convex approximation

The geometric constructions below produce finitely many scalar deformations, rather than a single spray entire in all its parameters. This section proves that these deformations suffice for CAP. The dimension shift by $\dim Y$ allows the deformation argument to retain the parameters of a relative chart on the whole compact.

4.1 Hypothesis and conclusion

Let Y be a connected complex manifold of dimension d . We use $\mathcal{O}(Q, Y)$ and $\mathcal{P}_N$ in the sense of Section 2. Whenever an identity is asserted on an open neighborhood, a holomorphic representative is chosen there. The neighborhood, the number of scalar families, and the families may depend on Q, W, f , but one neighborhood works for every scalar value. It is enough to verify the property for full-dimensional compact convex sets: any compact convex $Q \subset W$ has a full-dimensional compact convex thickening D with $Q \subset \text{int } D \subset D \subset W$.

Theorem 4.1. *Let $k \geq 1$. If Y satisfies $\mathcal{P}_{k+d}$, then for every compact convex set $Q \subset \mathbb{C}^k$, every map f holomorphic on a neighborhood of Q , and every $\epsilon > 0$, there is a holomorphic map $g : \mathbb{C}^k \rightarrow Y$ such that*

$$\sup_{z \in Q} d_Y(g(z), f(z)) < \epsilon,$$

where d_Y is the distance of any fixed Hermitian metric on Y . Consequently, if $\mathcal{P}_N$ holds for every $N \geq 1$, then Y has CAP and is Oka.

The zero-dimensional target case is immediate. Assume henceforth that $d > 0$. A property in a larger old dimension implies the corresponding property in a smaller dimension: extend the core independently of the extra coordinates, apply the larger-dimensional property to $Q \times \{0\}$, and restrict back to the original old coordinates.

The proof follows the convex-cut method of Kusakabe [50, proof of Theorem 2.2], with scalar propagation in place of a jointly entire dominating spray. Rank reduction makes the overlap matrix square, and Cartan splitting then gives approximation across each convex cut.

4.2 Relative charts on a whole old compact

Lemma 4.2. *If $Q \subset \mathbb{C}^k$ is compact convex and $f \in \mathcal{O}(Q, Y)$, then there are a convex Stein neighborhood $V \supset Q$, a number $r > 0$, and a holomorphic map*

$$G : V \times \mathbb{B}_r^d \rightarrow Y, \quad G(z, 0) = f(z),$$

such that $(z, p) \mapsto (z, G(z, p))$ is biholomorphic onto an open subset of $V \times Y$ containing the graph of the chosen representative $f|_V$. Its inverse has the form $(z, y) \mapsto (z, \Sigma_z(y))$.

Proof. Start with a local dominating spray with m parameters near Q [50, Lemma 2.3]. Reduce that neighbourhood to a convex Stein one. The differential at the zero section is

a surjective holomorphic bundle map from the trivial rank- m bundle to f^*TY . Split its kernel holomorphically. The complement has rank d and is isomorphic to f^*TY , hence is holomorphically trivial on this contractible Stein base. A holomorphic frame gives a map $J(z) : \mathbb{C}^d \rightarrow \mathbb{C}^m$. Restrict the spray to $J(z)p$.

Its parameter derivative is invertible along the zero section. At each $z_0 \in Q$, the relative inverse theorem supplies an old neighborhood V_{z_0} and a parameter ball on which the full map $(z, p) \mapsto (z, G(z, p))$ is injective and has invertible differential. Take a finite such cover of Q and a ball strictly smaller than all its parameter balls. Shrink the old domain to a convex Stein neighborhood contained in their union. If two points have the same image, their old coordinates agree; a member of the finite cover containing that coordinate then proves equality of their parameters. Thus the full map is injective and locally biholomorphic on this product. Its image is open, contains the whole core graph, and the local inverses agree there by uniqueness. Taking a further smaller old neighborhood and parameter ball leaves closure margins; in particular a later use of $\mathbb{B}_{2r}^d$ is obtained by choosing $2r$ inside the ball just constructed. $\square$

The chart is relative to z . The compact image $f(Q)$ need not lie in one coordinate chart of Y .

If a given dominating family $G_0(z, p)$ with $m > d$ parameters is accompanied by $T_0(z, p, u)$ entire in u , the same reduction replaces them by

$$G(z, q) = G_0(z, J(z)q), \quad T(z, q, u) = T_0(z, J(z)q, u).$$

On a sufficiently small fixed bounded q ball these are jointly defined over a neighborhood of all of Q , and T is still entire in its last variable. The endpoint $T(z, 0, u)$ is unchanged.

4.3 Scalar propagation with explicit choice order

For a compact convex $K \subset \mathbb{C}^k$ and a real affine function λ , consider the cut and approximation closure

$$K_0 = K \cap \{\lambda \leq 0\}, \quad \mathcal{A}_K = \overline{\mathcal{O}(K, Y)|_{K_0}}^{\mathcal{O}(K_0, Y)}.$$

The empty and full cuts are immediate. In this section assume K_0 is nonempty and proper. Fix a complete Hermitian metric on Y and its path distance; existence is recalled in the exhaustion argument below.

Lemma 4.3. *Suppose $f \in \mathcal{A}_K$, and suppose, on a neighborhood Ω of all of K_0 , there are holomorphic maps*

$$G : \Omega \times \mathbb{B}_r^d \rightarrow Y, \quad T : \Omega \times \mathbb{B}_r^d \times \mathbb{C} \rightarrow Y,$$

with $G(z, 0) = f(z)$, $T(z, p, 0) = G(z, p)$, and G a rank- d relative chart near $K_0 \times \{0\}$. Then

$$z \mapsto T(z, 0, \varphi(z)) \in \mathcal{A}_K \quad \text{for every } \varphi \in \mathcal{O}(K_0, \mathbb{C}).$$

The same conclusion holds with a local dominating m -parameter G , by the preceding rank reduction.

4.3.1 Fix the approximant, charts, and old buffers first

Fix a required final error $\epsilon > 0$. Reduce r so that G remains a relative chart on a neighborhood of $K_0 \times \mathbb{B}_r$. Since $f \in \mathcal{A}_K$, choose $g \in \mathcal{O}(K, Y)$ sufficiently close to f on K_0 that

$$\rho(z) = \Sigma_z(g(z)), \quad G(z, \rho(z)) = g(z), \quad |\rho| < r/8$$

is holomorphic near K_0 , and

$$\sup_{K_0} d_Y(T(z, \rho(z), \varphi(z)), T(z, 0, \varphi(z))) < e/3. \quad (4.1)$$

This uses continuity on the compact scalar-value set $\varphi(K_0)$ and continuity of the local inverse. It does not require derivative approximation of g .

Choose a bounded convex neighborhood Ω_0 of K_0 on which all these expressions exist and $|\rho| < r/8$. Independently apply Lemma 4.2 to g near the *whole* K , obtaining

$$F : V \times \mathbb{B}_R^d \rightarrow Y, \quad F(z, 0) = g(z), \quad \Psi_z = F_z^{-1}.$$

All chart neighborhoods and R are fixed now.

Choose $c > 0$, small enough and less than $\max_K \lambda/3$, so that

$$A = K \cap \{\lambda \geq c\}, \quad B = K \cap \{\lambda \leq 2c\}, \quad C = A \cap B$$

satisfy $B \subset \Omega_0$. These are compact convex sets; $A \cup B = K$, and the closures of $A \setminus B$ and $B \setminus A$ are separated by a positive λ gap. Thus they form a Cartan pair.

Choose compact convex neighborhoods K_0^+ and C^+ of K_0 and C , separated by a positive λ gap. They lie in the interior of a compact neighborhood B^+ of B contained in Ω_0 . All these sets lie in the required old chart domains; in particular ρ and φ are holomorphic near them. Let V_C be a fixed open neighborhood of C with $\overline{V}_C \subset \text{int } C^+$. These choices precede every polynomial approximation and every gluing radius. Their existence follows by compactness and the strict gap between $\lambda \leq 0$ and $\lambda \geq c$. This remains valid when K_0 has empty interior.

4.3.2 The overlap matrix is square and invertible

On a neighborhood of C^+ , consider

$$\mathcal{H}(z, p, u) = \Psi_z(T(z, p, u)).$$

There is an $h > 0$, with $h < \min(r/4, 1)$, such that this is holomorphic past

$$z \in C^+, \quad |p - \rho(z)| \leq h, \quad |u| \leq h.$$

The two relative charts are centered at the same approximant on the overlap:

$$\mathcal{H}(z, \rho(z), 0) = \Psi_z(G(z, \Sigma_z(g(z)))) = \Psi_z(g(z)) = 0.$$

Thus no error measuring $f - g$ remains in the transition at its center. Since both G_z and F_z are local biholomorphisms,

$$E(z) = \partial_p \mathcal{H}(z, \rho(z), 0) = d\Psi_z|_{g(z)} \circ dG_z|_{\rho(z)}$$

is an invertible $d \times d$ matrix. Set

$$L_A(z) = E(z)^{-1}.$$

This is a holomorphic matrix on a fixed convex neighborhood of C^+ . This step uses no new holomorphic bundle lift.

For Cartan splitting, fix the ordered pair (B, A) , the compact parameter set $D = \overline{\mathbb{B}}_1^d$, and the open transition domain

$$\Omega_C = V_C \times \mathbb{B}_{5/2}^d.$$

Let $\kappa_0 > 0$ be a sufficient uniform closeness threshold for this fixed domain and correction tolerance $1/2$. Choose

$$0 < \kappa < \min(\kappa_0, 1/4), \quad \Lambda = 1 + \|L_A\|_{C^+}.$$

A polynomial matrix L_B approximating L_A within $\eta \leq 1$ on C^+ has norm at most Λ there.

Derivatives of $\mathcal{H}$ on the fixed compact give $C_* \geq 1$ such that, whenever $|w| \leq 4\delta$, $4\delta\Lambda < h/2$, and $|u| \leq e_0 < h/2$,

$$|\mathcal{H}(z, \rho(z) + L_B(z)w, u) - w| \leq C_*(e_0 + \eta\delta + \delta^2), \quad z \in C^+. \quad (4.2)$$

Indeed the three terms come, respectively, from varying u , replacing $EL_A = I$ by EL_B , and the quadratic p remainder. One can take C_* to dominate the u derivative bound, four times the bound for E , and eight times the second p derivative bound times Λ^2 . This constant knows nothing about the later norm of L_B away from C^+ .

Also fix $P_*, U_* \geq 1$ bounding the parameter derivatives of T on

$$z \in K_0, \quad |p| \leq r/2, \quad u \in \varphi(K_0) + \overline{\mathbb{B}}_1.$$

These are finite Hermitian operator-norm bounds on a fixed compact.

4.3.3 The approximation and radius dependencies

Make the choices in the following order:

1. Choose $0 < \eta < \min(1, \kappa/(4C_*))$. Approximate L_A on C^+ by a polynomial matrix L_B to error η .
2. Record $L = \max(1, \|L_B\|_{B^+})$. It is finite but need not be controlled by η .
3. Choose $\delta > 0$ satisfying

$$8\delta < R, \quad 4\delta\Lambda < h/2, \quad \delta < r/(32L), \quad C_*\delta < \kappa/4, \quad P_*L\delta/2 < e/3. \quad (4.3)$$

4. Choose

$$0 < e_0 < \min(h/2, \delta\kappa/(4C_*)), \quad 0 < e_1 < \min(1, e/(3U_*)). \quad (4.4)$$

5. Choose a scalar polynomial u_B satisfying

$$\|u_B\|_{C^+} < e_0, \quad \|u_B - \varphi\|_{K_0^+} < e_1.$$

For the last step, choose a polynomial p with $\|p - \varphi\|_{K_0^+} < e_1/2$ and record

$$M_0 = \max(1, \|p\|_{K_0^+}), \quad M_C = \max(1, \|p\|_{C^+}).$$

A complex-affine function ℓ whose real part separates the two buffers sends them to disjoint planar compact convex sets. Their union has connected complement. A scalar polynomial q can therefore approximate 1 on $\ell(K_0^+)$ to error $e_1/(2M_0)$ and 0 on $\ell(C^+)$ to error e_0/M_C . Then $u_B = p \cdot (q \circ \ell)$ has the two required bounds. No norm bound on u_B between the buffers is needed.

4.3.4 One fixed-ball transition and its splitting

Consider the two families

$$F_A(z, w) = F(z, w), \quad F_B(z, w) = T(z, \rho(z) + L_B(z)w, u_B(z)).$$

For z near B and $|w| \leq 4\delta$, the old p argument of the second family has norm less than

$$r/8 + 4\delta L < r/4.$$

The scalar u_B may be large on the intervening part of B^+ ; *entire dependence in that scalar* is exactly what makes F_B defined there.

On the fixed overlap V_C , the normalized transition is

$$\tilde{\gamma}(z, \zeta) = \delta^{-1} \mathcal{H}(z, \rho(z) + \delta L_B(z)\zeta, u_B(z)), \quad |\zeta| < 4.$$

It satisfies

$$F(z, \delta\tilde{\gamma}(z, \zeta)) = F_B(z, \delta\zeta),$$

and the fixed-ball estimate

$$\|\tilde{\gamma} - \text{Id}\|_{V_C \times \mathbb{B}_4} \leq C_*(e_0/\delta + \eta + \delta) < 3\kappa/4. \quad (4.5)$$

There is no inverse constant chosen after δ : the domain of $\mathcal{H}$ was fixed first. The stronger first inequality in (4.3) also leaves an explicit margin for the image in the F chart.

Let $q = 3\kappa/4 < 1/4$. Vector-valued Cauchy estimates on unit complex line disks through points of $\mathbb{B}_{5/2}$, using only $\mathbb{B}_{7/2} \subset \mathbb{B}_4$, give

$$\|d_\zeta(\tilde{\gamma} - \text{Id})\|_{\mathbb{B}_{5/2}} \leq q < 1/2.$$

Integration on line segments gives injectivity on $\mathbb{B}_{5/2}$ in each fiber. Hence $(z, \zeta) \mapsto (z, \tilde{\gamma}(z, \zeta))$ is injective on the previously fixed domain Ω_C . No old derivative estimate is required, since the old coordinate is preserved.

Apply Forstnerič's Cartan splitting theorem [28, Lemma 2.1] on Ω_C to the ordered pair (B, A) and $D = \mathbb{B}_1^d$. Write the parameter components of the two fiber-preserving factors as α over A and β over B . They satisfy

$$\tilde{\gamma} \circ \beta = \alpha, \quad \|\alpha - \text{Id}\| < 1/2, \quad \|\beta - \text{Id}\| < 1/2,$$

on neighborhoods of $A \times \overline{\mathbb{B}}_1$ and $B \times \overline{\mathbb{B}}_1$, respectively, where the identity in these norms is the parameter component ζ . The threshold is the previously fixed κ_0 . Only a fixed correction tolerance is used; no linear bound in the transition error is required.

The cores

$$F(z, \delta\alpha(z, 0)), \quad T(z, \rho(z) + \delta L_B(z)\beta(z, 0), u_B(z))$$

agree near C . Shrink their old neighborhoods inside the original domains and so that their intersection lies in the agreement neighborhood. They glue to a map $g_e \in \mathcal{O}(K, Y)$. The output old neighborhood can depend on e and on the polynomials, which is allowed.

On K_0 the straight p and scalar segments stay inside the protected derivative compact, so

$$\sup_{K_0} d_Y(g_e, T(z, \rho(z), \varphi(z))) \leq P_* L \delta / 2 + U_* e_1 < 2e/3.$$

Combine with (4.1). This proves Lemma 4.3.

4.4 One invocation of property P and the exact intermediate cores

Choose $f \in \mathcal{A}_K$ and a rank- d relative chart

$$G : \Omega \times \mathbb{B}_{2r}^d \rightarrow Y$$

over a chosen holomorphic representative of f , with closure margins. Let D_0 be a full-dimensional compact convex set with

$$K_0 \subset \text{int } D_0 \subset D_0 \subset \Omega.$$

Apply $\mathcal{P}_{k+d}$ once to the holomorphic representative $(z, p) \mapsto G(z, p)$ on a neighborhood of the compact convex set

$$Q' = D_0 \times \overline{\mathbb{B}}_r^d \subset \mathbb{C}^{k+d}.$$

It supplies finitely many maps $T_j(z, p, u)$, jointly holomorphic on one old neighborhood of this whole product times the entire u line. Their core identity $T_j(z, p, 0) = G(z, p)$ holds on that open old neighborhood. Compactness gives a product $\Omega' \times \mathbb{B}_{r'}^d$ containing $K_0 \times \{0\}$ inside the common domain, with $r' > 0$. Reduce r' if necessary so that the same chart G and its inverse Σ_z are valid there. The full-dimensional thickening also makes the identity unambiguous if the property is initially formulated in terms of restrictions to compact sets.

For y in this common graph neighborhood, let

$$h_j(z, y, u) = T_j(z, \Sigma_z(y), u), \quad h_j(z, y, 0) = y.$$

For small $a = (a_1, \dots, a_q)$ their finite successive composition gives a local spray

$$P(z, a) = h_q(z, h_{q-1}(z, \dots h_1(z, f(z), a_1) \dots, a_{q-1}), a_q).$$

Choose a smaller graph tube whose closure over K_0 lies in the common domain of all h_j . At zero parameters every partial composition equals f . Starting with the first partial composition, continuity on the compact old set gives a parameter polydisc on which its image stays in that tube. Repeat with the next composition and shrink the polydisc each time. There are only finitely many compositions, so the resulting polydisc has positive radius and works for every intermediate image. Its strict containment persists on one old neighborhood of K_0 . Since $d_y h_j(z, y, 0) = \text{Id}$, the columns of $d_a P(z, 0)$ are exactly the supplied fresh directions. It is surjective on K_0 .

A graph inverse of P [50, proof of Lemma 2.5] represents every sufficiently close $\psi \in \mathcal{O}(K_0, Y)$ as

$$\psi(z) = P(z, a_1(z), \dots, a_q(z))$$

with all a_j uniformly small and holomorphic on a neighborhood of the entire K_0 . Indeed, Stein splitting of $\ker d_a P$ and the inverse theorem give a holomorphic map $a(z, y)$ on an open neighborhood $\mathcal{V}$ of the graph of f , with

$$P(z, a(z, y)) = y, \quad a(z, f(z)) = 0.$$

Compactness of the graph over K_0 supplies one uniform closeness tolerance for ψ and one small parameter bound. For any chosen holomorphic representative of such a ψ , the set

$$U_\psi = \{z : (z, \psi(z)) \in \mathcal{V}\} \cap \text{dom}(\psi)$$

is open and contains K_0 . The functions $a_j(z) = a_j(z, \psi(z))$ are holomorphic there. Their strict bounds on K_0 persist after an old shrinking. The tolerance is independent of ψ ; its old neighborhood need not be.

For the actual successive core maps f_j of this composition, with $f_0 = f$, consider the translated charts

$$\rho_j(z) = \Sigma_z(f_{j-1}(z)), \quad G_j(z, p) = G(z, \rho_j(z) + p), \quad \tilde{T}_j(z, p, u) = T_j(z, \rho_j(z) + p, u).$$

Uniform smallness of the a_j ensures $|\rho_j| < r'/4$ after a fixed reduction. A bounded p ball of radius less than $r'/4$ then works for all finitely many shifts, over one neighborhood of all of K_0 . Each G_j is still a relative chart, and each $\tilde{T}_j$ is entire in u . Its core is exactly f_{j-1} , and its endpoint with $p = 0$, $u = a_j(z)$ is exactly f_j .

If $f_{j-1} \in \mathcal{A}_K$, scalar propagation gives $f_j \in \mathcal{A}_K$. Induction yields $\psi \in \mathcal{A}_K$. Thus $\mathcal{A}_K$ is open in the uniform restriction topology. It is closed by construction and contains every constant map.

This restriction space is path connected. For $z_* \in K_0$, take a representative of a given map on a convex neighborhood V of K_0 . The maps

$$z \mapsto f(z_* + t(z - z_*)), \quad 0 \leq t \leq 1,$$

are holomorphic on V and form a continuous path in the uniform topology from a constant map to f . Constants can be joined along paths in the connected manifold Y . Hence

$$\mathcal{A}_K = \mathcal{O}(K_0, Y).$$

4.5 Direct convex-cut passage to CAP

We give the last approximation step explicitly. Suppose every real-affine cut $K_0 \subset K$ of compact convex sets in $\mathbb{C}^k$ has the density just established.

For a compact convex $Q \subset \mathbb{C}^k$ and a map f holomorphic near Q , choose a bounded full-dimensional real convex polytope P with

$$Q \subset \text{int } P \subset P \subset \text{dom}(f), \quad P = \bigcap_{j=1}^m \{\ell_j \leq 0\}.$$

Such a polytope exists by first taking a small convex thickening of Q and then a sufficiently fine finite circumscribed polytope. For a large cube $C_0 \supset P$, consider

$$K_j = C_0 \cap \bigcap_{i=j+1}^m \{\ell_i \leq 0\}, \quad 0 \leq j \leq m.$$

Then $K_0 = P$, $K_m = C_0$, and $K_{j-1} = K_j \cap \{\ell_j \leq 0\}$. Apply the cut approximation successively with arbitrarily small prescribed errors. The triangle inequality produces a map holomorphic near C_0 approximating f on Q to any prescribed error.

For the limit, use a complete Hermitian metric. Starting with the prescribed Hermitian metric h_0 and a nonnegative proper smooth exhaustion ϑ , take

$$h = (1 + |d\vartheta|_{h_0}^2)h_0.$$

The function ϑ is one-Lipschitz for its path distance, so bounded metric balls lie in compact exhaustion sublevels. Thus every Cauchy sequence has a limit. Since $h \geq h_0$, its distance dominates the prescribed distance. In the estimates below, d_Y denotes this complete distance, as in the preceding scalar-propagation argument.

Take cubes $C_n \subset \text{int } C_{n+1}$ exhausting $\mathbb{C}^k$. For a required error $e > 0$, the finite-cut argument gives $g_0 \in \mathcal{O}(C_0, Y)$ within $e/2$ of f on Q . The same argument applies to each inclusion $C_n \subset C_{n+1}$, using the finitely many supporting halfspaces of C_n . Choose

$$g_{n+1} \in \mathcal{O}(C_{n+1}, Y), \quad \sup_{C_n} d_Y(g_{n+1}, g_n) < e 2^{-n-2} \quad (n \geq 0).$$

Every fixed compact subset of $\mathbb{C}^k$ is contained in some C_n . The sequence is uniformly Cauchy there, and completeness gives a continuous locally uniform limit $g_\infty : \mathbb{C}^k \rightarrow Y$.

The limit is holomorphic. Near any source point, continuity of the limit gives a smaller relatively compact source neighborhood whose limiting image lies in one target chart. Locally uniform convergence places all sufficiently late maps in the same chart there. Their coordinatewise uniform limit is holomorphic. Finally,

$$\sup_Q d_Y(g_\infty, f) \leq \sup_Q d_Y(g_0, f) + \sum_{n=0}^{\infty} e 2^{-n-2} < e.$$

This proves Theorem 4.1.

5 Spectral transfer for toric and matrix presentations

5.1 The two quotient presentations

The two presentations below have different quotient actions but share the same deformation argument. After constructing finite families of rooted curves, we carry out the coefficient calculation in one spectral algebra. The toric and matrix gauges are treated separately; the normal equations and the passage to entire scalar families are then handled together.

Theorem 5.1. *Let X be a smooth complex projective threefold satisfying one of (i) or (ii).*

- (i) *X is rationally connected, meets the dense torus of a smooth projective toric n -fold V , and is the scheme-theoretic zero locus of $n - 3$ Cox-homogeneous equations on V .*
- (ii) *X has one of the following regular presentations:*

$$\text{Gr}(2, 5) \cap H_1 \cap H_2 \cap Q, \quad \text{Gr}(2, 6) \cap H_1 \cap \cdots \cap H_5,$$

or the double cover of the smooth threefold $Y = \text{Gr}(2, 5) \cap H_1 \cap H_2 \cap H_3$ given by $z^2 = Q(p)$ in $\text{Tot}(\det \mathcal{S}^)$. Here the H_i are Plücker hyperplanes, Q is a Plücker quadric, and $\mathcal{S}$ is the actual tautological bundle, pulled back to X in the double-cover case. For every $x \in X$, there are three free maps $u_i : \mathbb{P}^1 \rightarrow X$ with common root p , independent derivatives at p , and*

$$u_i(p) = x, \quad u_i^* \mathcal{S} = \mathcal{O}(-e_i)^{\oplus 2}, \quad e_i > 0.$$

Then X satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka.

A single equal-split very free curve through every point suffices in (ii). Equal splitting is open, and $H^1(u^*TX(-2p)) = 0$ makes the rooted first-derivative map submersive. Three nearby rooted maps therefore have independent derivatives. If the selected degrees differ, source covers unramified at the root raise them to a common positive column degree e .

Let $V = V_\Sigma$ be a smooth projective toric variety of dimension n . Write $s_0 = \#\Sigma(1)$ and $r_0 = s_0 - n$. A basis $L_1, \dots, L_{r_0}$ of $\text{Pic}(V)$ identifies each invariant prime-divisor class with an

integral character $\chi_i = (\chi_{i1}, \dots, \chi_{ir_0}) \in \mathbb{Z}^{r_0}$. For $\mu \in \mathbb{Z}^{r_0}$, write $L^\mu = \bigotimes_\alpha L_\alpha^{\mu_\alpha}$. The Cox ring is $\mathbb{C}[Z_1, \dots, Z_{s_0}]$, graded by $\deg Z_i = \chi_i$. Denote by B_Σ the ideal generated by $\prod_{i \notin \sigma(1)} Z_i$ for maximal cones σ , and write $\widehat{V} = \mathbb{C}^{s_0} \setminus V(B_\Sigma)$. The Cox quotient is the actual principal torus bundle

$$\widehat{V} \longrightarrow V, \quad G = (\mathbb{C}^*)^{r_0}, \quad (g \cdot Z)_i = g^{\chi_i} Z_i.$$

Indeed, for every smooth maximal cone the classes of the complementary r_0 invariant divisors form a basis of $\text{Pic}(V)$. On the open set where those r_0 coordinates are nonzero, a unique element of G makes them all equal to one. This identifies the open set with $\mathbb{A}^n \times G$ over the corresponding affine toric chart. The quotient description and the identification of Cox-graded pieces with spaces of sections are those of Cox [16, Theorem 2.1 and Proposition 1.1]. No finite stabilizer or stack quotient occurs here.

The scheme-theoretic condition concerns the zero scheme on V , or equivalently its inverse image in $\widehat{V}$; no assertion is needed about extra components of the affine equation ideal in the irrelevant locus. Smoothness and the stated codimension make the equation section of $\bigoplus_\lambda L^{\alpha_\lambda}$ regular. The entire direct sum is therefore the normal quotient. Choose a sufficiently high ample character h such that the complete monomial vector M_h of Cox degree h embeds V in projective space, and write $A_X = L^h|_X$. The zero set of these monomials in $\mathbb{A}^{s_0}$ is exactly $V(B_\Sigma)$. Here is the required combinatorial verification. Represent h by an invariant ample Cartier divisor. Its polytope has normal fan Σ ; the normal-fan and monomial embedding descriptions are recalled in [17, Lecture II, Theorem 1.2 and the discussion following (2.2)]. The vertex for a maximal cone σ supplies a monomial with exponent zero on its rays and strictly positive exponent on every other ray. Conversely, the zero exponents of any lattice-point monomial are the facets containing that point; their rays belong to the normal cone of its smallest face, hence lie in some maximal cone. Thus that monomial is divisible by the corresponding generator of B_Σ . The two inclusions of zero sets follow. In particular M_h detects the actual Cox irrelevant locus, including tuples with some zero coordinates.

In case (ii), the full-rank matrix space has right action

$$M \cdot g = Mg, \quad (M, z) \cdot g = (Mg, z \det g), \quad g \in \text{GL}_2.$$

Its quotient is respectively $\text{Gr}(2, n)$ or $\text{Tot}(H)$, where $H = \det S^*$. The complete normal characters are $(\det g)^{\alpha_\lambda}$, with

$$(a_\lambda) = (1, 1, 2), \quad (1, 1, 1, 1, 1), \quad (1, 1, 1, 2)$$

in the three cases. Plücker relations are identities among matrix minors and are not additional normal equations. All minors, together with z when present, form a projective value tuple Ξ of one character $\det g$; on X this is the stated projective embedding. In the special case the projective ambient is the cone over the Plücker Grassmannian, with coordinates (p, z) and vertex $[0 : 1]$. The stated equation scheme misses that vertex: on $p = 0$, the branch equation is $z^2 = 0$, whereas z is a unit in its projective chart. Thus the equation scheme lies in the open $\text{Tot}(H)$ and is exactly the indicated double cover.

In both cases write Z for the polynomial coordinate tuple and $R_g Z$ for its linear quotient action. Thus $R_g Z = (g^{\chi_i} Z_i)_i$ in the toric case and $R_g(M, z) = (Mg, z \det g)$ in the matrix case. Write c for the number of defining equations, indexed by $\lambda = 1, \dots, c$. The equation and value characters satisfy

$$Q_\lambda(R_g Z) = \vartheta_\lambda(g) Q_\lambda(Z), \quad \Xi(R_g Z) = \vartheta_h(g) \Xi(Z).$$

For the torus, $\Xi = M_h$, $\vartheta_\lambda(g) = g^{\alpha_\lambda}$, and $\vartheta_h(g) = g^h$. For matrices, $\vartheta_\lambda(g) = (\det g)^{\alpha_\lambda}$ and $\vartheta_h(g) = \det g$. Each coordinate has an actual source degree $d_i \geq 0$, each equation has degree $k_\lambda \geq 0$, and all entries of Ξ have one degree $d_h > 0$. The action preserves each degree block. In the matrix case the degrees are e for entries and $2e$ for z ; then $k_\lambda = 2e\alpha_\lambda$ and $d_h = 2e$.

5.2 Rooted coefficient maps and finite selectors

In case (i), fix an anchor $x \in X$ and a very free map $u : \mathbb{P}^1 \rightarrow X$ with $u(p) = x$, whose existence follows from [47, Definition–Theorem 1.1(6)]. For $t \neq p$, evaluation at t on the fixed-root Hom space is submersive: its obstruction is $H^1(u^*TX(-p-t)) = 0$. Its image therefore meets $X \cap V^\circ$, a dense open subset. Choose a nearby very free map meeting this open subset. No Cox coordinate section on this curve is identically zero. Consequently, with

$$D_\alpha = \deg u^*L_\alpha, \quad d_i = \chi_i \cdot D, \quad k_\lambda = \alpha_\lambda \cdot D,$$

every d_i is nonnegative. The integers D_α themselves need not be nonnegative. Every monomial in Q_λ has source degree k_λ , so $k_\lambda \geq 0$ as well. The fixed-root first-jet map is submersive by $H^1(u^*TX(-2p)) = 0$. On the open neighborhood of torus-meeting very free maps, three choices therefore have independent rooted derivatives. Their full degree vector D is the same. Coordinates of degree zero are constant in the rational source variable.

In the toric case the tangent sequence of the principal Cox quotient and the complete normal sequence are

$$0 \longrightarrow \mathcal{O}_X^{r_0} \xrightarrow{\mathbf{E}} \bigoplus_i L^{\chi_i}|_X \longrightarrow TV|_X \longrightarrow 0, \quad \mathbf{E}_{i\alpha} = \chi_{i\alpha} Z_i,$$

$$0 \longrightarrow TX \longrightarrow TV|_X \xrightarrow{dQ} \bigoplus_\lambda L^{\alpha_\lambda}|_X \longrightarrow 0.$$

The first is obtained by differentiating the free torus action in the local product charts. Pull back by a genuine selected curve and twist by $-p$. Since $H^0(\mathcal{O}(-p)) = H^1(\mathcal{O}(-p)) = 0$ and $H^1(u^*TX(-p)) = 0$, the complete coefficient derivative is surjective:

$$J_u : \bigoplus_i H^0(\mathcal{O}(d_i)(-p)) \longrightarrow \bigoplus_\lambda H^0(\mathcal{O}(k_\lambda)(-p)).$$

The ranks are $\sum_i d_i$ and $\sum_\lambda k_\lambda$; degree-zero summands have no positive coefficients.

For the matrix presentation, the vertical tangent bundle along a selected map is $\text{End}(u^*\mathcal{S}) = \mathcal{O}^{\oplus 4}$. The coordinate variation bundle is $\mathcal{O}(e)^{\oplus 2n}$, with an additional $\mathcal{O}(2e)$ in the special case. The infinitesimal vertical action is

$$B \longmapsto MB, \quad B \longmapsto (MB, z \text{ tr } B)$$

respectively; its matrix part is injective even when $z = 0$. Differentiating the principal quotient and the regular normal equations therefore gives exactly the same rooted surjectivity argument, since $\text{End}(u^*\mathcal{S})(-p) = \mathcal{O}(-1)^{\oplus 4}$ is acyclic. The complete coefficient ranks are

$$10e \longrightarrow 8e, \quad 12e \longrightarrow 10e, \quad 12e \longrightarrow 10e.$$

Its kernel is $H^0(u^*TX(-p))$. These are all positive coefficients of the stated equations; no normal row or matrix entry is discarded.

For either quotient, take three genuine rooted curves of one common degree vector with independent derivatives at the anchor. Freeness makes root evaluation submersive, and the relevant coordinate splittings persist on an open neighborhood of each chosen map. For matrices take $A_X = H|_X$; the toric choice was made above. For each selected map, root evaluation on its unrestricted Hom component is submersive. Embed that component in a projective variety and take the reduced closure of the evaluation graph in its product with X . Projection to X is proper, and near the selected point this graph is the original smooth Hom space. Cut by $\dim \text{Hom} - 3$ sufficiently ample hypersurfaces through that point, with

differentials independent on the evaluation kernel. The resulting slice is smooth of dimension three there, and its evaluation differential is an isomorphism. There is consequently a unique reduced integral component through the selected point, étale over X at that point. Its image contains a nonempty open subset of X , hence it dominates X . Since both dimensions are three, its generic fibre is zero-dimensional. It is thus proper and generically finite; its boundary in the graph compactification does not change this generic dimension or the selected local chart. Normalization and the finite normalization $\pi_a : T^a \rightarrow X$ in its function field give a proper birational map to T^a . It is an isomorphism near the selected point: it is quasi-finite there, and the target is normal, so Zariski's main theorem applies [63, Lemma 33.17.3(4)]. Finiteness of $T^a \rightarrow X$ follows from finiteness of normalization over the complex projective variety [61].

A high twist of $(\pi_a)_* \mathcal{O}_{T^a}$ by A_X is globally generated and separates the finite anchor fibre. A generic section θ is primitive for the function-field extension and separates the selected residue value from every other residue value over x . Indeed the section space spans the extension at the generic point; the finitely many proper intermediate fields and the finitely many unwanted residue equalities give proper linear conditions. The chosen twist is denoted by $L = A_X^{a_\theta}$, and $n = [\mathbb{C}(T^a) : \mathbb{C}(X)]$. In a local frame of L , the section θ is integral over $\mathcal{O}_X$, and its monic minimal polynomial is $t^n + c_1 t^{n-1} + \dots + c_n$. The coefficients lie in $\mathcal{O}_X$: they belong to $\mathbb{C}(X)$, are integral over $\mathcal{O}_X$, and X is normal. On changing the frame, c_j transforms as a section of L^j . The polynomials therefore define a closed subscheme of $\text{Tot}(L)$, whose algebra is

$$\mathcal{A} = \mathcal{O}_X \oplus L^{-1} \oplus \dots \oplus L^{-(n-1)};$$

multiplication is polynomial multiplication reduced by the monic relation in each frame. The frame transformations identify these algebras on overlaps. The morphism $\mathcal{A} \rightarrow (\pi_a)_* \mathcal{O}_{T^a}$ sends the degree-one local generator to the framed value of θ . It is injective: a relation of degree below n would contradict the minimal polynomial at the generic point. Its generic algebra is the field $\mathbb{C}(T^a)$ because θ is primitive. Thus $T = \text{Spec}_X \mathcal{A}$ is integral and finite locally free of rank n . Its normalization is T^a , since both have that function field and T^a is the normalization of X in it.

The selected point remains simple. Over the henselized normal local ring R of X at x , the normalization algebra splits as $R \times B'$, with $\theta = (t, \theta')$ and $\theta' - t$ a unit in B' . The generic norm of this unit and the norm of its inverse belong to R , so the norm is a unit. Monic division writes the minimal polynomial as $(z - t)q(z)$ with $q(t)$ this unit up to sign. Thus $p'(t)$ is a unit, and normalization is an isomorphism near that simple spectral point. No flatness of the remaining normalization algebra is needed.

Take three identical copies for each of the three selected derivatives. Their disjoint union $\pi : T \rightarrow X$ has rank m and carries a rational family of rooted curves. Its components are integral projective threefolds, and the chosen nine points are simple. After an old pullback and compatible line frames its full algebra is

$$A = \prod_{\alpha} \mathcal{O}_U[\theta_{\alpha}]/(p_{\alpha}), \quad p_{\alpha} \text{ monic.}$$

The copied families use identical regularizations, so each group of three retains the same rooted derivative.

Root normalization gives polynomial coefficients on the generic selector. In the toric case their root lines are $N_{\alpha} = \pi^* L_{\alpha}$, and the source line bundles are $\mathcal{O}(D_{\alpha}) \otimes N_{\alpha}$. In the matrix case, $u^* \mathcal{S}(e)$ is fibrewise trivial; evaluation at the root identifies its rank-two direct image with $\pi^* \mathcal{S}$. Thus the intrinsic root matrix is the tautological inclusion $\pi^* \mathcal{S} \rightarrow \mathbb{C}^n \otimes \mathcal{O}_T$, and the extra root coordinate is a section of $\pi^* H$.

There are finitely many rational positive coefficient sections, one for each of the stated coordinate and source-index blocks. For a rational section A_j of its coefficient bundle, the

local functions h such that hA_j is regular form a coherent denominator ideal; this definition is independent of a bundle frame. Intersect these finitely many ideals with the ideal of the closed complement of the genuine free étale locus. This coherent ideal is the unit ideal at each of the nine selected points. After a sufficiently high twist, global generation and avoidance of the nine linear vanishing conditions give a section d of $B_d = \pi^* A_X^{a_d}$ in that ideal and nonzero at those points. It clears every rational coefficient and vanishes on the entire nongenuine, nonfree and non-étale locus. The same section and framing are used on all three copies of each selector. Source substitution $v \mapsto d^{m+1}v$ leaves the root literal. With $L_s = B_d^{-(m+1)}$, the projective tooth source in the subspace convention is

$$p : E = \mathbb{P}_T(\mathcal{O} \oplus L_s) \longrightarrow T, \quad S \in \mathcal{O}_E(1), \quad T_0 \in p^* L_s \otimes \mathcal{O}_E(1).$$

For source index $j > 0$, each positive coefficient is multiplied by $d^{(m+1)j}$. If dA_j was regular before substitution, its new coefficient is $d^{(m+1)j-1}(dA_j)$. On a local Artin factor of length $\ell \leq m$ supported in $V(d)$, d belongs to its maximal ideal $\mathfrak{m}$ and $\mathfrak{m}^\ell = 0$. Since $(m+1)j-1 \geq m$ for every $j \geq 1$, that entire coefficient is zero in the Artin ring, not just in its residue field.

The toric coordinate Z_i^0 is a section of $p^* N^{\chi_i} \otimes \mathcal{O}_E(d_i)$. The matrix coefficients have types

$$a_j^0 \in \pi^* \text{Hom}(\mathcal{S}, \mathbb{C}^n \otimes \mathcal{O}_X) \otimes L_s^{-j}, \quad c_j^0 \in \pi^* H \otimes L_s^{-j}$$

when the scalar occurs. Consequently M^0 is a morphism $p^* \pi^* \mathcal{S} \rightarrow \mathbb{C}^n \otimes \mathcal{O}_E(e)$, while its minors and z^0 lie in $p^* \pi^* H \otimes \mathcal{O}_E(2e)$. Every equation vanishes as a regular section identity because it vanishes rationally on each integral selector component.

Let $\mathcal{V}$ and $\mathcal{W}$ be the full positive coordinate and equation coefficient bundles. Their precise source-line twists are

$$\mathcal{W} = \bigoplus_{\lambda} \bigoplus_{j=1}^{k_{\lambda}} \pi^* \mathcal{N}_{\lambda} \otimes L_s^{-j},$$

where $\mathcal{N}_{\lambda} = L^{\alpha_{\lambda}}|_X$ for the torus and $\mathcal{N}_{\lambda} = H^{\alpha_{\lambda}}|_X$ for matrices. In the toric case

$$\mathcal{V} = \bigoplus_i \bigoplus_{j=1}^{d_i} N^{\chi_i} \otimes L_s^{-j}.$$

For matrices it is

$$\bigoplus_{j=1}^e \pi^* \text{Hom}(\mathcal{S}, \mathbb{C}^n \otimes \mathcal{O}_X) \otimes L_s^{-j},$$

with $\bigoplus_{j=1}^{2e} \pi^* H \otimes L_s^{-j}$ added in the special case. These need not split into line bundles on T . The polynomial equation map based at Z^0 has

$$F : \mathcal{V} \longrightarrow \mathcal{W}, \quad F(0) = 0, \quad dF_0 = J.$$

Let $Z_{\text{bad}} \subset T$ be the reduced union of the entire zero divisor $V(d)$ and the rank-defect locus of J . The chosen $V(d)$ already contains the nongenuine, nonfree and non-étale loci. It has dimension at most two and misses the nine selected points. The actual value ideal of $\Xi(Z^0)$ on E is the unit ideal off $p^{-1}Z_{\text{bad}}$.

The controller can be constructed without splitting $\mathcal{V}$. If $r = \text{rank } \mathcal{W}$, let

$$\mathcal{D}_J = \bigwedge^r \mathcal{V} \otimes (\det \mathcal{W})^{-1}.$$

The alternating determinant and adjugate maps satisfy $Ja_J(\eta) = d_J(\eta) \text{Id}_{\mathcal{W}}$ in every bundle frame, so this is a polynomial bundle identity. Their determinant image is the full maximal-minor ideal $\mathfrak{f}_J$. For $r = 0$ the empty determinant is 1 and the adjugate map is zero.

Support annihilation on the proper tooth source gives

$$(\mathcal{I}_{Z_{\text{bad}}} \mathcal{O}_E)^{k_1} \subset (\Xi(Z^0))$$

for some k_1 ; see [62, Lemma 30.10.2]. Choose $k \geq \max(k_1, m)$ and $\mathcal{K} = \mathcal{I}_{Z_{\text{bad}}}^k \mathfrak{f}_J$. The map $\mathcal{I}_{Z_{\text{bad}}}^k \mathcal{D}_J \rightarrow \mathcal{K}$ is onto. Serre vanishing after a sufficiently high twist by $\pi^* A_X$ makes its map on global sections onto as well. Thus any section b of $\mathcal{K} \otimes \pi^* A_X^\ell$ has a lift η , and $C_0 = a_J(\eta)$ satisfies

$$JC_0 = b \text{Id}_{\mathcal{W}}, \quad b = B^T \Xi(Z^0). \quad (5.1)$$

The second identity is local on every projective source chart, including infinity; it follows from the support-power inclusion. Here B denotes a local coefficient column after choosing frames of the line bundles of b and $\Xi(Z^0)$; no global choice of B is asserted or needed. Different local columns may differ by relations among the entries of $\Xi(Z^0)$. Each nevertheless gives the same element b of their ideal. The later ideal comparisons will be proved on each such chart, and hence agree intrinsically on overlaps. Both identities survive arbitrary old base change. Since $\mathcal{I}_{Z_{\text{bad}}}^m$ vanishes in a local Artin ring of length at most m supported on the bad locus, b vanishes on that whole factor, not only on its residue field.

A single sufficiently high twist of $\mathcal{K}$ separates any six points outside Z_{bad} . Indeed choose a globally generated first twist and a very ample second twist. For each one of six points, multiply a section nonzero there by five hyperplanes vanishing at the other points and nonzero at that point. Increasing the twist also separates the fixed entire good anchor fibre; the preceding global lift to η remains available.

Call a characteristic value active if it is nonzero and occurs at exactly one geometric point; all other values are inactive. Let $\mathfrak{B} = H^0(T, \mathcal{K} \otimes \pi^* A_X^\ell)$ be the section parameter space. Three distinct inactive values have one of three witnesses: three disjoint pairs of good points with equal values; one bad zero point and two such pairs; or one good zero point and two such pairs. All points in a witness are distinct and lie over the same target point. These configuration schemes are locally closed subsets of fibre powers of T over X ; in the second case one factor lies in Z_{bad} . Since $T \rightarrow X$ is finite and $\dim Z_{\text{bad}} \leq 2$, their dimensions are at most 3, 2, 3, respectively.

The entries $b(t_i)$ lie in the same line $(A_X^\ell)_x$, so their equalities are independent of a frame choice. In the first case there are three pair-equality conditions; in the second there are two, since b already vanishes at the bad point; in the third there are two equalities and one good-point zero condition. Six-point separation makes these linear conditions on $\mathfrak{B}$ independent, of ranks 3, 2, 3. The corresponding incidence schemes over the configuration schemes therefore have dimensions at most $\dim \mathfrak{B}$. They have finitely many irreducible components. Remove from $\mathfrak{B}$ the closures of the images of the nondominant components. Each remaining dominant component has generic fibre dimension zero, and hence has zero-dimensional fibres over a nonempty open parameter set. Taking the intersection of these finitely many open sets gives a common choice of b . Each of its witness schemes is zero-dimensional of finite type, and thus has finitely many points; properness of the configuration schemes is not needed.

Choose b also so that all good anchor values are nonzero and pairwise distinct. The target images of its actual witnesses form a finite closed set not containing the anchor. Its complement is therefore a target neighborhood on which at most two distinct inactive values occur. The selected values are simple in the full characteristic polynomial, including all unselected factors.

Lemma 5.2. *Under either hypothesis of Theorem 5.1, the finite selector, coefficient map, controller and value-ideal data used below exist at every anchor. On the three-dimensional selector, the construction yields a finite locally free spectral algebra with the chosen points simple, a full coefficient map $F : \mathcal{V} \rightarrow \mathcal{W}$ whose differential J is surjective at those points, and*

a closed bad locus of dimension at most two. Regular sections b, C_0 satisfy

$$JC_0 = b \operatorname{Id}_{\mathcal{W}}, \quad b \in (\Xi(Z^0))$$

in every local projective-source frame. The value ideal of the baseline tuple is the unit ideal away from the bad locus. For any old base change, the pullback of this spectral algebra is finite locally free, and the pullbacks of the two displayed identities remain valid.

Proof. In the toric case, rational connectedness supplies a very free rooted curve through the anchor. Rooted jet submersivity supplies three nearby curves of the same degree vector with independent derivatives. The Euler sequence for the free Cox quotient and the regular normal sequence give the displayed surjective full coefficient derivative. In the matrix cases, the three curves are part of the hypothesis; the equal splitting makes the vertical bundle trivial, and the principal-quotient and regular-normal sequences give the same surjectivity. Thus every selector component contains a point at which J has full rank.

The proper generically finite slices and the monic primitive-element construction above give the finite locally free spectral algebra and preserve those simple points. The denominator section d contains the entire nongenuine, nonfree and non-étale locus. Hence neither $V(d)$ nor the rank-defect locus of J contains a selector component; their reduced union has dimension at most two. Off that union the baseline is a genuine quotient-valued curve, so its complete projective value ideal is the unit ideal. The support-power inclusion, the maximal-minor adjugate identity and Serre lifting then give the regular b and C_0 and the two displayed identities. Pullback preserves finite local freeness and carries these identities to the old parameter space. This proves all assertions. $\square$

5.3 The spectral coefficient algebra

5.3.1 The universal coefficient ring

The preceding finite selectors supply the class of algebras needed for the deformation calculation. Throughout this subsection, the *spectral product* is

$$A = \prod_{\alpha=1}^q R[\theta_\alpha]/(p_\alpha(\theta_\alpha)), \quad p_\alpha(T) \text{ monic of degree } n_\alpha, \quad m = \sum_{\alpha} n_\alpha, \quad (5.2)$$

over an arbitrary commutative $\mathbb{C}$ -algebra R . Neither R nor its finite factors need be reduced. The factors are disjoint; the chosen attachment b need not generate their product. After old framing, the preceding finite selector gives exactly this presentation, including when repeated components have identical defining polynomials. Every further old base change preserves this class.

All generic calculations below occur in one free universal polynomial model. They are descended there as polynomial identities and only then specialized to the actual old data. We never require a dense good subset of the actual old base. No result for arbitrary finite algebras outside the specified spectral class is imported into this proof.

Let

$$R_{\text{univ}} = \mathbb{C}[c_{\alpha j}, \beta_{\alpha j}, \omega_{\alpha j} : 1 \leq \alpha \leq q, 0 \leq j < n_\alpha], \quad p_\alpha(T) = T^{n_\alpha} + \sum_{j=0}^{n_\alpha-1} c_{\alpha j} T^j.$$

The corresponding universal data are

$$A_0 = \prod_{\alpha} R_{\text{univ}}[\theta_\alpha]/(p_\alpha), \quad b_\alpha = \sum_j \beta_{\alpha j} \theta_\alpha^j, \quad \eta_\alpha = \sum_j \omega_{\alpha j} \theta_\alpha^j. \quad (5.3)$$

The algebra A_0 is free of rank m over the integral domain R_{univ} . All of its tensor powers are also free. Every instance (5.2), together with specified elements $b, \eta \in A$, is obtained by a ring map $R_{\text{univ}} \rightarrow R$ and base change from (5.3). Equalities between coefficients of copied factors are simply imposed by that map.

For statements with an arbitrary weight use $r = \eta$. For the weighted handle statements, use $\rho = \eta$, $r = b^3\eta$ in the *same* ring. Arbitrary-weight polynomial identities may be specialized to $r = b^3\rho$ by substituting the monomial coordinates of $b^3\rho$ for the weight variables. The free universal ring is not replaced by a possibly nonflat ring of constrained weight data.

Lemma 5.3. *For $F = \text{Frac}(R_{\text{univ}})$ and an algebraic closure $\overline{F}$, one has $A_0 \otimes \overline{F} = \overline{F}^m$. Its attachment values $b_1, \dots, b_m$ are pairwise distinct and nonzero. The values of η are nonzero. Thus, for either $r = \eta$ or $r = b^3\eta$, the numbers*

$$\sigma_i = P'(b_i)^2 r_i, \quad P(z) = \det(z \text{Id} - \text{mult}_b),$$

are nonzero.

Proof. Each polynomial p_α has nonzero discriminant generically. To check all further nonvanishing conditions simultaneously, specialize the p_α to polynomials whose combined list of roots consists of m distinct nonzero complex numbers. Choose b_α equal to the residue class of θ_α and choose $\eta_\alpha = 1$. (In degree one, the constant representative of that residue class is used.) At this specialization the characteristic polynomial of b has m distinct nonzero roots and every weight value is nonzero. Hence the relevant discriminants and norms are nonzero polynomials in R_{univ} . Their generic values are nonzero, proving the claim. $\square$

Lemma 5.4. *An identity between polynomial operations on finitely many copies of A_0 or its tensor powers follows if it holds over $\overline{F}$. The same is true with finitely many polynomial variables adjoined, or with a target $R_{\text{univ}}[z]/(z^2)$. Rational expressions with denominators powers of the monic $P(z)$ may first be cleared and then descended. Every resulting identity survives every specialization $R_{\text{univ}} \rightarrow R$.*

Proof. Use the displayed monomial bases, and add independent scalar variables for every source coordinate. Each asserted identity becomes finitely many polynomial identities in a free module over a polynomial ring over the domain R_{univ} . That module injects into its scalar extension to $\overline{F}$. The quotient $R_{\text{univ}}[z]/(z^2)$ is free with basis $1, z$. Clearing P reduces the rational case to the polynomial case. Substitution in a polynomial identity is valid over every ring, including rings with nilpotents and zero divisors. $\square$

Remark 5.5. This lemma does not authorize testing an identity in $A_0/(b^2)$ after inverting b or its norm. That localization is zero. For example, a nonzero element of $\mathbb{C}[b]/(b^2)$ maps to zero there. The retained-evaluation and root statements below are established by *full-module identities with visible b^2 factors*, and only then reduced. Generic injectivity of the rational model is likewise never asserted after a bad specialization.

5.3.2 Polynomial Cauchy operators and their rational realization

The following formulas first specify operations in the universal model; specialization then gives them on every spectral product:

$$P(z) = \det(z \text{Id} - \text{mult}_b), \quad \Delta(u, v) = \frac{P(u) - P(v)}{u - v}, \quad (5.4)$$

$$K(u, v) = \frac{P'(v)^2 - P'(u)\Delta(u, v)}{u - v} + \frac{3}{2}P''(v)\Delta(u, v), \quad K_2(u, v) = \frac{K(u, v)}{u - v}. \quad (5.5)$$

All expressions are polynomials. Indeed the first numerator in (5.5) vanishes at $u = v$; the value of its first quotient on the diagonal is $-\frac{3}{2}P'(v)P''(v)$. The added term therefore makes $K(v, v) = 0$, proving the second divisibility.

Let tensor subscripts indicate the factors of $A \otimes_R A$, and let Tr_2 be the regular-algebra trace in the second factor. The operators are

$$\begin{aligned} \sigma &= P'(b)^2 r, & \mathfrak{t}(a) &= \text{Tr}(\sigma a), \\ D(a) &= \text{Tr}_2(K(b_1, b_2) r_2 a_2), & \delta &= D(1), \end{aligned} \quad (5.6)$$

$$B(a, c) = aD(c) + cD(a) - ac\delta - D(ac), \quad (5.7)$$

$$\Theta(a, c) = \text{Tr}_2(K_2(b_1, b_2) r_2 (a_1 - a_2)(c_1 - c_2)). \quad (5.8)$$

These operations have polynomial coefficients in the universal ring. Traces are traces of finite multiplication matrices; no trace pairing is inverted.

On the generic split algebra, use i, j for its m factors. The regular trace is the sum of coordinates. Direct evaluation gives

$$D(a)_i = \sum_{j \neq i} \frac{\sigma_j a_j}{b_i - b_j}, \quad B(a, c)_i = - \sum_{j \neq i} \frac{\sigma_j (a_i - a_j)(c_i - c_j)}{b_i - b_j}, \quad (5.9)$$

and

$$\Theta(a, c)_i = \sum_{j \neq i} \frac{\sigma_j (a_i - a_j)(c_i - c_j)}{(b_i - b_j)^2}. \quad (5.10)$$

The diagonal terms of K_2 need not vanish, but the two difference factors in Θ kill them. In particular, they must not be removed from the definition of an operator using K_2 without those factors.

On the localization where $P(z)$ is invertible, the rational realization is

$$R_a(z) = \mathfrak{t}((z - b)^{-1} a). \quad (5.11)$$

This is algebraically well defined since Cayley–Hamilton gives

$$(z - b)\Delta(z, b) = P(z), \quad R_a(z) = \frac{U_a(z)}{P(z)}, \quad U_a(z) = \text{Tr}(\sigma \Delta(z, b) a) \in R[z].$$

On the generic split algebra it is $R_a = \sum_i \sigma_i a_i / (z - b_i)$.

Lemma 5.6. *For every spectral product and all $a, c \in A$,*

$$R_a R_c = R_1 R_{ac} + R_{B(a, c)}. \quad (5.12)$$

Proof. Work first over $\overline{F}$. At $z = b_i$ the double poles of $R_a R_c - R_1 R_{ac}$ cancel. Its residue is

$$\sigma_i (a_i D(c)_i + c_i D(a)_i - D(ac)_i - a_i c_i \delta_i) = \sigma_i B(a, c)_i.$$

Thus $R_a R_c - R_1 R_{ac} - R_{B(a, c)}$ has no finite poles and tends to zero at infinity. It is zero. After multiplication by $P(z)^2$, this is the polynomial identity

$$U_a U_c - U_1 U_{ac} - P U_{B(a, c)} = 0. \quad (5.13)$$

Lemma 5.4 descends it, and specialization proves (5.12) over the stated arbitrary base. $\square$

5.3.3 Associativity from highest poles

For $j \geq 1$, let $e_j(a)$ denote a symbol linear in $a \in A$, and take

$$\mathcal{M}_d = R[\varepsilon] \oplus \bigoplus_{j=1}^d e_j(A[\varepsilon]), \quad \mathcal{M} = \bigcup_{d \geq 0} \mathcal{M}_d.$$

The first summand is the old scalar ring, not another copy of A . Products with constants are scalar products, and the other products are

$$e_j(a)e_k(c) = e_{j+k}(ac) + \varepsilon e_{j+k-1}(B(a, c)). \quad (5.14)$$

This is initially only a bilinear product on the displayed free module.

Theorem 5.7. *The product (5.14) is associative, commutative, and unital. It respects $\mathcal{M}_d \mathcal{M}_e \subset \mathcal{M}_{d+e}$ and commutes with arbitrary base change among spectral products. There is an algebra homomorphism*

$$\mathcal{M} \longrightarrow R[\varepsilon, z, P(z)^{-1}], \quad e_j(a) \longmapsto \varepsilon^j R_1^{j-1} R_a. \quad (5.15)$$

The homomorphism is injective in the generic universal model, but no injectivity is claimed after specialization.

Proof. Equation (5.12) proves preservation of every product (5.14), without presupposing associativity. Work now over $\overline{F}$. Suppose

$$c + \sum_{j=1}^d \varepsilon^j R_1^{j-1} R_{a_j} = 0, \quad c \in \overline{F}[\varepsilon], \quad a_j \in \overline{F}^m[\varepsilon].$$

At $z = b_i$, the coefficient of the highest possible pole $(z - b_i)^{-d}$ is $\varepsilon^d \sigma_i^d (a_d)_i$. The polynomial ring $\overline{F}[\varepsilon]$ is a domain and $\sigma_i \neq 0$, so $(a_d)_i = 0$. Doing this at each i gives $a_d = 0$. Descend through $d - 1, \dots, 1$ and then obtain $c = 0$. This proves injectivity on each finite block.

The image lies in an associative commutative algebra. For any three elements of bounded degree, their associator lies in a finite free block and its image is zero. Generic injectivity makes it zero over $\overline{F}$. Its old scalar coordinates are polynomials in the universal parameters and the source coordinates. Lemma 5.4 therefore makes it zero before specialization and over every R . Commutativity, the unit, the filtration, and base change are also immediate from the defining polynomial formulas. Finally, (5.13) already proves the rational homomorphism over every base, even where its generic injectivity has disappeared. $\square$

At $\varepsilon = 0$ the algebra is exactly $R \oplus vA[v]$.

A literal homogenized algebra. The symbol e_j is $R[\varepsilon]$ -linear, and the constant summand of $\mathcal{M}_d$ is $R[\varepsilon]$, embedded through the scalar unit of A . In particular, $\mathcal{M}_d$ is free of rank $1 + md$ over $R[\varepsilon]$. To homogenize without a presentation claim, introduce a formal variable ζ , unrelated to the later fresh-time coordinate, and take the Rees algebra

$$\mathcal{S} = \bigoplus_{d \geq 0} \mathcal{M}_d \zeta^d \subset \mathcal{M}[\zeta].$$

Write $\lambda = \zeta$, $Y_a = e_1(a)\zeta$, and $w = e_1(1)\zeta$. Since $B(1, a) = 0$, this algebra is generated by λ and the degree-one symbols Y_a , and its multiplication obeys

$$Y_a Y_c = w Y_{ac} + \varepsilon \lambda Y_{B(a, c)}.$$

Its degree- d component is, by its definition, the free module

$$R[\varepsilon]\lambda^d \oplus \bigoplus_{j=1}^d \{w^{j-1}Y_a\lambda^{d-j} : a \in A[\varepsilon]\}.$$

Indeed these summands are respectively $R[\varepsilon]\zeta^d$ and $e_j(A[\varepsilon])\zeta^d$ inside $\mathcal{M}_d\zeta^d$. Dehomogenization $\lambda = 1$ identifies each fixed homogeneous degree with $\mathcal{M}_d$, so the generic rational realization is faithful on that degree. This does not give injectivity on the direct sum of all degrees: $\lambda - 1$ maps to zero. We check each later homogeneous identity in its single finite free block and descend it as a polynomial identity before arbitrary old specialization.

Corollary 5.8. *The product $a \circ_t c = ac + tB(a, c)$ on $A[t]$ is associative, commutative, and unital.*

Proof. Expand the associator of $e_1(a), e_1(c), e_1(d)$ using (5.14). Its e_2 and e_1 coefficients are, respectively, the coefficients of t and t^2 in the associator of $\circ_t$. The free direct summands and powers of ε separate these coefficients. They vanish by Theorem 5.7; the constant coefficient is ordinary associativity in A . $\square$

In either presentation every coordinate monomial in Q_λ has the same actual source degree k_λ . For the torus, $d_i = \chi_i \cdot D$ and $k_\lambda = \alpha_\lambda \cdot D$; for matrices, entries have degree e , the scalar has degree $2e$, and $k_\lambda = 2ea_\lambda$. The same algebra gives its full raw residual in $\mathcal{M}_{k_\lambda}$, with zero scalar root; all its positive coefficients are retained.

Lemma 5.9. *The substitution $e_j(a) \mapsto (\varepsilon R_1)^{j-1} \varepsilon R_a$ is an algebra homomorphism into $R[\varepsilon, z, P(z)^{-1}]$ for every spectral product over an arbitrary old base.*

Proof. This is the homomorphism in Theorem 5.7; its product identity is the denominator-cleared polynomial equality (5.13), specialized before any evaluation is made. $\square$

5.3.4 The module identities

Proposition 5.10. *On every spectral product,*

$$bD(a) - D(ba) = \mathfrak{t}(a)1 - \sigma a, \quad (5.16)$$

$$D(a)D(c) - \delta D(ac) - D(B(a, c)) = -\sigma\Theta(a, c). \quad (5.17)$$

Both are identities in the full old module A .

Proof. In generic coordinates (5.9) immediately gives

$$b_i D(a)_i - D(ba)_i = \sum_{j \neq i} \sigma_j a_j = \mathfrak{t}(a) - \sigma_i a_i.$$

This proves (5.16) by polynomial descent.

For the defect let $V_i(a) = \sum_{j \neq i} \sigma_j a_j / (b_i - b_j)^2$ and use $x = z - b_i$. The Laurent expansion at this simple pole is

$$R_a(z) = \frac{\sigma_i a_i}{x} + D(a)_i - xV_i(a) + O(x^2).$$

Taking the constant coefficient of (5.12) gives

$$\begin{aligned} D(a)_i D(c)_i - \delta_i D(ac)_i - D(B(a, c))_i &= \sigma_i (a_i V_i(c) + c_i V_i(a) - V_i(ac) - a_i c_i V_i(1)) \\ &= -\sigma_i \Theta(a, c)_i, \end{aligned}$$

using (5.10). Both sides of (5.17) were defined by polynomial operations on A_0 . Descent therefore proves the full-module identity, with no attachment quotient involved. $\square$

The off-diagonal V_i in the proof is only notation for a Laurent coefficient. No globally defined V_i is used in the weighted argument. The following useful identities also hold:

$$B(b, a) = \mathfrak{t}(a)1 - \mathfrak{t}(1)a, \quad (5.18)$$

$$B(ba, c) = bB(a, c) + \mathfrak{t}(ac)1 - \mathfrak{t}(a)c, \quad (5.19)$$

$$\mathfrak{t}(B(a, c)) = 0. \quad (5.20)$$

The first two result by substituting (5.16) in the definition of B . The last is the coefficient of z^{-1} in (5.12) expanded at infinity. Its two product terms are $O(z^{-2})$, while $R_{B(a, c)} = z^{-1}\mathfrak{t}(B(a, c)) + O(z^{-2})$.

5.3.5 Handle jets and retained evaluation

From now on take $r = b^3\rho$. Only the first-order handle algebra $R[z]/(z^2)$ is needed. Its handle jet is

$$H_a(z) = -\mathrm{Tr}(P'(b)^2 b^2 \rho a) - z \mathrm{Tr}(P'(b)^2 b \rho a) \quad \text{in } R[z]/(z^2). \quad (5.21)$$

Lemma 5.11. *The assignment*

$$e_j(a) \longmapsto \varepsilon^j H_1(z)^{j-1} H_a(z) \quad (5.22)$$

is an algebra homomorphism into $R[\varepsilon, z]/(z^2)$.

Proof. In the generic universal model $b_i \neq 0$, so (5.21) is the Taylor truncation of R_a at $z = 0$. Truncating (5.12) gives

$$H_a H_c = H_1 H_{ac} + H_{B(a, c)} \pmod{z^2}.$$

Its coefficients are polynomial in the universal parameters and source coordinates. The free module $R_{\mathrm{univ}}[z]/(z^2)$ therefore permits descent by Lemma 5.4. The resulting identity proves preservation of (5.14) and all products. No identity in $A/(b^2)$ has been tested after localization. $\square$

On the full old module the retained linear map is

$$\Phi \left(c + \sum_{j=1}^d e_j(a_j) \right) = c + \sum_{j=1}^d (v + \varepsilon \delta)^{j-1} (a_j v + \varepsilon D(a_j)) \in A[\varepsilon, v]. \quad (5.23)$$

It need not be multiplicative there. Equations (5.14) and (5.17) give the exact defect

$$\Phi(e_i(a)e_j(h)) - \Phi(e_i(a))\Phi(e_j(h)) = \varepsilon^2(v + \varepsilon \delta)^{i+j-2} \sigma \Theta(a, h). \quad (5.24)$$

The sign is positive for image of the product minus product of the images.

Lemma 5.12. *Reduction of Φ modulo b^2 is the algebra homomorphism*

$$e_j(a) \longmapsto (v + \varepsilon D_r(1))^{j-1} (av + \varepsilon D_r(a)) \quad \text{in } (A/(b^2))[\varepsilon, v]. \quad (5.25)$$

Its value at $v = 0$ equals (5.22) followed by $z \mapsto b$ in $A/(b^2)$. In the full old module,

$$D_r(a) - \mathrm{ev}_b(H_a) = b^2 \mathcal{E}(a), \quad (5.26)$$

where

$$\mathcal{E}(a) = b D_\rho(a) - \mathrm{Tr}(P'(b)^2 \rho a) 1 + 3P'(b)^2 \rho a. \quad (5.27)$$

Here ev_b substitutes b in the degree-at-most-one representative; it is only old-linear as a map into the full A .

Proof. Since $\sigma = P'(b)^2 b^3 \rho$, the full-module defect (5.24) vanishes modulo b^2 , proving multiplicativity without any flatness of $A/(b^2)$.

For the root identity, (5.16) for weight ρ reads

$$D_\rho(bx) = bD_\rho(x) - \text{Tr}(P'(b)^2 \rho x) 1 + P'(b)^2 \rho x.$$

Apply it successively to $x = b^2 a, ba, a$. Since $D_r(a) = D_\rho(b^3 a)$, the result is

$$D_r(a) = b^3 D_\rho(a) - \text{Tr}(P'(b)^2 b^2 \rho a) 1 - b \text{Tr}(P'(b)^2 b \rho a) - b^2 \text{Tr}(P'(b)^2 \rho a) + 3b^2 P'(b)^2 \rho a.$$

Subtracting the displayed representative of $H_a(b)$ proves (5.26) before reduction. In particular the trace in (5.27) is an old scalar times 1. Only now reduce modulo b^2 : the generator roots agree, hence so do the two algebra homomorphisms on all of $\mathcal{M}$. $\square$

The square weight does not suffice uniformly for root agreement. For $r = b^2 \rho$, take $A = R = \mathbb{C}[s]$, $b = s$, and $\rho = 1$. Then $D_r = 0$, whereas the first-order Taylor handle of $e_1(1)$ has root $-2\varepsilon s \neq 0$ in $(\mathbb{C}[s]/(s^2))[\varepsilon]$.

Every descended identity has finitely many polynomial coordinate expressions and may therefore be evaluated on the later protected holomorphic parameter products. Raw coefficients and weights may depend holomorphically on ε after the source inverse. This is substitution in polynomial identities, not injectivity of the rational realization after specialization.

5.4 Moment weights and quotient gauges

Fix a compact convex $K \subset \mathbb{C}^N$ and a holomorphic core f near K , with anchor $a \in K$ and $f(a) = x$. Pull the preceding data back to one convex Stein neighborhood U of all of K . Oka–Grauert gives compatible frames of $f^* A_X$ and $f^* TX$, and of $f^* L_\alpha$ in the toric case or $f^* \mathcal{S}$ in the matrix case; the monic orders then have fixed full-module frames. The coordinate root f_i is one old tuple common to every label; in the matrix case it consists of (F, ζ) , with the scalar omitted when absent. Write A for the full rank- m algebra and $P(z) = \det(z - \text{mult}_b)$. As above, $r = b^3 \rho$ and $\sigma = P'(b)^2 b^3 \rho$; N continues to denote the arbitrary old source dimension.

All frames, splittings, controllers and coefficient modules below are fixed once on this Stein neighborhood U . The finite cover serves only to choose uniform radii for inverse charts and matrix inverses. In the toric case, the slice $C_s(Z - f) = 0$ and the coordinate map (ξ, Q) are global on U ; in the matrix case, the normalization by Λ and the tangent and normal coordinates are likewise global. Local inverse charts are therefore restrictions of one inverse and agree on their common domains. Thus the residual maps form a single holomorphic system in fixed module coordinates.

In the compatible source-line frame, v is the scalar affine tooth coordinate. For each coordinate i , the framed baseline has the form

$$Z_i^0 = f_i + \sum_{j=1}^{d_i} a_{ij}^0 v^j, \quad a_i^0 = (a_{ij}^0)_{j=1}^{d_i}.$$

For a positive coefficient list $\mathbf{c} = (c_j)_{j=1}^{d_i}$, write $\text{pol}_i(\mathbf{c}) = \sum_{j=1}^{d_i} c_j v^j$ and let pol act coordinate-wise. A degree-zero list is empty. The matrix-entry and covering-coordinate lists have degrees e and $2e$, respectively.

In the fixed tangent frame let $v_0 \in A^3$ be the rooted baseline derivative. The moment and residue maps are

$$(\mathcal{M}\rho)_\ell = -\text{Tr}(P'(b)^2 b^{2-\ell} \rho v_0), \quad \ell = 0, 1, \quad \mathcal{R}(\rho) = \text{Tr}(P'(b)^2 b^3 \rho v_0).$$

At the anchor, two selected columns from each of the three copied triples give an invertible 2-by-2 Vandermonde block, after changing the tangent basis to the three selected derivatives. Thus $\mathcal{M} : A \rightarrow \mathcal{O}_U^6$ has rank 6 there. For one selected triple with values b_i , let $Q_\alpha(z) = \prod_i (z - b_i)$. The vector supported on that triple, with entries

$$\kappa_i = \frac{b_i^2}{Q'_\alpha(b_i)P'(b_i)^2b_i^3},$$

has zero moments and residue equal to its selected derivative, by Lagrange interpolation. Prescribe units on all remaining Artin factors and correct their moments using six selected columns. Adding the three vectors κ_α avoids all nine coordinate-zero hyperplanes. Consequently the unit locus in the moment kernel is nonempty. The residue map on that kernel is surjective; its three-column determinant and the three unit conditions are simultaneous nonempty Zariski-open conditions. Choose three full-unit kernel weights with independent anchor residues.

In the fixed monomial frame, let I index six columns such that $\Delta = \det \mathcal{M}_I$ is nonzero at a . The map inc_I inserts those six coordinates into the fixed monomial frame of A . The adjugate gives

$$q = \Delta/\Delta(a), \quad D_0 = \text{inc}_I \text{adj}(\mathcal{M}_I)/\Delta(a), \quad \Pi = q \text{Id} - D_0 \mathcal{M}, \quad \mathcal{M} D_0 = q \text{Id}, \quad \mathcal{M} \Pi = 0.$$

Extend each chosen anchor weight as constant coordinates and apply Π . The resulting ρ_0 has zero moments on all of U and is a unit near a . Only the fixed complex number $\Delta(a)$ was divided.

5.4.1 The integral-character toric gauge

The full Euler matrix along the lifted old core is $E_{i\alpha} = \chi_{i\alpha} f_i$. Its rank is r_0 everywhere. Cartan B splits the holomorphic exact sequence and supplies an r_0 -by- s_0 matrix C_s with $C_s E = \text{Id}_{r_0}$. The affine slice $C_s(Z - f) = 0$ is transverse to the torus orbits. Its dimension is $3 + c$, and the complete normal coordinates are $\nu_\lambda = Q_\lambda$. A Stein splitting of the tangent-normal sequence extends the fixed three dual tangent coordinates to linear functionals ξ on this slice. Thus (ξ, ν) has an inverse chart $Z = \chi_s(\xi, \nu)$ near the whole graph of f . A finite old cover, parameter IFT, and central uniqueness give one chart on a common product neighborhood of $K \times \{0\}$. The old image need not lie in a single target chart.

Keep ε , the positive deviations y , and the weight variation $\delta\rho$ independent, with $\rho = \rho_0 + \delta\rho$. For raw lists a_{ij} the universal handle and retained expressions are

$$Z_{H,i} = f_i + \sum_{j=1}^{d_i} \varepsilon^j H_1^{j-1} H_{a_{ij}} \in J_2, \quad J_2 = \mathcal{O}_U[z]/(z^2),$$

$$\Phi(Z_i) = f_i + \sum_{j=1}^{d_i} (v + \varepsilon\delta)^{j-1} (v a_{ij} + \varepsilon D_r(a_{ij})), \quad \delta = D_r(1).$$

Let T_d be the positive coefficient map of this retained expression; it is an old-linear unipotent triangular operator, with top row the identity. Evaluation $\text{ev}_b : J_2 \rightarrow A$ uses the unique representative of degree at most one and is only old-linear.

Choose gauge jets γ_α near 1, and let $\Gamma_\alpha = \text{ev}_b(\gamma_\alpha)$. For every integral character, γ^μ and Γ^μ use multiplication and inversion in the jet algebra and the full label algebra, respectively. The simultaneous equations are

$$C_s((\gamma^{-\chi_i} Z_{H,i})_i - f) = 0, \quad T_{d_i} a_i = \Gamma^{\chi_i}(a_i^0 + y_i) \quad (d_i > 0). \quad (5.28)$$

At $\varepsilon = 0$ their central solution is $\gamma_\alpha = 1$ and $a = a^0 + y$. The upper derivative in the gauge variables is $-C_s \mathbf{E} = -\text{Id}_{r_0}$ in the full jet algebra; its derivative in a is zero. The lower diagonal block in a is the identity. Consequently parameter IFT, on a finite compact cover and with central uniqueness on overlaps, gives one holomorphic solution over a common product neighborhood of the whole old compact and the bounded coefficient and weight balls. Every label and jet nilpotent is retained among these finite old scalar coordinates. Shrink the domain so that all γ_α are jet units and every multiplication determinant of Γ_α is nonzero. For $d_i = 0$ there is no positive unknown and the raw coordinate is exactly f_i . Its normalized coordinate is $\gamma^{-\chi_i} f_i$; no gauge is required to equal one.

The normalized handle coordinates $C_i = \gamma^{-\chi_i} Z_{H,i}$ satisfy the slice equations. Their actual coordinates satisfy

$$C = \chi_s(\varepsilon j, \varepsilon n) = f + \varepsilon V, \quad j \in J_2^3, \quad n \in J_2^c.$$

All quotients by ε are ordinary Hadamard remainders in the still independent parameter, before any substitution involving q . After fixing the bounded coefficient balls, choose the ε disk so that the constant jet entries lie in the chart domain; its nilpotent entries are evaluated by finite Taylor expansion.

For jets $p_1, \dots, p_t$, let

$$\mathcal{C}_b(p_1, \dots, p_t) = \sum_{j_1 + \dots + j_t \geq 2} (p_1)_{j_1} \cdots (p_t)_{j_t} b^{j_1 + \dots + j_t - 2}.$$

Thus $\prod_i \text{ev}_b(p_i) - \text{ev}_b(\prod_i p_i) = b^2 \mathcal{C}_b(p_1, \dots, p_t)$; the product under the last evaluation is truncated in J_2 . Every exponent is nonnegative and the tail for one factor is zero. The tail operator is old-multilinear, and is zero if at most one argument has a positive jet coefficient. This observation also controls inverse gauges exactly. Write $\gamma_\alpha = 1 + \varepsilon \eta_\alpha$ and $\gamma_\alpha^{-1} = 1 + \varepsilon \xi_\alpha$ in J_2 . Since their truncated product is one,

$$\Gamma_\alpha^{-1} - \text{ev}_b(\gamma_\alpha^{-1}) = -\varepsilon^2 b^2 \Gamma_\alpha^{-1} \mathcal{C}_b(\eta_\alpha, \xi_\alpha). \quad (5.29)$$

Indeed the product of their evaluations equals $1 + \varepsilon^2 b^2 \mathcal{C}_b(\eta_\alpha, \xi_\alpha)$; multiplying this identity by the existing unit Γ_α^{-1} proves the formula. No attachment, smoothing parameter, or jet coefficient is cancelled.

Here is a finite formula for every character $\mu \in \mathbb{Z}^{r_0}$. List $|\mu_\alpha|$ signed factors for each α . Write these as $p_\ell = 1 + \varepsilon \theta_\ell \in J_2$ and $g_\ell = \Gamma_\alpha^{\pm 1} \in A$ for $1 \leq \ell \leq t$. For a positive factor set $\kappa_\ell = 0$; for a negative factor set $\kappa_\ell = -\Gamma_\alpha^{-1} \mathcal{C}_b(\eta_\alpha, \xi_\alpha)$. Then $g_\ell - \text{ev}_b(p_\ell) = \varepsilon^2 b^2 \kappa_\ell$. For $C = f + \varepsilon V \in J_2$, let

$$\begin{aligned} \mathcal{T}_{\mu,V} &= \sum_{\ell=1}^t \kappa_\ell \prod_{a < \ell} g_a \prod_{a > \ell} \text{ev}_b(p_a) \text{ev}_b(C) \\ &\quad + f \sum_{\substack{I \subset \{1, \dots, t\} \\ |I| \geq 2}} \varepsilon^{|I|-2} \mathcal{C}_b((\theta_a)_{a \in I}) \\ &\quad + \sum_{\substack{I \subset \{1, \dots, t\} \\ |I| \geq 1}} \varepsilon^{|I|-1} \mathcal{C}_b((\theta_a)_{a \in I}, V). \end{aligned}$$

Empty sums are zero. Finite product telescoping followed by the subset expansion of the p_ℓ gives the full-module identity

$$\Gamma^\mu \text{ev}_b(C) - \text{ev}_b(\gamma^\mu C) = \varepsilon^2 b^2 \mathcal{T}_{\mu,V}. \quad (5.30)$$

All attachment and smoothing exponents in this formula are nonnegative. Only the gauge units are inverted. It applies to every coordinate and every equation character, including negative characters, over all old Artin factors.

5.4.2 The ordered matrix gauge

On the whole old Stein neighborhood, compatible frames give the common root (F, ζ) and a holomorphic left inverse $\Lambda F = I_2$. The relative normalization is

$$(R, w) \mapsto (R(\Lambda R)^{-1}, w(\det(\Lambda R))^{-1}).$$

At the root its derivative is $(h, k) \mapsto (h - F\Lambda h, k - \zeta \operatorname{Tr}(\Lambda h))$. The scalar parts are omitted in the ordinary and genus-eight cases. Its dimension is $3 + c$, and the full normal coordinates are $\nu_\lambda = Q_\lambda$. A Stein splitting of its tangent-normal sequence supplies the three tangent coordinates ξ . Parameter IFT and central uniqueness on a finite compact old cover give one inverse chart $Z = \chi_s(\xi, \nu)$ along the whole old graph. The old image need not lie in a single target chart. As above, $\varepsilon, y, \delta\rho$ stay independent; the entrywise operators T_e, T_{2e} and H are the universal operators. The normalized pair (C, K_H) has chart coordinates $(\varepsilon j, \varepsilon n)$, and the scalar block is omitted when absent. All divisions by ε are Hadamard remainders in that independent scalar parameter.

Let $r_M = a^0 + y_M$ and, when present, $r_z = c^0 + y_z$ be the normalized positive lists. For raw matrix coefficients a , write

$$R_H = F + \sum_{j=1}^e \varepsilon^j H_1^{j-1} H_{a_j}, \quad G = \Lambda R_H, \quad \Gamma = \operatorname{ev}_b(G), \quad C = R_H G^{-1}.$$

Thus $\Gamma = I_2 + \mathcal{L}(a)$, where $\mathcal{L}$ is old-linear for fixed independent parameters and is zero at $\varepsilon = 0$. If $\mathcal{R}_{r_M}(B) = (r_{M,j} B)_j$, the raw coefficients are

$$a = (T_e - \mathcal{R}_{r_M} \mathcal{L})^{-1} r_M, \quad c = T_{2e}^{-1}((r_{z,j} \det \Gamma)_j). \quad (5.31)$$

The second formula is used only in the special model. The first operator is centrally the identity; the second is the existing unipotent triangular source inverse. The factor $\det \Gamma$ stays inside T_{2e}^{-1} . A common compact-uniform parameter disk makes both G and Γ units. Neither matrix multiplication nor the determinant multiplier is commuted through an old-linear source operator.

For the scalar handle let $W_H = \zeta + \sum_{j=1}^{2e} \varepsilon^j H_1^{j-1} H_{c_j}$ and $K_H = W_H(\det G)^{-1}$. The normalized retained pair has precisely the positive lists (r_M, r_z) . In particular its actual top equations are

$$a_e = r_{M,e} \Gamma, \quad c_{2e} = r_{z,2e} \det \Gamma. \quad (5.32)$$

Equal matrix-column degrees are essential here: a full constant GL_2 gauge preserves these actual source lines.

Lemma 5.13. *In $J_2 = \mathcal{O}_U[w]/(w^2)$ write*

$$G = I_2 + \varepsilon(A_0 + wA_1), \quad C = F + \varepsilon(V_0 + wV_1), \quad K_H = \zeta + \varepsilon(k_0 + wk_1).$$

Let $D(A_0, A_1)$ be the coefficient of t in $\det(A_0 + tA_1)$. The following identities hold in the entire finite spectral algebra, including nonreduced factors:

$$\begin{aligned} \det \Gamma - \operatorname{ev}_b(\det G) &= \varepsilon^2 b^2 \det A_1, \\ \operatorname{ev}_b(C) \Gamma - \operatorname{ev}_b(CG) &= \varepsilon^2 b^2 V_1 A_1, \\ \operatorname{ev}_b(K_H) \det \Gamma - \operatorname{ev}_b(K_H \det G) &= \varepsilon^2 b^2 \mathcal{T}_z, \end{aligned}$$

where

$$\mathcal{T}_z = \zeta \det A_1 + k_1 \operatorname{Tr} A_1 + \varepsilon(k_1 D(A_0, A_1) + (k_0 + bk_1) \det A_1).$$

Proof. Expand $\det(I_2 + \varepsilon(A_0 + bA_1))$ and compare with the degree-below-two representative of $\det G$; the missing term is $\varepsilon^2 b^2 \det A_1$. Ordinary ordered matrix multiplication gives the second identity. For the third, the product of the evaluations of K_H and $\det G$ differs from the evaluation of their truncated product by

$$b^2 \varepsilon k_1 (\varepsilon \operatorname{Tr} A_1 + \varepsilon^2 D(A_0, A_1)).$$

Add the first identity multiplied by $\operatorname{ev}_b(K_H) = \zeta + \varepsilon(k_0 + bk_1)$. No attachment or smoothing factor has been cancelled. $\square$

5.4.3 The common retained root

The full-module root identity of Lemma 5.12 is $D_r(c) - h_c = b^2 \mathcal{E}(c)$, where $h_c = \operatorname{ev}_b(H_c)$ and $\mathcal{E}$ is the explicit operator in (5.27). For $j \geq 1$, the finite telescope

$$\mathfrak{R}_j(c) = \delta^{j-1} \mathcal{E}(c) + \mathcal{E}(1) h_c \sum_{t=0}^{j-2} \delta^{j-2-t} h_1^t + \mathcal{C}_b(H_1^{[j-1]}, H_c)$$

satisfies $\delta^{j-1} D_r(c) - \operatorname{ev}_b(H_1^{j-1} H_c) = b^2 \mathfrak{R}_j(c)$. Here $H_1^{[j-1]}$ means $j-1$ repeated arguments of $\mathcal{C}_b$; empty sums are zero. Set $\mathfrak{R}_i = \sum_{j=1}^{d_i} \varepsilon^{j-1} \mathfrak{R}_j(a_{ij})$. In the toric case the gauge tail is $\mathcal{T}_i = \mathcal{T}_{\chi_i, V_i}$; in the matrix case it is $V_1 A_1$ for the matrix block and $\mathcal{T}_z$ for the scalar block. In the latter notation V_1 is the coefficient of the nonconstant handle term, and $\mathfrak{R}$ is taken entrywise in that block. The normalized retained constant tuple c_{ret} satisfies

$$c_{\text{ret}} - f = \varepsilon(A_j(j) + A_n(n) + b^2 h), \quad h = R_{\Gamma^{-1}}(\mathfrak{R} - \varepsilon \mathcal{T}). \quad (5.33)$$

For matrices the displayed action means right multiplication by Γ^{-1} on the matrix block and multiplication by $(\det \Gamma)^{-1}$ on the scalar block. For a zero-degree toric coordinate $\mathfrak{R}_i = 0$, and the same identity still holds. Indeed the raw retained root differs from evaluation of its raw handle by $\varepsilon b^2 \mathfrak{R}$, while the preceding gauge tails compare the raw handle with the action on the normalized handle. Applying the inverse gauge proves the formula. The chart Hadamard formula expresses V using

$$\int_0^1 \partial_\xi \chi_s(t\varepsilon j, t\varepsilon n) dt, \quad \int_0^1 \partial_\nu \chi_s(t\varepsilon j, t\varepsilon n) dt.$$

Composition with ev_b gives the old-linear maps A_j, A_n . They are not asserted to be A -linear.

5.5 The sufficient full residual system

In the universal source algebra retain every positive coefficient of every equation:

$$E_{\text{raw}} = (\operatorname{pos} Q_\lambda(Z))_\lambda, \quad g = \mathcal{U} E_{\text{raw}}, \quad \mathcal{U} = \operatorname{diag}_\lambda(\operatorname{mult}_{\vartheta_\lambda(\Gamma)^{-1}} T_{k_\lambda}).$$

Here $g_H = \gamma$ for the torus and $g_H = G$ for the matrix gauge. The scalar roots are $Q_\lambda(f) = 0$. The displayed order means that T_{k_λ} acts first and multiplication by the scalar equation character acts second. The operator is invertible on the protected domain. The exact handle homomorphism and multihomogeneity give

$$n = R E_{\text{raw}} = R \mathcal{U}^{-1} g, \quad (R E_{\text{raw}})_\lambda = \vartheta_\lambda(g_H)^{-1} \sum_{j=1}^{k_\lambda} \varepsilon^{j-1} H_1^{j-1} H_{(E_{\text{raw}})_\lambda j}. \quad (5.34)$$

A zero-degree row contributes the zero normal jet. For fixed parameter values R is old-linear on the full residual module, even though its coefficients depend holomorphically on the gauges.

The retained multiplication defect has the form $\Phi(XY) - \Phi(X)\Phi(Y) = \varepsilon^2 b^3 \Omega(X, Y)$, where, for $X = x_0 + \sum_i e_i(a_i)$ and $Y = y_0 + \sum_j e_j(c_j)$,

$$\Omega(X, Y) = \sum_{i,j} (v + \varepsilon \delta)^{i+j-2} P'(b)^2 \rho \Theta(a_i, c_j).$$

For any fixed parenthesized coordinate monomial $W = W_1 W_2$, the recursion

$$\Omega_W = \Omega(W_1(Z), W_2(Z)) + \Omega_{W_1} \Phi(W_2(Z)) + W_1(\Phi Z) \Omega_{W_2}$$

starts with zero on single coordinates. It proves exactly $\Phi(Q_\lambda(Z)) - Q_\lambda(\Phi Z) = \varepsilon^2 b^3 \Omega_{Q_\lambda}$ for every equation. Write pos_k for the list of coefficients of $v, \dots, v^k$ in an ordinary polynomial over A , and set

$$(\Omega_+)_\lambda = \text{pos}_{k_\lambda}(\vartheta_\lambda(\Gamma)^{-1} \Omega_{Q_\lambda}).$$

All terms in each equation have source degree k_λ , so this list belongs to its full equation block; no row is omitted. The source inverses give, as identities of polynomial coordinate tuples,

$$Z_y = f + \text{pol}(a^0 + y), \quad Z_{\text{ret}} = R_{\Gamma^{-1}} \Phi(Z) = Z_y + \Delta c, \quad \Delta c = c_{\text{ret}} - f.$$

Here Δc is constant in v , but is a tuple over the full label algebra A , not just over the old scalar ring. Since $Q_\lambda(Z)$ has zero scalar root, the definition of g and the retained multiplication defect give, in each equation block,

$$g_\lambda = \text{pos}_{k_\lambda}(\vartheta_\lambda(\Gamma)^{-1} \Phi(Q_\lambda(Z))) = \text{pos}_{k_\lambda} Q_\lambda(Z_{\text{ret}}) + \varepsilon^2 b^3 (\Omega_+)_\lambda.$$

In the first equality the positive-list operator is precisely T_{k_λ} followed by the equation-character multiplier, in that order. The second equality uses multihomogeneity for the ordinary quotient action, after applying Φ ; it does not commute a gauge multiplier through an old-linear source operator. The constant-displacement Taylor operator T_c has components

$$(T_c h)_\lambda = \text{pos}_{k_\lambda} \left(\int_0^1 dQ_\lambda(Z_y + t \Delta c)[h] dt \right).$$

This is a polynomial expression in $Z_y, \Delta c$ and is A -linear in h , because it uses ordinary multiplication in $A[v]$. Thus $\text{pos}_{k_\lambda} Q_\lambda(Z_y) = F_\lambda(y)$ and the polynomial Taylor identity give

$$g - F(y) = \varepsilon T_c(A_j(j) + A_n(n)) + \varepsilon b^2(T_c h + \varepsilon b \Omega_+).$$

Only the slot T_c is used to pull b^2 outside a map. Substitution of (5.34) gives the full residual identity

$$(\text{Id} - \varepsilon P_*)g = F(y) + \varepsilon Q_* j + \varepsilon b^2 H_*, \tag{5.35}$$

with $P_* = T_c A_n R \mathcal{U}^{-1}$, $Q_* = T_c A_j$ and $H_* = T_c h + \varepsilon b \Omega_+$. The left matrix is centrally the identity and uniformly invertible after a common shrinking.

The differential $\mathbf{L}_s$ of gauge normalization is

$$h \longmapsto h - \mathbf{E} C_s h$$

for the torus, and

$$(h, k) \longmapsto (h - \mathbf{F} \Lambda h, k - \zeta \text{Tr}(\Lambda h))$$

for matrices, with the scalar block omitted when absent. The first raw coefficient is zero in every degree-zero coordinate. Write $\Pi_s = d\xi_f \circ \mathbf{L}_s$ and $v_0 = \Pi_s(a_{\text{first}}^0)$. The source inverse is holomorphic in the still independent parameter ε and has central value $a^0 + y$. Hadamard's

formula therefore gives $a = a^0 + y + \varepsilon a_*$ in every full coefficient block. Substitution into the handle, before any q -substitution, gives

$$Z_H = f + \varepsilon H_{a_{\text{first}}^0 + y_{\text{first}}} + \varepsilon^2 W_H^*,$$

with W_H^* holomorphic: the correction to the first coefficient is multiplied by ε , and every term with source index $j \geq 2$ already contains ε^j . Both gauge normalizations are holomorphic near the central unit. Their first derivatives are the displayed $\mathbf{L}_s$, so the normalized handle has expansion

$$C = f + \varepsilon \mathbf{L}_s H_{a_{\text{first}}^0 + y_{\text{first}}} + \varepsilon^2 C_*.$$

Applying the holomorphic tangent chart and dividing by the independent ε gives its central first divided jet exactly as

$$j_\ell(s, y, \delta\rho, 0) = -\text{Tr}(P'(b)^2 b^{2-\ell}(\rho_0 + \delta\rho)(v_0 + \Pi_s y_{\text{first}})), \quad \ell = 0, 1.$$

The old-scalar matrix Π_s commutes with the regular trace; expanding $\rho = \rho_0 + \delta\rho$ therefore separates the constant, linear and bilinear terms in this central expression. The constant term is zero because $\mathcal{M}\rho_0 = 0$. The old-linear map T has components

$$(Ty)_\ell = -\text{Tr}(P'(b)^2 b^{2-\ell} \rho_0 \Pi_s y_{\text{first}}), \quad \ell = 0, 1.$$

Hadamard's formula for $j(\varepsilon) - j(0)$ now gives the full identity

$$j = \mathcal{M}\delta\rho + Ty + \Psi_2(y, \delta\rho) + \varepsilon Z_*, \quad (\Psi_2(y, \delta\rho))_\ell = -\text{Tr}(P'(b)^2 b^{2-\ell} \delta\rho \Pi_s y_{\text{first}}).$$

In particular Ψ_2 is exactly bilinear, and $Z_* = \int_0^1 (\partial_\varepsilon j)(s, y, \delta\rho, t\varepsilon) dt$ is holomorphic on a common smaller product neighborhood. All these expansions use the fixed monomial frame of the finite free A -module and the two scalar coordinates of J_2 . They neither divide by a label element nor discard nilpotent coordinates.

Use the two adjugate identities, with $\mathbf{v}$ a 6-component auxiliary vector, and substitute

$$y = q^2 b C_0 u, \quad \delta\rho = q D_0 \mathbf{v}, \quad \varepsilon = q^2 \tau.$$

Denote this substitution by

$$\Sigma(s, u, \mathbf{v}, \tau) = (s, q(s)^2 b(s) C_0(s) u, q(s) D_0(s) \mathbf{v}, q(s)^2 \tau).$$

It is a holomorphic map between the fixed finite-dimensional module coordinates. After shrinking the product domain once, its image lies in the common domain on which H_* and Z_* were constructed. Set $H_*^\Sigma = H_* \circ \Sigma$ and $Z_*^\Sigma = Z_* \circ \Sigma$. These maps remain holomorphic where $q = 0$, where $b = 0$ or is a zero divisor, and on every nonreduced label factor, since they arise by composition rather than division.

The A -polynomial Taylor formula is $F(y) = Jy + B(y)(y, y)$, with A -bilinear displayed slots. Using $JC_0 = b \text{Id}$ and $\mathcal{M}D_0 = q \text{Id}$ gives the identities

$$Jy = q^2 b^2 u, \quad B(y)(y, y) = q^4 b^2 B(q^2 b C_0 u)(C_0 u, C_0 u), \quad \mathcal{M}\delta\rho = q^2 \mathbf{v}.$$

The old-linearity of T pulls out the old scalar q^2 , not the label element b . Bilinearity of Ψ_2 similarly gives $\Psi_2(y, \delta\rho) = q^3 \Psi_2(b C_0 u, D_0 \mathbf{v})$. Consequently, as identities before any cancellation,

$$F + \varepsilon b^2 H_* = q^2 b^2 G_1, \quad j = q^2 G_2,$$

$$G_1 = u + q^2 B(q^2 b C_0 u)(C_0 u, C_0 u) + \tau H_*^\Sigma,$$

$$G_2 = \mathbf{v} + T(b C_0 u) + q \Psi_2(b C_0 u, D_0 \mathbf{v}) + \tau Z_*^\Sigma.$$

At an old point s_0 , these equalities hold in the local analytic ring of the joint parameter product at $(s_0, 0, 0, 0)$, with the finite free spectral algebra pulled back from U . They follow term by term from $JC_0 = b \text{Id}$, $\mathcal{M}D_0 = q \text{Id}$, old-linearity of T , and bilinearity of the displayed Taylor terms. In particular the factors $q^2 b^2$ and q^2 occur as identities in this joint parameter ring. The central Jacobian is

$$\begin{pmatrix} \text{Id} & 0 \\ T \text{mult}_b C_0 & \text{Id} \end{pmatrix},$$

at $(u, \mathbf{v}, \tau) = (0, 0, 0)$ over every old point. Since the displayed formulas give G_1, G_2 directly as holomorphic functions, this is their ordinary derivative also when $q = 0$, at controller zero, and on nonreduced label fibres. The finite parameter IFT gives one branch of this fixed system near the whole old compact. On an overlap, both local constructions use the same fixed frames and solve the same equations $G_1 = G_2 = 0$; the inverse charts used upstream agree as explained above. After restricting to a common uniqueness ball, central uniqueness therefore identifies the branches. The minimum of finitely many parameter radii is positive. Choose the $u, \mathbf{v}$ balls inside the previous coefficient balls before shrinking the τ disk. The branch satisfies $u, \mathbf{v} = O(\tau)$ uniformly, since it is centrally zero. Multiplying the normalized equations back into (5.35), and then using the invertible $\mathcal{U}$, gives

$$g = j = E_{\text{raw}} = n = 0. \quad (5.36)$$

Thus rational evaluation of the source algebra satisfies every actual defining equation in the full finite spectral algebra before passage to a projective quotient.

5.6 The projective value ideal and actual source infinity

With $r = b^3 \rho$, let

$$M_0 = r_1 K(b_2, b_1), \quad \mathcal{N} = K(b_1, b_2) r_2 \quad \text{in } A \otimes A.$$

For $|v| \geq \eta > 0$ the finite-matrix exterior expressions are

$$Y_c^{\text{ext}} = vc + \varepsilon \text{Tr}_2((1 - \varepsilon M_0/v)^{-1} \mathcal{N} c_2), \quad \zeta^{\text{ext}} = Y_1^{\text{ext}}.$$

Their retained versions are $Y_c^{\text{ret}} = vc + \varepsilon D_r(c)$ and $\zeta^{\text{ret}} = v + \varepsilon \delta$. The inverse identity gives

$$Y_c^{\text{ext}} - Y_c^{\text{ret}} = \varepsilon^2 b^3 C_c, \quad C_c = \frac{1}{v} \text{Tr}_2((1 - \varepsilon M_0/v)^{-1} \rho_1 K(b_2, b_1) \mathcal{N} c_2).$$

The attachment is pulled through the first, A -linear tensor factor. The reversed order of the arguments of the first kernel is essential. One sufficiently small ε disk makes these inverses uniform on a protected compact old parameter product and all $|v| \geq \eta$.

For any coordinate of actual source degree d_i , let

$$Z_i^{\text{ext}} = f_i + \sum_{j=1}^{d_i} (\zeta^{\text{ext}})^{j-1} Y_{a_{ij}}^{\text{ext}}.$$

A finite telescope gives $Z_i^{\text{ext}} - \Phi(Z_i) = \varepsilon^2 b^3 \Omega_i$, with

$$\Omega_i = \sum_{j=1}^{d_i} \left((\zeta^{\text{ext}})^{j-1} C_{a_{ij}} + C_1 Y_{a_{ij}}^{\text{ret}} \sum_{t=0}^{j-2} (\zeta^{\text{ext}})^{j-2-t} (\zeta^{\text{ret}})^t \right).$$

At infinity $h_\infty = 1/v$, the expressions $h_\infty Y_c^{\text{ext}}$ and $h_\infty \zeta^{\text{ext}}$ are holomorphic, and $C_c = h_\infty \hat{C}_c$. Every term of $h_\infty^{d_i} \Omega_i$ therefore has a factor $h_\infty^{d_i-j+2}$, of exponent at least two. Thus this is a

literal identity of sections in the actual degree- d_i frame, including infinity. For $d_i = 0$ the sums are empty.

On the branch (5.36), write $u = \tau\hat{u}$. The positive coefficient equations and (5.33) give the identity of actual coordinate sections

$$Z^{\text{ext}} = R_\Gamma Z^0 + \tau b \mathcal{D},$$

$$\mathcal{D} = R_\Gamma(q^2 \text{pol}(C_0 \hat{u}) + q^2 b h) + q^4 \tau b^2 \Omega.$$

For coordinate i , the homogeneous form of $\text{pol}_i(\mathbf{c})$ is $\sum_j c_j T_0^j S^{d_i-j}$, and the constant displacement is multiplied by S^{d_i} . The quotient action preserves these actual degree blocks. This is diagonal character multiplication for toric coordinates and ordered right multiplication for matrices. The scalar block in the special case has degree $2e$ throughout. All attachment exponents in the displayed formula are nonnegative.

A degree-zero coordinate occurs only in the toric case. Its raw and exterior values are exactly f_i , while the root identity gives

$$f_i = \Gamma^{x_i} f_i + \tau b q^2 b \Gamma^{x_i} h_i. \quad (5.37)$$

Thus the same coordinate comparison holds without imposing a unit gauge on that coordinate.

The projective value tuple Ξ is polynomial in the coordinates, homogeneous for both the source degree d_h and its scalar quotient character ϑ_h . Finite product telescoping gives

$$\Xi(Z^{\text{ext}}) = \vartheta_h(\Gamma) \Xi(Z^0) + \tau b \mathcal{D}_h.$$

The actual certificate (5.1) then gives

$$\Xi(Z^{\text{ext}}) = (\vartheta_h(\Gamma) \text{Id} + \tau \mathcal{D}_h B^T) \Xi(Z^0). \quad (5.38)$$

This equality is used in each of the local frames of (5.1), with its own local column B . The comparison matrices need not glue: invertibility on each chart gives equality of the two intrinsic value ideals on that chart. The comparison matrices are centrally the identity. A finite cover of the protected compact exterior in the projective tooth source gives one common disk on which they are invertible. Consequently the entire value ideals agree in full local rings, including nilpotents and source infinity. Indeed an invertible matrix over the local ring has an inverse over that ring and gives both ideal inclusions, not just equality of their geometric zero sets. The disk is chosen for this fixed finite cover; the conclusion is used only on its protected exterior neighborhood. Every entry of the comparison matrix is regular in the same local ring. At every controller-zero point, its value at the central branch parameter is the identity; throughout the common disk it remains invertible. Substitution of the implicit branch preserves regularity. The ideal equality therefore persists on the controller-zero locus, after nonreduced specialization, and in the source-infinity chart. At every geometric factor with $b_i \neq 0$, the baseline tuple is nonzero.

For toric coordinates the nonzero value tuple is equivalent to Cox admissibility. For the special matrix model, retain the complete ideal $(p(M), z)$. If the branch equation holds and $(p(M), z)$ is the unit ideal in a local ring, its quotient by $(p(M))$ has $z^2 = 0$. A nonzero local ring cannot have its unit ideal generated by a nilpotent element, so $(p(M))$ is also the unit ideal. This proves rank-two admissibility at every later use of an actual solution.

The rational source homomorphism gives

$$R_c(z) = \text{Tr}(\sigma c(z - b)^{-1}) = U_c(z)/P(z), \quad Z_i^{\text{res}} = f_i + \sum_{j=1}^{d_i} (\varepsilon R_1)^{j-1} \varepsilon R_{a_{ij}}.$$

All $Q_\lambda(Z^{\text{res}})$ vanish by (5.36). On a characteristic cluster of length $k \geq 2$, write $b = \lambda + n$, $n^k = 0$, and $P(z) = (z - \lambda)^k S(z)$. Then $P'(b) = kn^{k-1}S(b)$, so $P'(b)^2 = 0$. Only simple factors contribute to these traces. In their geometric coordinates,

$$v_i = \varepsilon \sigma_i / (z - b_i), \quad \varepsilon R_c = \sum_i v_i c_i, \quad \varepsilon R_1 = \sum_i v_i.$$

Bounds on multiplication matrices bound all character values, without choosing uniformly conditioned eigenbases. The lifted core f on a protected compact anchor neighborhood lies in the admissible coordinate open set: the Cox open set or the rank-two matrix open set. It therefore has a positive Euclidean distance from its closed complement. Choose $\eta > 0$ with $m\eta < 1/2$ so that, for all coordinates, the estimates $\sum_{j=1}^{d_i} B(m\eta)^j$ place the full perturbed tuple within half that distance. If every $|v_i| < \eta$, the rational tuple is then admissible. Zero-degree coordinates contribute no perturbation.

If some $|v_i| \geq \eta$, its factor is simple, $\sigma_i \neq 0$ and $b_i \neq 0$. For a distinct simple second factor,

$$(M_0)_{ij} = -\frac{\sigma_i}{b_i - b_j}, \quad (\mathcal{N})_{ij} = \frac{\sigma_j}{b_i - b_j}.$$

For a nonsimple second cluster $b_i - b_2$ is a unit, $\Delta(b_i, b_2) = 0$, and $\mathcal{N} = \sigma_2 / (b_i - b_2) = 0$ on that whole cluster. The diagonal kernel is zero. Substituting $z = b_i + \varepsilon \sigma_i / v_i$ into the convergent exterior expression therefore gives exactly $Y_c^{\text{ext}}(i, v_i) = \varepsilon R_c(z)$. Equation (5.38) makes its projective value tuple nonzero. In the toric case this gives admissibility directly; in the special matrix case the already solved branch equation and the preceding local-ring argument give rank two. First choose the coefficient bounds, then η , and only then shrink the smoothing disk for the exterior inverses and all ideal matrices. This proves nonvanishing whenever $P(z) \neq 0$ near the anchor.

It remains to check the actual top coefficients at active roots. For $d_i > 0$ take $\widehat{Z}_i = a_{i,d_i}$, while for $d_i = 0$ take $\widehat{Z}_i = f_i$; use the analogous baseline tuple $\widehat{Z}^0$. The top row of T_d is the identity. The source inverses therefore give

$$\widehat{Z} = R_\Gamma \widehat{Z}^0 + \tau b \mathcal{D}^\infty.$$

On every positive-degree block the correction is the corresponding top row of $R_\Gamma q^2 C_0 \widehat{u}$. For a degree-zero toric coordinate it is $\Gamma^{x_i} q^2 b h_i$, by (5.37). In particular the actual matrix and scalar equations are $a_e = (a_e^0 + y_{M,e})\Gamma$ and $c_{2e} = (c_{2e}^0 + y_{z,2e}) \det \Gamma$. In the source-infinity chart, trivialize the actual source line of degree d_h . Restriction of the local certificate (5.1) to its section at infinity gives $b = (B^\infty)^T \Xi(\widehat{Z}^0)$: its restriction sends each homogeneous baseline value to its actual top coefficient. Thus B^∞ is a restricted local column, not a separately postulated global object. It gives

$$\Xi(\widehat{Z}) = (\vartheta_h(\Gamma) \text{Id} + \tau \mathcal{D}_h^\infty (B^\infty)^T) \Xi(\widehat{Z}^0).$$

One common shrinking makes this matrix invertible. Thus the actual top tuple is nonzero at every simple factor with $b_i \neq 0$. In the special case its branch equation is the highest coefficient of the already solved degree- $4e$ equation; the top matrix is therefore rank two by the same local-ring argument. This uses actual top coefficients, not congruences modulo powers of b .

The exact monic clearing is

$$\widetilde{Z}_i = f_i P^{d_i} + \sum_{j=1}^{d_i} \varepsilon^j U_1^{j-1} U_{a_{ij}} P^{d_i-j}.$$

Its formal degree is md_i and its formal top coefficient is f_i , since each positive term has degree at most $md_i - j$. No common factor is removed. At a simple root $\lambda = b_\ell$, with $P_\lambda = P/(z - \lambda)$, its value is

$$\tilde{Z}_i(\lambda) = (\varepsilon P_\lambda(\lambda) \sigma_\ell)^{d_i} \hat{Z}_i(\ell).$$

For $d_i = 0$ this means the actual constant f_i . Near the anchor, q and the full weight ρ are units. For small fixed $\tau \neq 0$, the scalar in parentheses is nonzero at every active value. The coordinate multipliers come from a central source cocharacter:

$$c \mapsto (c^{D_1}, \dots, c^{D_{r_0}}) \quad \text{or} \quad c \mapsto c^e I_2.$$

Indeed $d_i = \chi_i \cdot D$ in the toric case; the matrix and scalar weights are e and $2e$. Negative D_α are harmless since $c = \varepsilon P_\lambda(\lambda) \sigma_\ell \neq 0$. Consequently $\Xi(\tilde{Z}(\lambda)) = c^{d_h} \Xi(\tilde{Z}(\ell)) \neq 0$. No active value is a raw common-zero site.

The ordinary projective vector $\Xi(\tilde{Z})$ has one common positive formal degree md_h . Its formal top coefficient is $\Xi(f)$, so it represents the embedded old core at infinity. It satisfies the equations of the closed embedded X . For completeness this last assertion holds over the entire old domain, not only on the protected anchor neighborhood. At $z = \infty$ the rational coordinate tuple is $f + O(1/z)$; on an old neighborhood of any given old point it is therefore admissible for sufficiently large $|z|$. There every Q_λ vanishes, so its quotient lies in X . Every homogeneous equation of the embedded X , evaluated on $\Xi(\tilde{Z})$, is a polynomial in z with holomorphic old coefficients and vanishes on this open set. Its coefficients therefore vanish identically. The claimed equation identities follow throughout $U \times \mathbb{P}^1$.

The preceding alternatives exhaust the compactified fresh source. Where $P(z) \neq 0$, either all $|v_i| < \eta$ and the distance estimate gives admissibility, or some $|v_i| \geq \eta$ and the exterior ideal identity gives it. At a zero of P , an active simple root is admissible by the top-coefficient calculation; the remaining inactive roots are precisely the possible raw sites. At $z = \infty$ the top value is $\Xi(f) \neq 0$. The rational notation U_c/P is used only on the first region; the displayed monic clearing extends the homogeneous tuple to every root and to infinity. All gauge inverses and multiplication determinants depend only on the bounded old parameter product, not on z , and were made units before this evaluation. Consequently the cleared tuple is globally defined on $U \times \mathbb{P}^1$, with its possible failures to represent a projective point confined to the raw zero set. The raw zero set is closed in $U \times \mathbb{P}^1$, misses the whole core, and has no fibre equal to a line. It is consequently finite and proper over U . Near the anchor only inactive characteristic values can be raw sites, and there are at most two by the threefold incidence argument.

In the marked source coordinate $t = 1/z$, the derivative of the rational profile is

$$\partial_t \varphi_\tau|_0 = q^2 \tau \operatorname{Tr}(P'(b)^2 b^3 \rho_0 v_0) + O(\tau^2).$$

The corrections of the raw lists and weights are $O(\tau)$. The three chosen kernel weights have independent anchor residues and $q(a) = 1$, so their derivative determinant has a nonzero coefficient of τ^3 . One sufficiently small fixed nonzero τ gives three actual independent derivatives at the anchor.

5.6.1 Removing the finite raw sites

It remains to turn the rational source profiles into entire scalar families without shrinking the old domain. The following argument applies to any closed complex submanifold $Y \subset \mathbb{P}^r$; in the toric case the chosen ample Cox-monomial vector provides this projective realization.

Theorem 5.14. *Suppose that U is a finite-dimensional Stein manifold and that $Z_0, \dots, Z_r$ are homogeneous polynomials in $[S : T]$ of one common degree $D \geq 1$, with holomorphic old*

coefficients. Suppose they satisfy the equations of Y , their value at $[1 : 0]$ is a nowhere zero tuple representing $f : U \rightarrow Y$, and near some $a \in U$ their raw common-zero set has at most two distinct points in each source fibre. Then there is a holomorphic $H : U \times \mathbb{C} \rightarrow Y$ with $H(s, 0) = f(s)$ whose fresh derivative at $(a, 0)$ is the derivative of the original source profile in $t = T/S$. The whole domain U remains unchanged.

Proof. Let $B \subset U \times \mathbb{P}^1$ be the reduced raw common-zero set. It is closed analytic and misses $U \times \{0\}$. No source fibre can be the whole line, so its fibres are finite. Properness of the projection makes $B \rightarrow U$ a finite analytic map.

The distinct-triple locus in $B \times_U B \times_U B$ is denoted by C° , and its closure in that finite fibre product by C . Write $\pi : C \rightarrow U$ for the projection. The closure is the locally finite union of irreducible components not contained in any pairwise diagonal, so it is analytic and finite over U . Continuity gives $\pi(C) \subset \overline{\pi(C^\circ)}$, while closedness of the finite projection gives the reverse inclusion. Hence $B_3 := \pi(C) = \overline{\{s : \#B_s \geq 3\}}$. The two-site bound on a neighborhood of the anchor excludes it from this closure; a two-site count only on the anchor fibre would not do so. Cartan's theorem for the ideal of B_3 gives $d \in \mathcal{O}(U)$ vanishing on B_3 and satisfying $d(a) = 1$. Replace the tuple homogeneously by $Z_i(s, S, d(s)T)$. If $d \neq 0$, the substitution is an automorphism of the source line and the new raw fibre has at most two sites. If $d = 0$, the tuple is $S^D(Z_i(s, 1, 0))_{i=0}^r$; its only possible site is $[0 : 1]$. The whole core remains regular. Denote its new finite raw set by $\tilde{B}$. It has at most two points in every fibre.

Apply Lemma 5.15 below to obtain an avoiding section $R(s, t)$ with $R(s, t) = t + O(t^2)$ along the whole zero section. Composition with the new projective tuple is holomorphic on $U \times \mathbb{C}$ and lies in Y . At a , $d(a) = 1$ and the section has derivative one, so the specified fresh derivative is unchanged. Nothing has been deleted from U . $\square$

Lemma 5.15. *Let $B \subset U \times \mathbb{P}^1$ be closed analytic, finite over a finite-dimensional Stein manifold U , with at most two points per fibre and disjoint from the zero section. There is a holomorphic $R : U \times \mathbb{C} \rightarrow \mathbb{P}^1$ satisfying $R(s, t) \notin B_s$ and $R(s, t) = t + O(t^2)$ along $U \times \{0\}$.*

Proof. The open complement $E = (U \times \mathbb{P}^1) \setminus B \rightarrow U$ is a holomorphic submersion. We explain the finite-map stratification used in the section theorem. On every closed reduced irreducible base piece W , first reduce the restricted finite space and take its finite direct image algebra A_W . Let r be its generic rank. Remove the singular locus of W , the coherent nonfree locus detected by the rank Fitting ideal, and the support of the finite direct image of the relative Kähler differentials. The latter is proper because the generic reduced finite algebra is separable in characteristic zero. Add the intersections of distinct base components. On what remains the finite map is étale, hence has a constant number of distinct geometric points. Repeat on the reduced closed exceptional set, treating its components of largest dimension first and including all smaller-dimensional components in the next exceptional set. Each removed exceptional locus is proper on the component under consideration, so dimension strictly drops. More explicitly, choose relatively compact open sets $U_j \Subset V_j \Subset U$ with U_j increasing to U . Local finiteness of analytic components and compactness of $\overline{U_j}$ leave only finitely many components relevant on a neighborhood of that compact set. The preceding construction, followed by restriction to U_j , gives a finite descending chain of closed reduced analytic subsets

$$U_j = X_{j,0} \supset X_{j,1} \supset \cdots \supset X_{j,r_j} = \emptyset.$$

The differences have pure dimension, are smooth, and have their closure boundaries in the next member of the chain: singular points, component intersections, rank changes and branch points were all placed there. There are at most $\dim U + 1$ dimension levels; the strata need not be connected. On each connected component of a difference the restricted finite reduced

map is étale with constant rank 0, 1 or 2. Reduction is repeated at each stage, so this counts actual sites, not fibre nilpotents.

Locally on a stratum, its zero, one or two sites are holomorphic labelled sections. A holomorphic Möbius transformation places them at fixed standard points, so E restricts to a holomorphic fibre bundle with fibre $\mathbb{P}^1$, $\mathbb{C}$ or $\mathbb{C}^*$. All three are elliptic. For the pullback to $U \times \mathbb{C}$, use the relatively compact exhaustion $U_j \times \{|t| < j\}$ and its finite closed chain $X_{j,k} \times \{|t| < j\}$. Its strata are the preceding smooth strata times the disk, with the same local trivializations. This is the exhaustion by finite stratified holomorphic fibre bundles required by the section theorem, not merely a locally finite partition.

There is a positive continuous function $\delta(s)$ such that the closed affine disk of radius $\delta(s)$ misses B_s . To see this, use the closed finite set and the regular zero section to find safe disks on old neighborhoods, then take a continuous positive minorant of their radii. Let $\chi : [0, \infty) \rightarrow [0, 1]$ equal one near zero and vanish on $[1, \infty)$. The section

$$R_0(s, t) = \chi(|t|/\delta(s))t$$

is continuous and avoids B_s : its nonzero value lies in the safe disk, and its zero value is the safe core. It equals t on an open tube around the entire closed complex submanifold $U \times \{0\}$ and is holomorphic there. Apply Forstnerič's stratified section theorem with finite-jet interpolation, Theorem 1.4(C) and Corollary 1.6 of [30], to the pullback submersion over the reduced Stein manifold $U \times \mathbb{C}$. The compact approximation set is empty, the closed interpolation submanifold is $U \times \{0\}$, and the jet order is one. The cutoff section is holomorphic on a neighborhood of that entire submanifold, as required. The resulting holomorphic section R therefore satisfies $R(s, 0) = 0$ and $\partial_t R(s, 0) = 1$ for every $s \in U$. $\square$

5.7 Completion

Proof of Theorem 5.1. At each anchor, Lemma 5.2 supplies the finite selector, the surjective coefficient differential, the controller identities, and the unit value ideal off the fixed-base bad locus. Pullback to the old parameter space preserves the finite locally free spectral algebra and the controller identities. The construction then gives three rational profiles on one neighborhood of the entire old compact, with regular exact core and independent anchor derivatives. Their globally written homogeneous tuples have at most two raw sites on an anchor neighborhood. Theorem 5.14 therefore supplies three entire fresh-scalar families on the same old domain, preserving the whole core and the three anchor derivatives. Independence persists near that anchor. A finite anchor cover of the compact, followed by intersection of the finitely many whole-old domains, gives $\mathcal{P}_N$. This holds in every source dimension. Theorem 4.1 then gives CAP in all dimensions, and hence the Oka property. $\square$

6 Fano models in smooth toric ambient varieties

Theorem 5.1 applies to the actual smooth threefold in each presentation below. The equations are global and have the expected codimension, and the threefold meets the ambient dense torus. Rational connectedness follows in each case from the Fano theorem [9, 48] after the anticanonical class has been checked. Lemma 5.2 then constructs at every anchor the coefficient-rank, controller, finite spectral-algebra and value-ideal data used by the common theorem.

6.1 Ordinary complete intersections and Verra threefolds

Theorem 6.1. *Every smooth quartic threefold in $\mathbb{P}^4$ satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka.*

Proof. The smooth hypersurface is irreducible and is contained in no coordinate hyperplane, so it meets the dense torus of $\mathbb{P}^4$. Its equation is a regular hypersurface equation, and adjunction gives $-K_X = \mathcal{O}_X(1)$. Apply Theorem 5.1. $\square$

The quartic result of Guo and Xie [41, Theorem 1.1] also applies, in its broader range of intrinsic dimensions at least three. The proof here supplies the three-dimensional case within the common complete-intersection construction.

Theorem 6.2. *Every smooth complete intersection $X_{2,3} \subset \mathbb{P}^5$ satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka.*

Proof. The two equations have independent differentials on X . Their homogeneous ideal contains no linear form, so X is contained in no coordinate hyperplane. Irreducibility then gives a dense-torus point in $\mathbb{P}^5$. Adjunction gives $-K_X = \mathcal{O}_X(1)$. Theorem 5.1 applies regardless of the rank of the containing quadric. $\square$

Every smooth prime Fano threefold of genus four has this complete-intersection model, including the rank-five containing quadric; see [53, Section 4, immediately before Proposition 4.2]. Thus this theorem treats every member of family 1.3.

Corollary 6.3. *Every smooth complete intersection of three quadrics in $\mathbb{P}^6$ satisfies $\mathcal{P}_N$ for all $N \geq 1$ and is Oka.*

Proof. Its regular homogeneous ideal contains no linear form, so irreducibility gives a dense-torus point. Adjunction gives $-K_X = \mathcal{O}_X(1)$. Theorem 5.1 therefore applies. $\square$

Theorem 6.4. *Every smooth hypersurface of bidegree $(2, 2)$ in $\mathbb{P}^2 \times \mathbb{P}^2$ satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka.*

Proof. The product is a smooth projective toric fourfold, and the given hypersurface is a regular divisor. Adjunction gives $-K_X = \mathcal{O}_X(1, 1)$, which is ample. In particular X is irreducible. It cannot be contained in a coordinate boundary divisor: an irreducible divisor so contained would equal that divisor, whereas its bidegree is $(2, 2)$ rather than $(1, 0)$ or $(0, 1)$. Thus X meets the product torus and Theorem 5.1 applies. $\square$

These are the Verra presentations of family 2.6; see [13, Lemma 12.3].

6.2 Double covers in smooth toric projective bundles

We use the convention

$$\mathbb{P}_B(E) = \text{Proj}_B \text{Sym}(E^\vee), \quad \mathcal{O}_{\mathbb{P}_B(E)}(-1) \hookrightarrow p^*E;$$

thus points of a fibre are one-dimensional subspaces of the fibre of E . In the quotient convention the same space is $\mathbb{P}_B(E^\vee)$.

Proposition 6.5. *Let B be a smooth projective toric variety, let L be an equivariant line bundle on B , and let $s \in H^0(B, L^{\otimes 2})$ have a nonempty smooth zero divisor. Let*

$$A = \mathbb{P}_B(\mathcal{O}_B \oplus L), \quad \xi = c_1(\mathcal{O}_A(1)).$$

Here $p : A \rightarrow B$ is the bundle projection. There are fibre coordinates

$$u \in H^0(A, \mathcal{O}_A(1)), \quad v \in H^0(A, \mathcal{O}_A(1) \otimes p^*L).$$

The double cover associated to (L, s) is the scheme-theoretic hypersurface

$$X = V(v^2 - p^* s u^2) \subset A, \quad [X] = 2\xi + 2p^* c_1(L). \quad (6.1)$$

The ambient variety is smooth, projective and toric. The hypersurface is smooth and meets the ambient dense torus. Moreover

$$K_X = \pi^*(K_B + L), \quad \pi = p|_X.$$

Proof. The two components of the tautological inclusion give u and v . On $u = 0$, the coordinate v is invertible in a local fibre chart, where the equation in (6.1) is a unit. Consequently its intersection with $u = 0$ is empty as a scheme; there is no component or infinitesimal structure at infinity. The chart $u \neq 0$ is the total space of L . In its fibre coordinate $w = v/u$, the equation is $w^2 = s$. This is exactly the finite flat cover with algebra $\mathcal{O}_B \oplus L^{-1}$, whose multiplication $L^{-2} \rightarrow \mathcal{O}_B$ is given by s .

The equation is a nonzero section of a line bundle on the integral smooth ambient variety, hence is an effective Cartier divisor. In a local frame of L its differential is $2w dw - ds$, which is nonzero at every point of X : when $w = 0$, this follows from the smoothness of the branch divisor. Thus the equation is regular and defines the asserted smooth hypersurface. The nonempty reduced branch also makes this cover integral: in a rational trivialization of L , the rational function representing s has odd valuation along a branch component, so it is not a square in $\mathbb{C}(B)$.

The equivariant splitting makes A a toric projective bundle by [18, Proposition 7.3.3]. Over the dense torus T_B , the line bundle is equivariantly trivial, and the locus $uv \neq 0$ is a torus of dimension $\dim B + 1$. The nonzero section s cannot vanish identically on T_B . Choose $b \in T_B$ with $s(b) \neq 0$; its two inverse images have $uv \neq 0$. This proves the dense-torus condition without a generality assumption on s . Finally, $K_A = -2\xi + p^*(K_B - L)$, and adjunction proves the formula for K_X . $\square$

The section u is nowhere zero on X , so $\mathcal{O}_A(1)|_X \simeq \mathcal{O}_X$. Its degree on every rational curve in X is zero. Thus the degree-zero coordinate case of the toric spectral construction is used in these cover models.

For $B = \prod_{i=1}^r \mathbb{P}^{n_i}$ and $L = \mathcal{O}_B(d_1, \dots, d_r)$, the Cox degrees in $\text{Pic}(A) = \bigoplus_i \mathbb{Z}H_i \oplus \mathbb{Z}\xi$ are

$$\deg x_{ij} = H_i, \quad \deg u = \xi, \quad \deg v = \xi + \sum_i d_i H_i, \quad \deg(v^2 - su^2) = 2\xi + 2 \sum_i d_i H_i. \quad (6.2)$$

The characteristic open set requires each block $(x_{i0}, \dots, x_{in_i})$ and the pair (u, v) to be nonzero. The resulting $(\mathbb{C}^*)^{r+1}$ action is free there. In particular:

- Family 2.2 has $B = \mathbb{P}^1 \times \mathbb{P}^2$ and $L = \mathcal{O}_B(1, 2)$. Its ambient has dimension four, its equation has degree $(2, 4, 2)$, and $-K_X = \pi^* \mathcal{O}_B(1, 1)$.
- Family 3.1 has $B = (\mathbb{P}^1)^3$ and $L = \mathcal{O}_B(1, 1, 1)$. Its ambient has dimension four, its equation has degree $(2, 2, 2, 2)$, and $-K_X = \pi^* \mathcal{O}_B(1, 1, 1)$.
- The quartic double solid has $B = \mathbb{P}^3$ and $L = \mathcal{O}_B(2)$; the sextic double solid has the same base and $L = \mathcal{O}_B(3)$. Their equations have degrees $(4, 2)$ and $(6, 2)$, respectively, and their anticanonical lines are $\pi^* \mathcal{O}_B(2)$ and $\pi^* \mathcal{O}_B(1)$.

The all-member descriptions of 2.2 and 3.1 are given by [12, Section 3, opening paragraph] and [13, Corollary 8.2, proof], respectively. All these displayed threefolds are Fano, so the usual Fano rational-connectedness theorem supplies rational connectedness.

For a sextic double solid, the geometric motivation can also be seen from the following construction. Blowing up the genus-two curve lying over a general line in $\mathbb{P}^3$ makes the pencil of planes through that line into a K3 fibration. Recent results on Oka K3 surfaces [66] suggest holomorphic flexibility in this picture and motivate the search for Oka geometry on the threefold itself. This observation supplies intuition, not a proof; no passage of the Oka property through the fibration or the blowdown is used below.

Corollary 6.6. *Every smooth sextic double solid satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka.*

Proof. Use $B = \mathbb{P}^3$ and $L = \mathcal{O}(3)$ in Proposition 6.5. The displayed anticanonical line is ample, so the Fano rational-connectedness theorem and Theorem 5.1 apply. $\square$

Corollary 6.7. *Every smooth quartic double solid satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka.*

Proof. Use $B = \mathbb{P}^3$ and $L = \mathcal{O}(2)$ in Proposition 6.5 and apply Theorem 5.1. $\square$

Corollary 6.8. *Every smooth member of Mori–Mukai family 2.2 satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka.*

Proof. Its all-member presentation above is a double cover of $\mathbb{P}^1 \times \mathbb{P}^2$ with $L = \mathcal{O}(1, 2)$ and ample $-K_X = \pi^*\mathcal{O}(1, 1)$. Proposition 6.5 and Theorem 5.1 apply. $\square$

Corollary 6.9. *Every smooth member of Mori–Mukai family 3.1 satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka.*

Proof. Its all-member presentation above is a double cover of $(\mathbb{P}^1)^3$ with $L = \mathcal{O}(1, 1, 1)$ and ample $-K_X = \pi^*\mathcal{O}(1, 1, 1)$. Proposition 6.5 and Theorem 5.1 apply. $\square$

6.3 The double cover of the point blowup

Let $B = \text{Bl}_p \mathbb{P}^3$, write H for the hyperplane pullback and E for the exceptional plane, and set $A_0 = H - E$. Projection from p identifies

$$B = \mathbb{P}_{\mathbb{P}^2}(\mathcal{O} \oplus \mathcal{O}(-1))$$

in the lines convention. Its fibre hyperplane class is H , and A_0 is pulled back from $\mathbb{P}^2$. Use Cox coordinates z_1, z_2, z_3, t, e , with

$$\deg z_i = A_0, \quad \deg t = H, \quad \deg e = H - A_0 = E, \quad [t : ez_1 : ez_2 : ez_3] \in \mathbb{P}^3.$$

The characteristic open set is $z \neq 0, (t, e) \neq (0, 0)$. Since $-K_B = 2H + 2A_0$, the cover line is $L = H + A_0 = 2H - E$. In the fourfold $\mathbb{P}_B(\mathcal{O} \oplus L)$ the additional coordinates have degrees ξ and $\xi + H + A_0$. Thus in the ordered basis (A_0, H, ξ) the complete degree list is

$$\deg z_i = (1, 0, 0), \quad \deg t = (0, 1, 0), \quad \deg e = (-1, 1, 0), \quad \deg u = (0, 0, 1), \quad \deg v = (1, 1, 1).$$

Every section of $2L$ is of the form

$$s = t^2 q_2(z) + te q_3(z) + e^2 q_4(z),$$

where q_i has ordinary degree i . The cover equation is

$$v^2 - (t^2 q_2(z) + te q_3(z) + e^2 q_4(z))u^2 = 0, \quad \deg = (2, 2, 2). \quad (6.3)$$

This also explains the negative entry in the degree of e ; it must not be replaced by a nonnegative product grading. For the downstairs quartic $F(t, Z) = t^2 q_2(Z) + t q_3(Z) + q_4(Z)$, substitution $Z = ez$ gives $F(t, ez) = e^2 s$. Thus s is its strict-transform equation of class $4H - 2E$, rather than the divisor pullback of class $4H$.

Every smooth member of family 2.8 is the cover of B branched in a smooth member of $|-K_B|$; see [55, Introduction and Sections 2–3]. Both possibilities for its restriction to E , a smooth conic and a union of two distinct lines, occur in that description. Equation (6.3) and Proposition 6.5 treat both. In particular, no smoothness of the exceptional quadric or transversality of the branch with the toric boundary is required. The line $H + A_0$ is ample: B is the graph of projection in $\mathbb{P}^3 \times \mathbb{P}^2$, and this line is the restricted Segre line. Hence $-K_X = \pi^*(H + A_0)$ is ample.

Theorem 6.10. *Every smooth double cover of $\mathrm{Bl}_p \mathbb{P}^3$ with smooth anticanonical branch divisor satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka.*

Proof. The presentation (6.3) has exactly one regular equation in a smooth projective toric fourfold. The threefold meets its dense torus, and its anticanonical line is ample, as checked above. Theorem 5.1 therefore applies, including when the branch conic on E is reducible and the exceptional double-cover quadric is singular. $\square$

6.4 The double flag as a regular codimension-two intersection

Set $B = \mathbb{P}^2 \times (\mathbb{P}^2)^*$ and $Y = V(g) \subset B$, where $g = \sum_{i=0}^2 x_i y_i$. Then Y is the smooth flag threefold and $-K_Y = \mathcal{O}_Y(2, 2)$. Given any smooth branch divisor $S \in |\mathcal{O}_Y(2, 2)|$, its section lifts to $F \in H^0(B, \mathcal{O}_B(2, 2))$, because

$$0 \longrightarrow \mathcal{O}_B(1, 1) \xrightarrow{g} \mathcal{O}_B(2, 2) \longrightarrow \mathcal{O}_Y(2, 2) \longrightarrow 0, \quad H^1(B, \mathcal{O}_B(1, 1)) = 0.$$

Let $\mathcal{A} = \mathbb{P}_B(\mathcal{O} \oplus \mathcal{O}(1, 1))$. The actual double cover of Y is

$$X = V(g, v^2 - Fu^2) \subset \mathcal{A}. \quad (6.4)$$

In the basis (H_1, H_2, ξ) its Cox degrees are

$$\deg x_i = (1, 0, 0), \quad \deg y_i = (0, 1, 0), \quad \deg u = (0, 0, 1), \quad \deg v = (1, 1, 1);$$

the equation degrees are $(1, 1, 0)$ and $(2, 2, 2)$. The ambient is a smooth projective toric fivefold.

Indeed, g cuts out the smooth irreducible projective bundle $\mathbb{P}_Y(\mathcal{O}_Y \oplus \mathcal{O}_Y(1, 1))$. On it the second equation is the nonzero Cartier equation of Proposition 6.5; its local proof does not require a toric base. Thus the two equations form a regular sequence and define exactly the smooth cover. The same unit-ideal argument excludes every point and scheme structure at $u = 0$. Replacing F by $F + gG$, where G has degree $(1, 1)$, changes neither the ideal nor the resulting scheme.

The flag meets the base torus, for example at $x = (1, 1, 1)$, $y = (1, 1, -2)$. Since Y is irreducible, its torus intersection is dense. A nonzero branch section is nonzero at some point of that intersection, and the two cover points have $uv \neq 0$. This verifies the ambient dense-torus hypothesis for every smooth branch member. Finally $-K_X = \pi^* \mathcal{O}_Y(1, 1)$ is ample. The two all-member presentations of family 2.6 are recorded in [12, Section 5, opening paragraph] and [13, Lemma 12.3, proof].

Theorem 6.11. *Every smooth double cover of the flag threefold branched in a smooth anticanonical divisor satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka. No condition excluding ruling fibres contained in the branch divisor is imposed.*

Proof. The two equations in (6.4) give the exact regular codimension-two model in the smooth toric fivefold $\mathcal{A}$. The preceding checks establish smoothness, rational connectedness and intersection with its dense torus. Apply Theorem 5.1. $\square$

6.5 Weighted models and smooth toric resolutions

For the weighted models below, one can replace the weighted ambient by a smooth projective toric resolution without changing the target or adding equations. The reason is an explicit Cartier equation that is nonzero on the whole singular locus. This is checked separately below; merely knowing that the final intersection misses the singular locus would not justify arbitrary pullbacks of non-Cartier equations.

Choose a projective toric resolution $r : \tilde{P} \rightarrow P$ that is an isomorphism over the smooth locus. Such a resolution follows by star subdivisions preserving the smooth cones; see [18, Theorem 11.1.9 and the following discussion]. The resolution is projective over P ; since P is projective, $\tilde{P}$ is projective as well.

1. For the sextic double solid in $P = \mathbb{P}(1, 1, 1, 1, 3)$, the equation $w^2 - F_6(x)$ is a section of the Cartier line $\mathcal{O}_P(6)$. The only singular point has all $x_i = 0$, and the equation is nonzero there. Its pullback is therefore a unit at every point above that singular point, and its zero scheme is exactly the original smooth threefold in the isomorphism locus.
2. For the quartic double solid in $P = \mathbb{P}(1, 1, 1, 1, 2)$, the same argument applies to the Cartier section $Q_4 = w^2 - F_4(x)$ of $\mathcal{O}_P(4)$ and the unique weight-two singular point.
3. For the degree-one model in $P = \mathbb{P}(1, 1, 1, 2, 3)$, the section

$$Q_6 = z^2 - y^3 - A_4(x)y - B_6(x)$$

belongs to the Cartier line $\mathcal{O}_P(6)$. The singular locus consists of the two coordinate points with only y or only z nonzero. At these points its values are, respectively, $-y^3$ and z^2 , both nonzero. Thus r^*Q_6 has no zeros over either singular point. Notice that the point of the hypersurface with $x = 0$ has both $y, z \neq 0$ and is already in the smooth locus, since $\gcd(2, 3) = 1$; it is not removed by this argument.

4. For the double cover of a smooth quadric, use $P = \mathbb{P}(1, 1, 1, 1, 1, 2)$ and the two Cartier equations

$$q_2(x) = 0, \quad w^2 - F_4(x) = 0.$$

The second is nonzero at the unique singular point. Therefore their common pulled-back zero scheme lies wholly in the isomorphism locus and equals the given threefold. The pulled-back quadric may contain exceptional divisors by itself; the quartic equation excludes all of them from the intersection.

5. For the graph presentations of families 2.1 and 2.3, take $r \times \text{id}_{\mathbb{P}^1}$ on the respective weighted ambient times $\mathbb{P}^1$. The equation $Q_6 = 0$ or $Q_4 = 0$ remains a Cartier equation nonzero above every ambient singularity. The graph equation

$$G = \ell_1(x)t_0 - \ell_0(x)t_1$$

is a section of the rank-one reflexive sheaf $\mathcal{F} = \mathcal{O}_P(1) \boxtimes \mathcal{O}_{\mathbb{P}^1}(1)$. Its pullback has a natural regular section in

$$\tilde{\mathcal{F}} = ((r \times \text{id})^* \mathcal{F})^{**}.$$

This rank-one reflexive sheaf is invertible on the smooth ambient; it is torus-equivariant because the original sheaf and the morphism are. Over the isomorphism locus it agrees with

$\mathcal{F}$, together with the section G . The common zero scheme of the two lifted sections thus equals the original graph, including its exceptional divisor over the elliptic centre. It has no other points or scheme structure: the first section is a unit on the ambient exceptional locus, and both sections agree with the original equations elsewhere.

In all cases the resulting ambient is smooth and the target is the same smooth threefold with exactly the original number of equations and the original codimension. Sections of the resulting equivariant line bundles are Cox-homogeneous polynomials, by the section identification recalled in Section 5. At every common zero the local equations therefore form a regular sequence and have independent normal differentials. For the graph, this can also be checked directly: off the centre the incidence equation has nonzero $\mathbb{P}^1$ differential, and above the centre it is a nonzero combination of the independent centre conormals.

These models also meet the ambient dense torus. The double-solid covers are surjective over their projective bases and are not contained in the ramification divisor. The degree-one hypersurface is irreducible and is contained in no coordinate hyperplane, as the nonzero z^2 and y^3 terms show. A smooth quadric is contained in no coordinate hyperplane, and its nonempty branch divisor is proper. In the graph cases, both independent linear forms ℓ_0, ℓ_1 are nonzero on a dense open of the irreducible source, so both t_0, t_1 are nonzero on a dense open of the graph. Taking these finitely many nonempty opens together gives a torus point. The resolution preserves that torus.

These are case-specific arguments for the displayed equations. They do not assert that the strict transform of an arbitrary weighted complete intersection is globally cut out by the same number of equations on an arbitrary resolution.

Theorem 6.12. *Every smooth double cover of a smooth quadric threefold branched in $|\mathcal{O}_{Q^3}(4)|$ satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka.*

Proof. Write $Q^3 = \{q_2(x) = 0\} \subset \mathbb{P}^4$. Every branch section lifts to an ambient quartic F_4 , by the restriction sequence $0 \rightarrow \mathcal{O}_{\mathbb{P}^4}(2) \rightarrow \mathcal{O}_{\mathbb{P}^4}(4) \rightarrow \mathcal{O}_{Q^3}(4) \rightarrow 0$. The double cover is

$$X = \{q_2(x) = 0, w^2 = F_4(x)\} \subset \mathbb{P}(1, 1, 1, 1, 1, 2).$$

Its smoothness and the smooth branch give independent equation differentials. If every x_i vanished, the second equation would force $w = 0$, which is impossible. Thus the entire threefold lies in the free locus. Weighted adjunction gives $-K_X = \mathcal{O}_X(1)$, the ample hyperplane pullback under the finite cover. The construction in Subsection 6.5 realizes this same two-equation zero scheme in a smooth projective toric ambient variety and verifies its dense-torus intersection. Theorem 5.1 applies. $\square$

The quartic and double-quadric presentations together cover all smooth genus-three members of family 1.2; see [14, Appendix B, Table 1, entry 1.2].

Theorem 6.13. *Every smooth del Pezzo threefold of index two and degree one satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka.*

Proof. For every such smooth threefold the weighted model is

$$U = \{z^2 = y^3 + A_4(x)y + B_6(x)\} \subset \mathbb{P}(1, 1, 1, 2, 3),$$

where $x = (x_0, x_1, x_2)$ has weight one. The weighted model and its nonzero square and cube coefficients for the arbitrary smooth member are given in [38, Section 3, Equation and construction of U]. Completing the square and cube gives the displayed normalization. If

$x \neq 0$, the weighted stabilizer is trivial. At $x = 0$, the equation forces $y, z \neq 0$, and their coprime weights two and three again give trivial stabilizer. The half-anticanonical base point is therefore included in the free locus. Smoothness gives the regular hypersurface equation, and weighted adjunction gives $-K_U = 2H$, where $H = \mathcal{O}_U(1)$ is ample. Subsection 6.5 gives the same hypersurface in a smooth projective toric fourfold, including the base point, and checks that it meets the dense torus. Theorem 5.1 proves the assertion, including every member of family 1.11. $\square$

6.6 The elliptic-centre graph models

The incidence equation must describe the actual blowup, including its exceptional divisor. The regular centre ideals below give precisely this graph through their Rees algebras.

Corollary 6.14. *Let $U = \{z^2 = y^3 + A_4(x)y + B_6(x)\} \subset \mathbb{P}(1, 1, 1, 2, 3)$ be smooth, and let C be a smooth intersection of two members of $|\mathcal{O}_U(1)|$. Then $\text{Bl}_C U$ is Oka and satisfies $\mathcal{P}_N$ for every $N \geq 1$.*

Proof. For independent forms ℓ_0, ℓ_1 cutting out C , the blowup is scheme-theoretically the graph

$$X = \{z^2 - y^3 - A_4(x)y - B_6(x) = 0, \ell_1(x)t_0 - \ell_0(x)t_1 = 0\} \subset \mathbb{P}(1, 1, 1, 2, 3) \times \mathbb{P}^1.$$

The centre ideal is a regular sequence, so its Rees algebra has exactly this incidence relation. The graph is smooth, and the two equation differentials are independent: away from C the incidence differential has a nonzero projective-line component; over C it is a nonzero combination of the independent centre differentials. At $x = 0$ the equation forces $y, z \neq 0$, whose weights have gcd one; elsewhere a weight-one coordinate is nonzero. Thus X lies in the free product locus. With $H = \mathcal{O}_X(1, 0)$ and $P = \mathcal{O}_X(0, 1)$, adjunction gives $-K_X = H + P$. Both are nef and $6H + P$ is ample, by the degree-six weighted embedding and the graph construction. Hence $H + P = \frac{1}{6}(6H + P) + \frac{5}{6}P$ is ample. Rational connectedness follows from the Fano theorem. The two-section resolution construction in Subsection 6.5 preserves the actual graph and its dense-torus intersection. Hence Theorem 5.1 applies. The all-member model for family 2.1 is recorded in [2, Section 4.3]. $\square$

Theorem 6.15. *Let V be a smooth quartic double solid and let C be a smooth intersection of two members of $|H|$, where H is the hyperplane pullback from $\mathbb{P}^3$. Then $\text{Bl}_C V$ satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka. This covers every smooth member of Mori–Mukai family 2.3.*

Proof. With $C = V \cap \{\ell_0 = \ell_1 = 0\}$, its ideal is a regular sequence. The Rees algebra therefore gives the actual blowup as

$$X = \{W^2 - F_4(x) = 0, \ell_1(x)t_0 - \ell_0(x)t_1 = 0\} \subset \mathbb{P}(1, 1, 1, 1, 2) \times \mathbb{P}^1.$$

Off C , the incidence equation has a nonzero differential in the $\mathbb{P}^1$ direction. Over C , its differential is $t_0 d\ell_1 - t_1 d\ell_0$, a nonzero combination of independent conormal vectors on V . Together with the quartic equation this proves the required codimension and differential independence. Some weight-one coordinate is nonzero everywhere on V , so the graph lies in the free product locus, including its exceptional divisor.

For $H = \mathcal{O}_X(1, 0)$ and $P = \mathcal{O}_X(0, 1)$, adjunction gives $-K_X = H + P$. Both classes are nef, and $2H + P$ is ample: the weighted degree-two coordinates $(x_i x_j, W)$ embed V , and X is a closed subvariety of its product with $\mathbb{P}^1$. Thus $H + P = \frac{1}{2}(2H + P) + \frac{1}{2}P$ is ample. The Fano rational-connectedness theorem applies to the actual blowup. Subsection 6.5 realizes

this graph as the exact two-equation zero scheme in a smooth toric resolution and verifies the dense-torus condition. Theorem 5.1 therefore applies. Finally, every smooth member of family 2.3 has this model, with a smooth centre, by [12, Section 2, opening paragraphs]. $\square$

7 Balanced curves on the matrix models

The scalar transfer for matrices requires equal splitting of the actual tautological rank-two bundle. The following rooted construction supplies it at every point.

7.1 Balanced curves through a prescribed point

Here a full component means an irreducible component of the entire stable-map space, not a subfamily obtained by fixing a root or imposing an incidence condition. For such a component M , let $M^{(1)}$ be the closure of its free irreducible-domain locus with one source marking in $\overline{M}_{0,1}(X, \alpha)$. The one-pointed family in the following statement is this component, with its proper evaluation map to X .

Theorem 7.1. *Let X be a smooth rationally connected complex projective variety and E a rank-two vector bundle on X . Let M be a full irreducible component of $\overline{M}_{0,0}(X, \alpha)$ whose general point parametrizes a free map with irreducible domain. Assume that evaluation from a resolution of the proper one-pointed family over M has connected fibres, that E is slope-semistable with respect to α , and that $c_1(E) \cdot \alpha$ is odd. For every $x \in X$ there is a very free map $u : \mathbb{P}^1 \rightarrow X$ with*

$$u(0) = x, \quad u^*E \simeq \mathcal{O}(e)^{\oplus 2}$$

for some integer e . If $\det E$ is ample, then $e > 0$.

Proof. Let N be the normalization of $M^{(1)}$. A resolution $R \rightarrow M^{(1)}$ factors through a proper surjection $R \rightarrow N$. For each $x \in X$, the fibre N_x is the image of the connected fibre R_x , hence is connected. Thus M belongs to the connected-evaluation locus used in [54, §3 and Corollary 3.4]. Write $c_1(E) \cdot \alpha = 2c + 1$. That restriction theorem [54, Corollary 3.4] gives a nonempty open $M^\circ \subset M$ of free maps satisfying

$$v^*E \simeq \mathcal{O}(c+1) \oplus \mathcal{O}(c).$$

The semistability here is measured against the actual curve class α : $\mu_\alpha(F) = c_1(F) \cdot \alpha / \text{rank } F$. It is not semistability for an unrelated fixed polarization.

To construct the maximal-degree lines in a genuine family, take a smooth irreducible open H in the parameterized morphism-space component lying over M° , and write

$$p : Y = \mathbb{P}^1 \times H \longrightarrow H, \quad g : Y \longrightarrow X, \quad \mathcal{A} = g^*E.$$

These are the actual source curves, not the fibres of the coarse stable-map forgetful morphism. Every fibre of $\mathcal{A} \otimes \text{pr}_1^* \mathcal{O}(-c-1)$ is $\mathcal{O} \oplus \mathcal{O}(-1)$, so its h^0 is one and its h^1 is zero. Cohomology and base change therefore make

$$\mathcal{L} = p_*(\mathcal{A} \otimes \text{pr}_1^* \mathcal{O}(-c-1))$$

a line bundle. Its evaluation map defines the line subbundle

$$F_0 = p^* \mathcal{L} \otimes \text{pr}_1^* \mathcal{O}(c+1) \hookrightarrow g^*E.$$

Indeed, on each source fibre this map is the nowhere-vanishing inclusion of the unique maximal-degree summand. Its image is intrinsic under reparameterization and defines an algebraic map $\lambda : Y \rightarrow \mathbb{P}_X(E)$ of lines. Freeness makes g smooth, by $H^1(v^*T_X(-t)) = 0$, so it is dominant.

For general $q \in X$, the maximal-degree lines of members of M° through q cannot all agree. Otherwise [54, Lemma 2.6] supplies a saturated rank-one subsheaf $L \subset E$. In this rank-one situation the descent can also be seen directly: the closure of $\lambda(Y)$ has one geometric point over the general point of X , hence is generically finite of degree one in characteristic zero. It gives a rational line in E , whose saturation is L . The composite $F_0 \rightarrow g^*E \rightarrow g^*(E/L)$ vanishes wherever that rational line is defined. A general member avoids the fixed locus of codimension at least two where L or E/L is not locally free. Indeed, on $Y = \mathbb{P}^1 \times H$ that inverse image has dimension at most $\dim H - 1$, and its image under the proper projection p is a proper closed subset of H . On a curve avoiding it, v^*L and $v^*(E/L)$ are vector bundles. The preceding composite vanishes on a dense open of the curve and therefore everywhere. Thus v^*L is a line subbundle containing $\mathcal{O}(c+1)$, contrary to semistability since $c+1 > (2c+1)/2$. The incidence of pairs of pointed members with equal evaluation and distinct maximal-degree lines has constructible dense image in X . It therefore covers a dense open $U \subset X$. At every point of U , at least two maximal-degree lines are available, so one can avoid any prescribed line in that fibre. The full-component hypothesis is used here only for the unrestricted tooth family M .

Fix $x \in X$. Smooth projectivity and rational connectedness give a very free map u with $u(p) = x$; see [19, Theorem 2.40]. Write $u^*E = \mathcal{O}(a) \oplus \mathcal{O}(b)$ with $a \geq b$. Suppose $a > b$. For any $q_0 \neq p$, evaluation at q_0 on the space of maps fixing $p \mapsto x$ is smooth at u , because

$$H^1(\mathbb{P}^1, u^*T_X(-p - q_0)) = 0.$$

Within the open set where very freeness persists and the splitting gap does not increase, choose a deformation with $u(q_0) \in U$. Relabel its splitting by (a, b) and stop if the summands are equal. Otherwise choose a tooth $v \in M^\circ$ through $q = u(q_0)$ whose maximal-degree line differs from that of u at q_0 .

Glue the two sources at these points to obtain $f : C = C_0 \cup C_1 \rightarrow X$, retaining p on the handle C_0 . The family

$$\pi : \text{Bl}_{(q_0, 0)}(\mathbb{P}^1 \times \mathbb{A}^1) \longrightarrow \mathbb{A}^1$$

has central fibre C and a section σ through p . For $j = 1, 2$, normalization gives

$$0 \longrightarrow f^*T_X(-jp) \longrightarrow u^*T_X(-jp) \oplus v^*T_X \longrightarrow T_{X,q} \longrightarrow 0.$$

The component H^1 groups vanish, and evaluation from the free tooth onto $T_{X,q}$ is surjective. Consequently both $H^1(C, f^*T_X(-p))$ and $H^1(C, f^*T_X(-2p))$ vanish. The first vanishing makes the relative morphism scheme, with σ mapped constantly to x , smooth over $\mathbb{A}^1$ at f . An étale-local section through f therefore supplies a family F of maps extending f and preserving the root. The second vanishing and semicontinuity ensure that the maps on nearby smooth fibres are very free.

Choose auxiliary sections σ_H, σ_T through smooth points of the central handle and tooth. They exist after a further étale base change, since the source family is smooth at these points. Set

$$\mathcal{B} = F^*E \otimes \mathcal{O}(-(b+1)\sigma_H - (c+1)\sigma_T).$$

Its central restrictions are

$$\mathcal{B}|_{C_0} = \mathcal{O}(a - b - 1) \oplus \mathcal{O}(-1), \quad \mathcal{B}|_{C_1} = \mathcal{O} \oplus \mathcal{O}(-1).$$

Both have zero H^1 . Their spaces of sections evaluate at the node onto the two distinct maximal-degree lines, which span the rank-two node fibre. Normalization thus gives $H^1(C, \mathcal{B}|_C) = 0$. On a smooth fibre, the twisting line has degree $-b - c - 2$, so

$$H^1(\mathbb{P}^1, F_t^* E(-b - c - 2)) = 0.$$

If $F_t^* E = \mathcal{O}(A) \oplus \mathcal{O}(B)$ with $A \geq B$, then

$$B \geq b + c + 1, \quad A + B = a + b + 2c + 1, \quad 0 \leq A - B \leq a - b - 1.$$

The gap has decreased while the root and very freeness are preserved. Repeating this operation terminates with equal summands. An automorphism of the source sends the retained root to 0. Finally, ampleness of $\det E$ gives $2e = \deg u^* \det E > 0$. $\square$

7.2 The genus-eight tautological bundle

Corollary 7.2. *Let*

$$X = \mathrm{Gr}(2, V) \cap \mathbb{P}(W), \quad \dim V = 6, \quad \dim W = 10,$$

be any smooth dimensionally transverse complex linear section, and let $\mathcal{S}$ be the actual tautological subbundle restricted to X . Every $x \in X$ lies on a very free map $u : \mathbb{P}^1 \rightarrow X$ such that

$$u(0) = x, \quad u^* \mathcal{S} \simeq \mathcal{O}(-e)^{\oplus 2} \quad \text{for some } e > 0.$$

Proof. Let $H = \mathcal{O}_X(1)$ and $E = \mathcal{S}^*|_X$. The Grassmannian model gives

$$\mathrm{Pic}(X) = \mathbb{Z}H, \quad -K_X = H, \quad H^3 = 14, \quad \det E = H.$$

We first check stability of E . A nonzero section of $\mathcal{S}$ would give a fixed vector contained in every two-plane represented by X . Then X would lie in the linear $\mathbb{P}^4$ of such two-planes. Its intersection with the actual linear section X is linear, forcing X itself to be linear, contrary to $H^3 = 14$. Thus $H^0(X, \mathcal{S}) = 0$. A destabilizing rank-one subsheaf would extend, after taking its double dual, to a nonzero map $\mathcal{O}_X(kH) \rightarrow E$ with $k \geq 1$. Since $E(-H) \simeq \mathcal{S}$, the case $k = 1$ is excluded. For $k > 1$, use

$$E(-kH) = \mathcal{S}(-(k-1)H) \hookrightarrow V \otimes \mathcal{O}_X(-(k-1)H).$$

Hence E is stable with respect to H^2 and, since the numerical curve space has dimension one, with respect to every nonzero effective curve class.

We next construct an odd-degree very free component. Every such threefold contains an actual line ℓ ; the all-member existence statement is recorded in [52, §2.1, before Lemma 2.1.4]. The Fano rational-connectedness theorem [48] supplies rational connectedness of X . Attach sufficiently many very free maps to distinct points of the line handle, precomposing each tooth by a degree-two source map so that every tooth has even H -degree. The comb theorem [19, Theorem 2.39(b)] smooths a subcomb retaining the handle to a very free map. Its degree is one plus a sum of even degrees and is therefore odd. Possible loss of teeth does not change this parity.

For completeness, this map deforms to an embedding of the same degree. On a smooth irreducible finite-type open $\mathcal{H}$ of its full morphism-space component, consisting of very free maps, two-point evaluation is smooth for distinct source points, by $H^1(v^* T_X(-s-t)) = 0$. The coincident-image incidence in $\mathcal{H} \times ((\mathbb{P}^1)^2 \setminus \Delta)$ consequently has dimension at most $\dim \mathcal{H} + 2 - 3 = \dim \mathcal{H} - 1$. First-jet evaluation is likewise smooth by $H^1(v^* T_X(-2s)) = 0$,

so the ramification incidence has dimension at most $\dim \mathcal{H} + 1 - 3 = \dim \mathcal{H} - 2$. Their constructible projections have proper closures. Outside these closures the maps are injective immersions and hence, by properness, embeddings.

Let M be the resulting full free-curve component, of odd degree. Its general member is very free and evaluation is dominant. The connected-fibre theorem [8, Theorem 1.3] applies because $-K_X = H$ is very ample. Its conic-image alternative is excluded by odd degree, and Picard rank one excludes a del Pezzo fibration. Thus the general fibre of the proper evaluation $M^{(1)} \rightarrow X$ is irreducible. In characteristic zero its geometric generic fibre is reduced and irreducible; equivalently, $\mathbb{C}(X)$ is relatively algebraically closed in $\mathbb{C}(M^{(1)})$. A resolution $R \rightarrow M^{(1)}$ does not change this function field. In its Stein factorization

$$R \longrightarrow S = \operatorname{Spec}_X(\operatorname{ev}_R)_* \mathcal{O}_R \longrightarrow X,$$

the first arrow has connected fibres and the second is finite. The relative algebraic-closure statement makes the finite arrow birational. Since X is normal, that arrow is an isomorphism, so all fibres of $R \rightarrow X$ are connected. Theorem 7.1 now gives $u^*E = \mathcal{O}(e)^2$ through every prescribed x , with $e > 0$.

Finally, the isomorphism $u^*\mathcal{S} \simeq \mathcal{O}(-e)^2$, followed by the actual tautological injection into $V \otimes \mathcal{O}$, gives a 6×2 matrix of degree- e binary forms with rank two at every source point. Its Plücker minors satisfy the five defining linear equations of X . Thus the splitting provides the actual equal-column-degree tautological lift. $\square$

7.3 Gushel–Mukai threefolds

Let X be a smooth complex Gushel–Mukai threefold and let $G = \operatorname{Gr}(2, V)$, where $\dim V = 5$. Its two presentations are

$$\begin{aligned} \text{ordinary: } & X = G \cap H_1 \cap H_2 \cap Q, \\ \text{special: } & \pi : X = \{z^2 = q(P)\} \longrightarrow Y = G \cap \mathbb{P}^6. \end{aligned}$$

Here P denotes Plücker coordinates; in the special case Y and its branch divisor $B = Y \cap \{q = 0\}$ are smooth, and z has the same weight as P [20, Propositions 2.22 and 2.30]. The Gushel bundle $\mathcal{U}$ is the restriction of the tautological bundle in the ordinary case and its pullback by π in the special case. With $H = \det \mathcal{U}^*$, one has

$$-K_X = H, \quad H^3 = 10, \quad \operatorname{Pic}(X) = \mathbb{Z}H \quad [20, \text{Lemma 2.29}].$$

The full anticanonical system is very ample in both cases. For the special model, $\pi_* \mathcal{O}_X = \mathcal{O}_Y \oplus \mathcal{O}_Y(-H_Y)$ gives

$$H^0(X, H) = H^0(Y, H_Y) \oplus \mathbb{C}z.$$

It embeds X as a quadric section of the cone over Y in $\mathbb{P}^7$; only the subsystem omitting z gives the double cover.

Proposition 7.3. *For every $x \in X$ there is a very free map $u : \mathbb{P}^1 \rightarrow X$ with $u(p) = x$ and $u^*\mathcal{U} \simeq \mathcal{O}(-e)^{\oplus 2}$ for some $e > 0$. Its actual matrix presentation has surjective rooted equation derivative, also after augmentation by the three geometric root-derivative coordinates. These assertions include every ramification anchor in the special model.*

Proof. Let $E = \mathcal{U}^*$. A nonzero section of $\mathcal{U}$ would be a fixed vector of V contained in every Gushel two-plane. Its image would then lie in the Schubert $\mathbb{P}^3$ of such planes and, being closed of dimension three, equal that space. This contradicts its degree, ten in the ordinary case and five in the special case. Thus $H^0(X, \mathcal{U}) = 0$. A destabilizing rank-one subsheaf of E would extend,

after taking its double dual, to a nonzero map $\mathcal{O}_X(kH) \rightarrow E$ with $k \geq 1$. Since $E(-H) \simeq \mathcal{U}$, the preceding vanishing excludes $k = 1$; for $k > 1$ use $\mathcal{U}(-(k-1)H) \hookrightarrow V \otimes \mathcal{O}_X(-(k-1)H)$. Hence E is stable with respect to every nonzero effective curve class, since the numerical curve space has dimension one.

There is an anticanonical line on every such X . For its associated data (V_6, V_5, A) , smoothness excludes decomposable vectors in A [20, Theorem 3.16]; the corank-two EPW locus $Y_A^{\geq 2}$ is a nonempty projective surface [21, §2.2]. Its hyperplane section $Y_A^{\geq 2} \cap \mathbb{P}(V_5)$ is nonempty. With the notation of [21, §2.2], this is precisely the scheme $Y_{A, V_5}^{\geq 2}$, obtained by base change from $\mathbb{P}(V_6)$ to $\mathbb{P}(V_5)$. The morphism of [21, Theorem 4.7(d)] maps $F_1(X)$ onto this scheme. Away from $\Sigma_1(X)$ it is an isomorphism; over $\Sigma_1(X) \cap Y_{A, V_5}^{\geq 2}$ it is the double cover described there, branched over $\Sigma_1(X) \cap Y_{A, V_5}^3$. In particular, the exceptional stratum does not remove points from its image: the corresponding plane conic is a pair of lines or a double line. This proves $F_1(X) \neq \emptyset$ for both ordinary and special threefolds, without a general-member restriction.

The Fano rational-connectedness theorem [48] and [19, Theorem 2.40] provide very free maps through prescribed points. Attach sufficiently many such maps to distinct points of a line, precomposing every tooth by a degree-two source map. By [19, Theorem 2.39(b)], a subcomb retaining the line smooths to a very free map. Its degree is odd, irrespective of lost teeth. The two-point and first-jet incidence argument in the proof of Corollary 7.2 deforms this map to an embedding of the same degree. Let M be the resulting full free-curve component. Since $-K_X$ is very ample, [8, Theorem 1.3] applies: odd degree excludes conic images, and Picard rank one excludes a del Pezzo fibration. The general evaluation fibre is therefore irreducible. Applying the function-field and Stein-factorization argument in the proof of Corollary 7.2 to this proper one-pointed component gives connected fibres after resolution. All hypotheses of Theorem 7.1 hold for E , and it supplies the asserted map through every x , with $e > 0$ since $\det E = H$ is ample.

The actual inclusion $u^*\mathcal{U} \hookrightarrow V \otimes \mathcal{O}$ gives a 5×2 matrix $M(s, t)$ of degree- e binary forms, of rank two everywhere. Its root value can be any prescribed frame of $\mathcal{U}_x$. In the special model the additional coordinate is $Z \in H^0(\mathbb{P}^1, \mathcal{O}(2e))$. The equations are

$$\begin{aligned} \text{ordinary: } & \ell_1(\wedge^2 M) = \ell_2(\wedge^2 M) = q(\wedge^2 M) = 0, \\ \text{special: } & \ell_j(\wedge^2 M) = 0 \ (1 \leq j \leq 3), \quad Z^2 = q(\wedge^2 M). \end{aligned}$$

A frame change $g \in \mathrm{GL}_2$ acts on Z by $\det g$. Thus these are exactly the two GM matrix presentations of Theorem 5.1. Its rooted coefficient calculation applies with vertical bundle $\mathrm{End}(u^*\mathcal{U})(-p) = \mathcal{O}(-1)^4$:

$$J : \mathbb{C}^{10e} \twoheadrightarrow \mathbb{C}^{8e} \quad \text{or} \quad J : \mathbb{C}^{12e} \twoheadrightarrow \mathbb{C}^{10e}, \quad \ker J \simeq H^0(u^*T_X(-p)).$$

The first case is ordinary and the second special. Since $H^1(u^*T_X(-2p)) = 0$, differentiation at p from this kernel is onto $T_x X \otimes T_p^* \mathbb{P}^1$. The augmented ranks are $10e \rightarrow 8e+3$ and $12e \rightarrow 10e+3$, respectively. No division by $Z(p)$ occurs, so this includes ramification anchors. Openness of equal splitting and very freeness, together with this smooth root-derivative map, supplies three nearby rooted maps of the same degree with independent geometric derivatives. $\square$

Corollary 7.4. *Every smooth ordinary or special Gushel–Mukai threefold and every smooth prime Fano threefold of genus eight satisfies $\mathcal{P}_N$ for every $N \geq 1$ and is Oka.*

Proof. Every smooth prime Fano threefold of genus eight has the stated Grassmannian model by [6, Theorem 1.2] and [7, Theorem 1.1(a)]. The ordinary and special Gushel–Mukai presentations are those of [20, Sections 2.4–2.5]. The genus-eight and Gushel–Mukai balancing results give an actual equal-split very free curve through each prescribed anchor. Their model

presentations are regular with the complete normal equations listed in Theorem 5.1. The first-jet and coefficient calculations there supply three free equal-split rooted curves with independent derivatives. Case (ii) of that theorem applies to each actual model. $\square$

8 Completion of the classification argument

The preceding sections place fourteen family labels under the common spectral transfer theorem by verifying its hypotheses on their geometric models. The remaining ninety-one are supplied by external theorems. We now verify the hypotheses of those theorems on the classified models and join the two groups into a proof for all 105 families.

8.1 The classified families

We use the classification labels of [2, Table 6.1]. The numbers of families of Picard rank $1, \dots, 10$ are respectively

$$17, 36, 31, 13, 3, 1, 1, 1, 1, 1.$$

Thus there are 105 families. Throughout this section a family means all its smooth complex members. In particular, a centre occurring in a blowup description satisfies the position conditions in the classification; an arbitrary curve of the same degree and genus is not being substituted for that centre.

Three families have two presentations: quartic threefolds and double covers of smooth quadric threefolds for 1.2, ordinary and special Gushel–Mukai threefolds for 1.5, and Verra and double-flag threefolds for 2.6. Both presentations must be treated, giving 108 presentations of 105 families, with each pair counted as one family.

The families requiring the model-specific arguments used in this paper are

$$\begin{aligned} &1.1, 1.2, 1.3, 1.4, 1.5, 1.7, 1.11, 1.12, \\ &2.1, 2.2, 2.3, 2.6, 2.8, 3.1. \end{aligned}$$

The remaining 91 families have the following two kinds of external proofs. Published uniform-rationality and cubic results cover 84 families. Published rationality results, together with Banecki’s preprint theorem, cover the seven families

$$1.6, 1.8, 1.9, 1.10, 2.18, 3.2, 3.4.$$

We give the arguments for these 91 families before joining them to the results of the preceding sections. This division concerns the proofs used here; it does not assert that the corresponding geometric properties are mutually exclusive.

8.2 Uniformly rational families

A smooth algebraic variety is uniformly rational if each point has a Zariski neighbourhood isomorphic to a Zariski open subset of affine space. Arzhantsev–Kaliman–Zaidenberg [3, Theorem 1.2] recall that blowing up a uniformly rational variety along a smooth closed centre of codimension at least two preserves uniform rationality. The same authors prove that every complete uniformly rational variety of positive dimension is algebraically elliptic [3, Theorem 1.3]. Algebraic ellipticity supplies a holomorphic dominating spray, so the subelliptic Oka principle [27, Theorem 1.1] gives the full Oka property. We retain uniform rationality through the blowup constructions and apply algebraic ellipticity to the final complete variety.

Products and projectivizations of algebraic vector bundles preserve uniform rationality: intersect a Zariski trivializing neighbourhood with an affine-space chart of the base and take an affine chart in the projective-space fibre. Products use the same product-chart argument.

The varieties $\mathbb{P}^3$, the smooth quadric Q^3 , and the flag threefold are uniformly rational: translated big cells cover the two homogeneous varieties. Every smooth del Pezzo threefold V_4 of degree four is uniformly rational by [3, Example 4.8(1)]. This applies to every smooth intersection of two quadrics, not just a general pencil. The pencil contains two independent smooth quadrics: Bertini gives smoothness away from the base, and independence of the differentials of the two defining quadrics gives smoothness along the base. The unique smooth complex del Pezzo threefold V_5 of degree five is covered by affine spaces by [4, Proposition 4(a)]. Finally, every smooth complex del Pezzo surface is uniformly rational. Such a surface is $\mathbb{P}^1 \times \mathbb{P}^1$ or is obtained from $\mathbb{P}^2$ by successive blowups at points, so the assertion follows from the same blowup theorem. We write S_d for a del Pezzo surface of degree d in the product rows of the classification; in particular $S_8 = \mathbb{F}_1$, whereas $(\mathbb{P}^1)^3$ occurs separately as family 3.27.

These facts immediately treat 1.14, 1.15, 1.16, 1.17, whose models are $V_4, V_5, Q^3, \mathbb{P}^3$. For Picard rank two, the classification gives smooth-centre blowups of $\mathbb{P}^3$ for

$$2.4, 2.9, 2.12, 2.15, 2.25, 2.27, 2.28, 2.30, 2.33, 2.35;$$

of Q^3 for

$$2.7, 2.13, 2.17, 2.21, 2.23, 2.29, 2.31;$$

of V_4 for 2.10, 2.16, 2.19; and of V_5 for 2.14, 2.20, 2.22, 2.26. Families 2.24 and 2.36 are projective bundles over $\mathbb{P}^2$, family 2.32 is the flag threefold, and family 2.34 is $\mathbb{P}^1 \times \mathbb{P}^2$. Hence all these 28 families are uniformly rational.

For Picard rank three, the corresponding smooth-centre blowup bases are $\mathbb{P}^3$ for

$$3.6, 3.12, 3.14, 3.18, 3.25, 3.26;$$

Q^3 for 3.10, 3.15, 3.19, 3.20; $\mathbb{P}^1 \times \mathbb{P}^2$ for 3.3, 3.5, 3.8, 3.21, 3.22; the flag threefold for 3.7, 3.13, 3.24; family 2.35 for 3.11, 3.16, 3.23, 3.29, 3.30; and family 2.36 for 3.9. Families 3.17 and 3.31 are projective bundles over $\mathbb{P}^1 \times \mathbb{P}^1$, while 3.27 and 3.28 are respectively $(\mathbb{P}^1)^3$ and $\mathbb{P}^1 \times \mathbb{F}_1$. This proves uniform rationality for these 28 families.

All thirteen rank-four families follow in the same way. Families 4.1, 4.3, 4.8, 4.13 are smooth-centre blowups of $(\mathbb{P}^1)^3$; 4.5 is a blowup of $\mathbb{P}^1 \times \mathbb{P}^2$; 4.6 is a blowup of $\mathbb{P}^3$; and 4.7 is a blowup of the flag threefold. Families 4.2, 4.4, 4.9, 4.12 are smooth-centre blowups of 3.31, 3.19, 3.25, 2.33, respectively. The remaining two are 4.10 = $\mathbb{P}^1 \times S_7$ and the blowup 4.11 of $\mathbb{P}^1 \times \mathbb{F}_1$ along the indicated smooth curve. Family 5.1 is the blowup of 2.29 along three disjoint exceptional fibres, and 5.2 is the blowup of 3.25 along the two indicated disjoint exceptional fibres. For disjoint centres these are successive smooth-centre blowups. Finally,

$$5.3, 6.1, 7.1, 8.1, 9.1, 10.1$$

are respectively $\mathbb{P}^1 \times S_d$ for $d = 6, 5, 4, 3, 2, 1$.

Several descriptions in these lists require checking that the construction is on a smooth model for every member. Here are the checks, which also avoid any appeal to genericity. For 2.24, the equation of bidegree $(1, 2)$ is linear in one $\mathbb{P}^2$ factor. Its three coefficients have no common zero on the other factor. If they vanished at a base point, the full $\mathbb{P}^2$ fibre would be contained in the hypersurface. The two base derivatives on that fibre would be two linear forms and would have a common zero, producing a singular point. The coefficient map is therefore everywhere surjective, and its rank-two kernel has the hypersurface as its projectivization.

The same argument applies to the equation of tridegree $(1, 1, 1)$ for 3.17, viewed as linear in its $\mathbb{P}^2$ factor.

For 3.3 and 4.1, an equation linear in one $\mathbb{P}^1$ factor is

$$u_0 f_0 + u_1 f_1 = 0.$$

At a common zero of f_0, f_1 , dependence of df_0, df_1 would give $[u_0 : u_1]$ for which the total differential vanishes. Smoothness of the hypersurface thus makes the pencil base smooth of codimension two. The displayed incidence hypersurface is its blowup, with bases $\mathbb{P}^1 \times \mathbb{P}^2$ and $(\mathbb{P}^1)^3$, respectively. The vertex resolutions in 3.9 and 3.31 are the smooth models

$$\mathbb{P}_{\mathbb{P}^2}(\mathcal{O} \oplus \mathcal{O}(2)), \quad \mathbb{P}_{\mathbb{P}^1 \times \mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(1, 1)),$$

respectively; the further centre in 3.9 is away from the vertex, and 4.2 is a further smooth-centre blowup of the second model. No blowup theorem for a smooth base is being applied directly to a singular cone. For 3.13, the proof of [13, Lemma 7.4] realizes every smooth member as a blowup of the flag threefold along a curve whose two projections are isomorphisms onto smooth conics. The strict-transform centres in 3.16, 3.23, 4.4 and the exceptional-fibre centres in 3.29, 3.30, 4.9, 4.12, 5.1, 5.2 are smooth in their indicated smooth models. These are the actual classified centres.

We have therefore proved the full Oka property for 81 families: 1.14–1.17; 2.4, 2.7, 2.9, 2.10, 2.12–2.17, 2.19–2.36; 3.3, 3.5–3.31; and every family of rank four through ten.

8.3 Cubic and rational families

Kaliman–Zaidenberg [46, Theorem 1.1] prove that every smooth cubic hypersurface of dimension at least two is algebraically elliptic. In particular every member of 1.13 is Oka. The two other cubic cases are 2.5, the blowup of a smooth cubic threefold along its classified smooth plane elliptic curve, and 2.11, its blowup along a line.

To apply the blowup result, let Y be an algebraically elliptic variety, with a dominating algebraic spray $s : E \rightarrow Y$. For a smooth affine variety T and an algebraic map $f : T \rightarrow Y$, the bundle f^*E has finitely many global generators. Their evaluation is a fibrewise surjection $T \times \mathbb{A}^m \rightarrow f^*E$, and composition with s gives an algebraic spray with core f that dominates at every point of T . Thus Y satisfies algebraic Ell_1 . Kusakabe [49, Corollary 1.5] proves that the blowup of such a variety along any smooth closed algebraic centre is Oka. This treats every classified centre in 2.5 and 2.11. The conclusion used here is holomorphic Oka; neither algebraic ellipticity of these blowups nor blowup invariance for arbitrary holomorphic Oka manifolds is required. Together with the preceding subsection, this accounts for the 84 published-theorem cases.

The four rank-one families 1.6, 1.8, 1.9, 1.10 have genera 7, 9, 10, 12, respectively. Kuznetsov–Prokhorov [51, Theorem 1.1(2)] prove rationality in genera 7 and 12 when a rational point exists. Over $\mathbb{C}$ this condition is automatic. For genus 9, their criterion is $F_3(X)(\mathbb{C}) \neq \emptyset$, where $F_3(X)$ is the Hilbert scheme of degree-three genus-zero curves used in [51, Theorem 1.1(3)(b)]. For every smooth member over an algebraically closed field, this scheme is isomorphic to $\text{Pic}^0(\Gamma)$ for a smooth genus-three curve Γ [51, Theorem 9.4]. It is nonempty. For genus 10, the criterion is nonemptiness of $X(\mathbb{C})$ and of the conic Hilbert scheme $F_2(X)(\mathbb{C})$ [51, Theorem 1.1(3)(c)]. The latter scheme is $\text{Pic}^0(\Gamma)$ for a smooth genus-two curve [51, Theorem 9.9]. These conditions hold for every smooth complex member. In particular the genus-12 argument is not confined to the Mukai–Umemura threefold.

Family 2.18 is a double cover of $\mathbb{P}^1 \times \mathbb{P}^2$ branched in bidegree $(2, 2)$. Projection to $\mathbb{P}^2$ is a conic bundle. Locally on the base its fibres have equations

$$w^2 = a(y)x_0^2 + b(y)x_0x_1 + c(y)x_1^2,$$

where a, b, c have degree two on $\mathbb{P}^2$, so its discriminant has degree four. The finite flat double cover and the product projection give flatness; smoothness of the total space, relative Picard rank one, and relative anticanonical ampleness give the conic-bundle hypotheses. Prokhorov [59, Corollary 5.6.1] therefore makes every member rational. This result allows reducible discriminants and nonreduced conic fibres; a general smooth discriminant is unnecessary. Every member of 3.4 is a blowup of such a member along a smooth conic fibre, hence is birational to it and is rational as well.

For 3.2, the base is $T = \mathbb{P}_s^1 \times \mathbb{P}_t^1$. The classified model is a quadratic divisor in

$$\mathbb{P}_T(\mathcal{O} \oplus \mathcal{O}(-1, -1)^{\oplus 2}).$$

The bidegrees of the entries of its symmetric conic matrix are

$$\begin{pmatrix} (2, 3) & (1, 2) & (1, 2) \\ (1, 2) & (0, 1) & (0, 1) \\ (1, 2) & (0, 1) & (0, 1) \end{pmatrix}.$$

Consequently its determinant has bidegree $(2, 5)$. A full $\mathbb{P}^2$ fibre cannot occur: the two base derivatives would then be two quadrics on that fibre, with a common zero at which the total space is singular. Every fibre is therefore a plane conic of pure dimension one, including a reducible conic or a double line. The total space is Cohen–Macaulay and the base is regular; the dimension equality $3 = 2 + 1$ gives flatness by miracle flatness [60, Lemma 10.128.1]. Generic smoothness then makes the determinant nonzero. The Fano condition supplies relative anticanonical ampleness, and the relative Picard rank is one. The base-point-free ruling pencil with members $\mathbb{P}_s^1 \times \{t\}$ has intersection number two with the discriminant. Prokhorov [59, Proposition 5.6], whose bound is three, proves rationality of every member of 3.2.

Finally, Banecki [5, Corollary 1.9] proves that every smooth rational complex projective variety is algebraically elliptic. We use the statement in arXiv:2411.17892v4, dated 3 August 2026; this is a preprint, not a published theorem. All seven varieties just treated are smooth, rational, and projective, so this result and the holomorphic Oka principle prove the full Oka property for 1.6, 1.8, 1.9, 1.10, 2.18, 3.2, 3.4. This is the only use of that preprint in the classification argument. The preceding internal arguments and published external theorems prove the result for the other 98 families. Thus the exact preprint dependency consists of the seven conclusions listed above.

8.4 The remaining families and the main theorem

The remaining fourteen families are covered for all smooth members. Corollary 6.6 gives 1.1, the sextic double solids. Theorems 6.1 and 6.12 give the quartic and double-quadric presentations of 1.2. Theorem 6.2 and Corollary 6.3 give 1.3, the quadric–cubic intersections in $\mathbb{P}^5$, and 1.4, the three-quadric intersections in $\mathbb{P}^6$, including the rank-five containing quadric in the former. Corollary 7.4 covers both ordinary and special presentations of 1.5 and the genus-eight Grassmannian model 1.7. All these cases use the same scalar-family property over a neighborhood of the entire old compact.

Theorem 6.13 and Corollary 6.7 give 1.11 and 1.12, the index-two del Pezzo threefolds of degrees one and two. The elliptic centres in 2.1 and 2.3 are the smooth complete intersections of two half-anticanonical divisors on these threefolds. Corollary 6.14 and Theorem 6.15 apply to the actual blowups; see also [2, Section 4.3] for 2.1. Both weighted-graph constructions apply the common theorem to the actual blowups; no birational substitute or Oka invariance under blowup is used.

Corollaries 6.8 and 6.9 cover 2.2 and 3.1, the smooth double covers with branch degrees $(2, 4)$ on $\mathbb{P}^1 \times \mathbb{P}^2$ and $(2, 2, 2)$ on $(\mathbb{P}^1)^3$, respectively. For 2.6, Theorems 6.4 and 6.11 cover both

the $(2, 2)$ hypersurfaces in $\mathbb{P}^2 \times \mathbb{P}^2$ and the smooth double covers of the flag threefold branched in anticanonical surfaces. The double-flag proof uses its exact two-equation zero scheme in a smooth toric projective bundle, without using deformation invariance. Theorem 6.10 gives 2.8, the double covers of $V_7 = \mathrm{Bl}_p \mathbb{P}^3$ branched in smooth anticanonical surfaces, including the reducible exceptional conic case.

The finite-dimensional constructions establish $\mathcal{P}_N$ for every finite N with the actual holomorphic map on an open neighbourhood of the whole old compact and an entire fresh parameter. Theorem 4.1 gives CAP_k from $\mathcal{P}_{k+3}$. All conclusions concern the actual smooth targets; the toric presentations do not invoke birational invariance.

Proof of Theorem 1.1. Let X be a smooth complex Fano threefold. According to the argument used, the classification divides the 105 family labels into 81 uniformly rational families, three cubic families, seven rational families, and the fourteen families treated above. Uniform rationality settles the first group; the cubic and algebraic Ell_1 arguments settle the second; the stated version of Banecki’s theorem settles the third; and the family theorems above settle the fourth. Each argument applies to X itself. Both presentations of 1.2, 1.5, 2.6 are included, and $81 + 3 + 7 + 14 = 105$. $\square$

Use of artificial intelligence

This paper developed through collaborative exploration with GPT-6-Astra. The authors brought the problem and the guiding ideas, notably the use of the classification of smooth Fano threefolds and a technique of Wang–Xie–Zhao, together with several key technical steps. Through repeated exchanges, GPT-6-Astra helped develop and test proof strategies, identify gaps, locate references, and draft and revise the manuscript; several arguments and formulations took shape in this process. The authors reviewed and revised the resulting material and take full responsibility for the mathematical content, the references, and the final text.

Funding

Song-Yan Xie received partial support from the National Key R&D Program of China (grants 2023YFA1010500 and 2021YFA1003100), the National Natural Science Foundation of China (grants 12288201 and 12471081), and the Xiaomi Young Talents Program.

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